

Unit 3

DYNAMICS

66 items: 34 from official papers (173 marks), 32 from the booklet

Syllabus sections in this unit

- 3.1 Momentum and Newton's laws of motion
- 3.2 Non-uniform motion
- 3.3 Linear momentum and its conservation

Unit 3: Dynamics**1. M/J 26/P22/Q6/b(i,ii)**

- (b) An α -particle travelling with a speed of $5.60 \times 10^6 \text{ m s}^{-1}$ collides with a stationary gold nucleus of mass 197 u. After the collision, the α -particle is travelling with a speed of $5.38 \times 10^6 \text{ m s}^{-1}$ in the opposite direction to its initial motion.

No external forces act on this system.

- (i) Show that the momentum of the α -particle before the collision is $3.72 \times 10^{-20} \text{ kg m s}^{-1}$.

[2]

- (ii) Use conservation of momentum to calculate the speed of the gold nucleus after the collision.

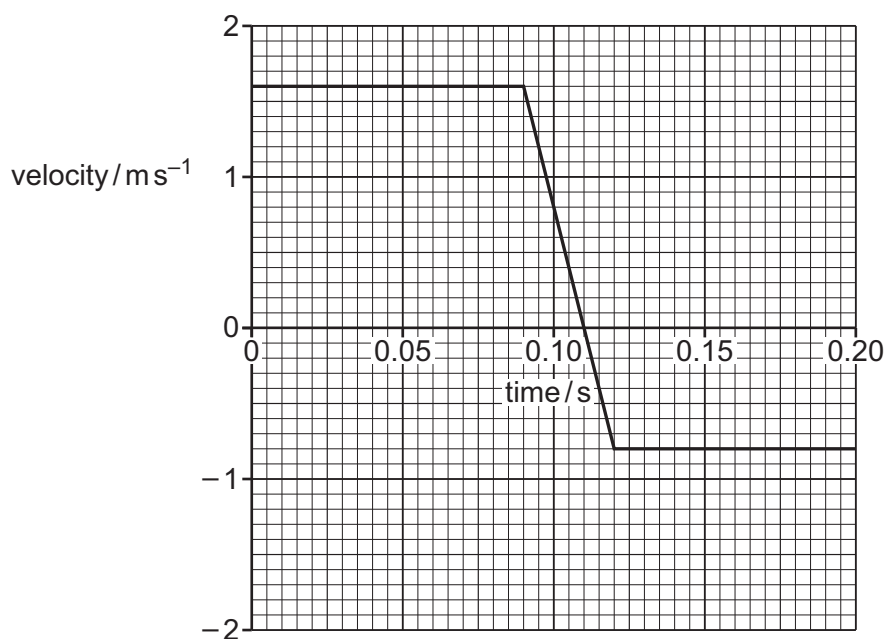
speed = ms⁻¹ [2]

2. M/J 26/P23/Q1/a,b(ii),c(i,iii)**1 (a)** Define momentum.

[1]

- (b)** A ball X of mass 0.15 kg moves along a horizontal frictionless surface. The ball collides elastically with ball Y that is initially stationary. Balls X and Y have equal diameters but different masses.

Figure 1.1 shows the variation with time of the velocity of ball X before, during and after the collision.

**Figure 1.1**

- (ii)** Determine the magnitude of the change in momentum of ball X due to the collision.

change in momentum = kg m s⁻¹ [2]

- (c) (i)** By considering relative speeds, explain why the velocity of ball Y after the collision must be +0.80 m s⁻¹.

[2]

- (iii)** Calculate the mass of ball Y.

mass = kg [2]

3. M/J 26/P24/Q1/b(iii),c(i)

- (b)** A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of 15 m s^{-1} . The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m.

Calculate:

- (iii)** the resultant force on the decelerating car.

force = N [2]

- (c)** The initial speed of the car in **1(b)** is now doubled to 30 m s^{-1} . The brakes of the car are then applied and the car comes to rest. The resultant force on the car remains the same as in **1(b)(iii)**.

Describe and explain the effect, if any, of doubling the initial speed on:

- (i)** the magnitude of the acceleration of the car

[1]

4. M/J 26/P24/Q4/a,b

- 4 An object is held at rest just below the surface of a liquid of uniform density as shown in Figure 4.1.

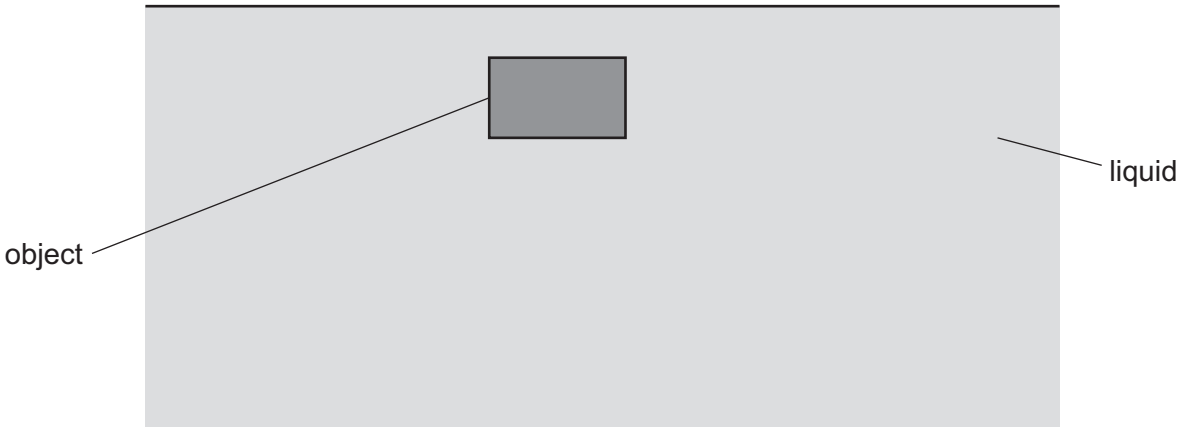


Figure 4.1

The object is released and accelerates downwards until it reaches terminal velocity.

The magnitude of the weight of the object is W .

The magnitude of the upthrust acting on the object is U .

The magnitude of the drag force acting on the object is D .

- (a) By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the magnitude of the three forces acting on the object as it accelerates.

Table 4.1

force	increases	stays the same	decreases
W			
D			
U			

[3]

- (b) State an expression, in terms of W , D and U , that shows the relationship between the forces acting on the object when the object is at its terminal velocity.

[1]

5. MAR 26/P22/Q1/a

- 1 (a) Define force.

[1]

6. MAR 26/P22/Q2/a,b

- 2** (a) State the principle of conservation of momentum.

[2]

- (b) A stationary boat of mass 160 kg floats on water. The mast of the boat is vertical. A sack of mass 23 kg is thrown at the boat, as shown in Figure 2.1.

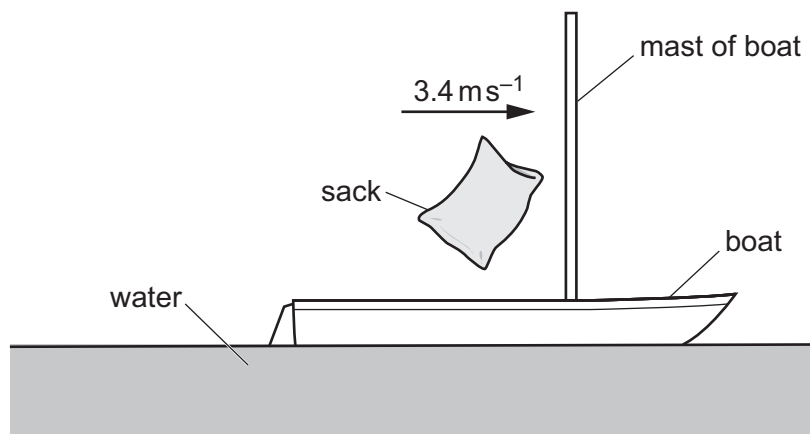


Figure 2.1

Immediately before colliding with the mast of the boat, the sack is travelling horizontally with a speed of 3.4 ms^{-1} .

The sack collides with the mast and remains in contact with it, without rebounding.

Assume that resistive forces acting on the boat during the collision are negligible.

- (i) Calculate the horizontal speed v of the sack and boat immediately after the collision.

$$v = \quad \text{ms}^{-1} \quad [2]$$

- (ii) The total kinetic energy of the sack and boat immediately before the collision is E_1 . The total kinetic energy of the sack and boat immediately after the collision is E_2 .

Determine the ratio $\frac{E_2}{E_1}$.

$$\text{ratio} = \quad [3]$$

- (iii) State and explain whether the collision between the sack and boat is elastic or inelastic.

[1]

7. O/N 25/P21/Q2/b(i),c(ii)

- (b) A small steel ball of radius r and mass m falls vertically at terminal speed v through oil.

The viscous drag force D that acts on the ball is given by

$$D = 6\pi\eta rv$$

where η is a property of the oil called its viscosity.

- (i) On Fig. 2.1, draw labelled arrows from the ball to show the directions of the **three** forces that act on the ball as it falls.



Fig. 2.1

[3]

- (c) The oil in (b) has a density of 920 kg m^{-3} and a viscosity of 4.7 in SI units.

The steel ball has a mass of $2.4 \times 10^{-3} \text{ kg}$ and a radius of $4.2 \times 10^{-3} \text{ m}$.

- (ii) Determine the terminal speed v of the ball.

$$v = \text{ms}^{-1} \quad [3]$$

8. O/N 25/P22/Q1/b(iii)

- 1 Scientists are investigating the variation in air pressure at different locations on a mountain.

(b)

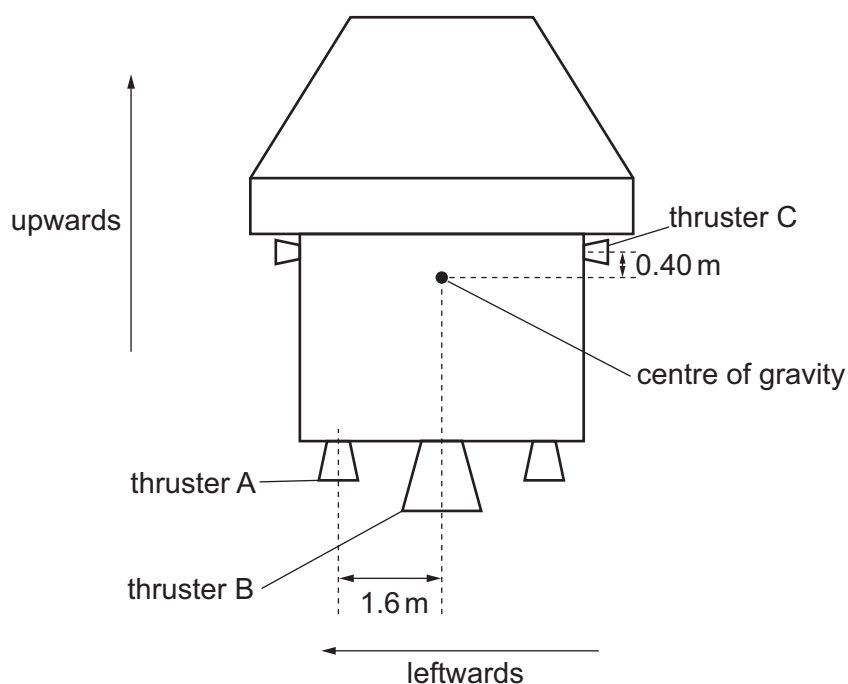
- (iii) The balloon has weight 19 N.

Calculate the magnitude of the initial acceleration of the balloon.

$$\text{acceleration} = \text{ms}^{-2} \quad [3]$$

9. O/N 25/P22/Q2/b,c

- 2 A spacecraft in deep space uses jets of hot gas from its thrusters to change its velocity. Fig. 2.1 shows a side view of the spacecraft and some of its thrusters.

**Fig. 2.1** (not to scale)

Thruster A is a distance of 1.6 m leftwards from the centre of gravity of the spacecraft. Thruster C is a distance of 0.40 m upwards from the centre of gravity of the spacecraft.

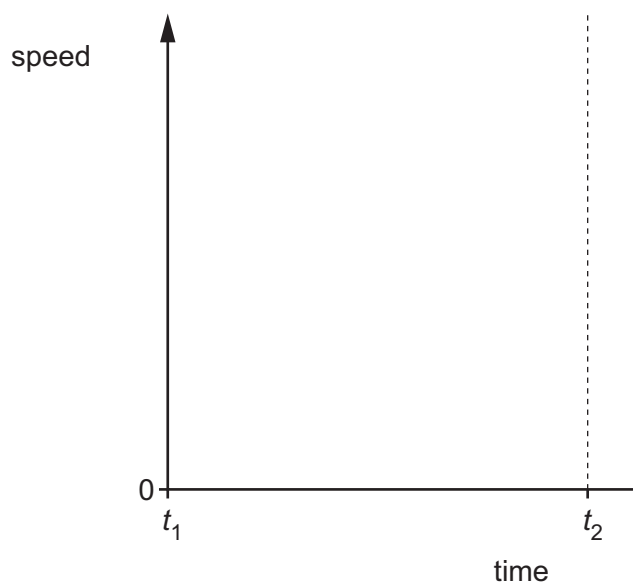
Thrusters A and B can produce forces on the spacecraft in the upwards direction only.

Thruster C can produce a force on the spacecraft in the leftwards direction only.

All the thrusters shown produce forces entirely in the same plane as the centre of gravity.

- (b) Thrusters A and C are now switched off and the spacecraft is stationary. Thruster B is activated at time t_1 , producing a constant force on the spacecraft until the fuel runs out at time t_2 . As the fuel is used, the total mass of the spacecraft decreases.

On Fig. 2.2, sketch the variation of speed of the spacecraft with time from t_1 to t_2 .

**Fig. 2.2**

- (c) The spacecraft now splits apart into a carrier and a payload as shown in Fig. 2.3.

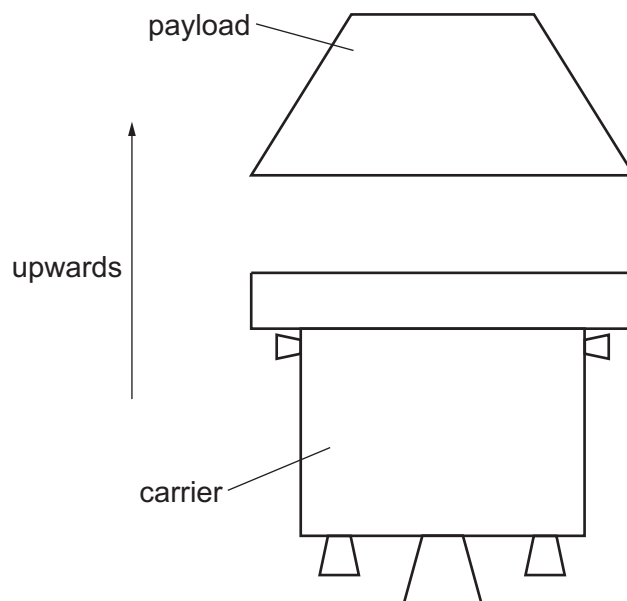


Fig. 2.3

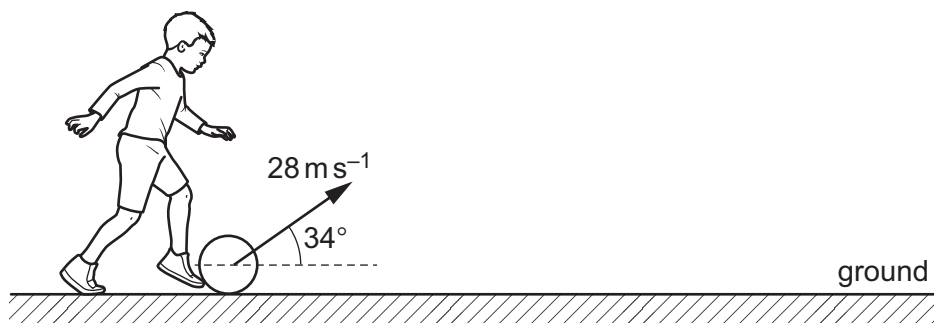
During the split, an average force of 5500 N acts on the payload for a time of 0.36 s. The velocity of the payload increases by 8.5 m s^{-1} in the upwards direction.

The combined mass of the carrier and payload is $2.5 \times 10^3 \text{ kg}$.

- (i) State the principle of conservation of momentum. [2]
- (ii) Show that the mass of the payload is 230 kg. [2]
- (iii) Calculate the magnitude of the change in velocity of the carrier. ms^{-1} [3]
- change in velocity =

10. O/N 25/P24/Q1/b

- 1 A child kicks a ball so that it leaves horizontal ground with a velocity of 28 m s^{-1} at an angle of 34° to the horizontal, as shown in Fig. 1.1.

**Fig. 1.1**

Air resistance is negligible. The ball leaves the ground at time $t = 0$.

- (b) The total change in momentum of the ball between leaving the ground at $t = 0$ and landing on the ground at $t = 3.2 \text{ s}$ is 13 kg m s^{-1} .

- (i) Define momentum.

[1]

- (ii) Calculate the force that acts on the ball while it is in the air.

force =

N [2]

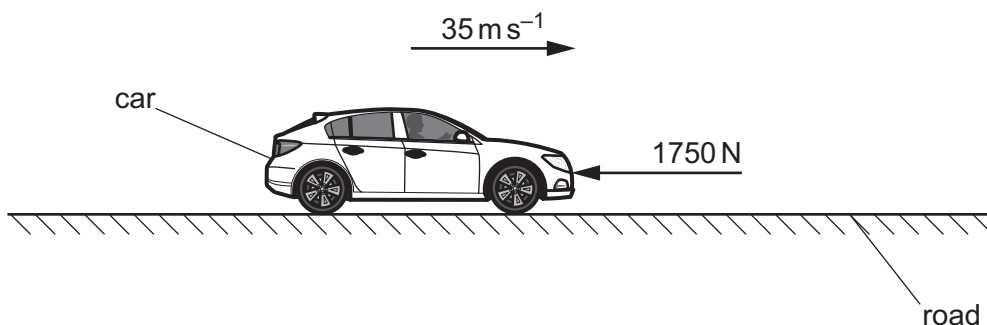
- (iii) Determine the mass of the ball.

mass =

kg [1]

11. M/J 25/P21/Q3/c(i)

- (b) An electric car is powered by a motor. The car is travelling at a constant speed of 35 m s^{-1} along a straight horizontal road, as shown in Fig. 3.1.

**Fig. 3.1**

There is a total resistive force of 1750 N acting on the car.

- (i) Calculate the power transmitted to the wheels of the car by the motor.

power = W [2]

- (ii) Calculate the useful work done by the motor when the car travels a distance of 17 km .

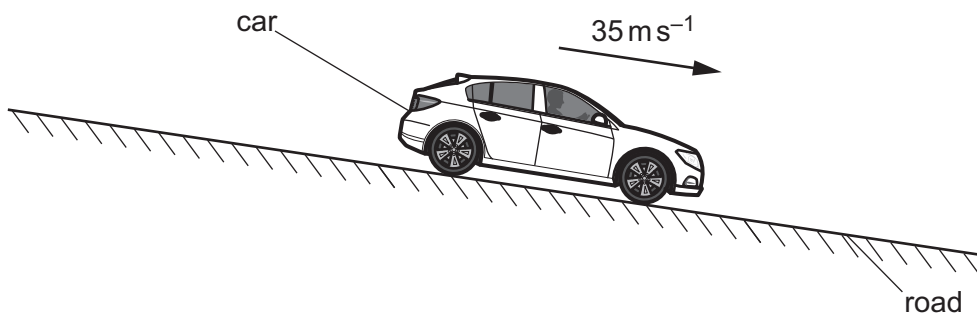
work done = J [2]

- (iii) The potential difference (p.d.) across the motor has a constant value of 600 V and the motor has an efficiency of 85% .

Calculate the current in the motor.

current = A [3]

- (c) The car in (b) now reaches a slope, as shown in Fig. 3.2.

**Fig. 3.2**

The car continues down the slope at the same speed as in (b).

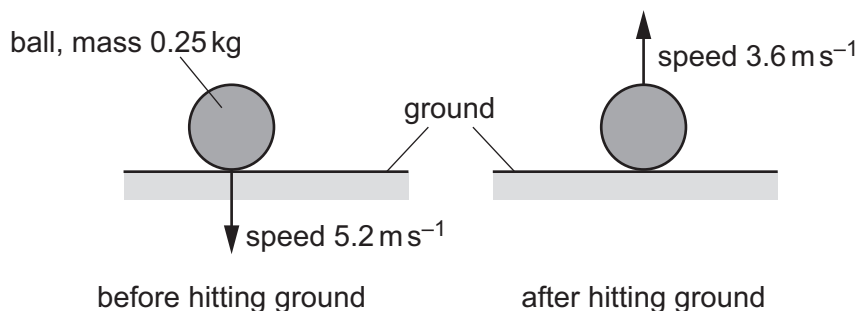
State and explain the effect, if any, of the slope on:

- (i) the air resistance acting on the car

[1]

12. M/J 25/P22/Q4/a,b

- 4** A small ball is dropped from rest from height h_1 above the ground and falls vertically downwards. The ball collides with the ground and bounces back vertically upwards, reaching a maximum height h_2 . Fig. 4.1 shows the ball just before and just after hitting the ground.

**Fig. 4.1**

The ball has mass 0.25 kg and is in contact with the ground for a time of 0.18 s . Just before the ball hits the ground, it has speed 5.2 m s^{-1} . Just after it leaves the ground, it has speed 3.6 m s^{-1} . Air resistance acting on the ball is negligible.

- (a)** State and explain whether the collision is elastic or inelastic.

[1]

- (b) (i)** Calculate the change in momentum of the ball during the collision with the ground.

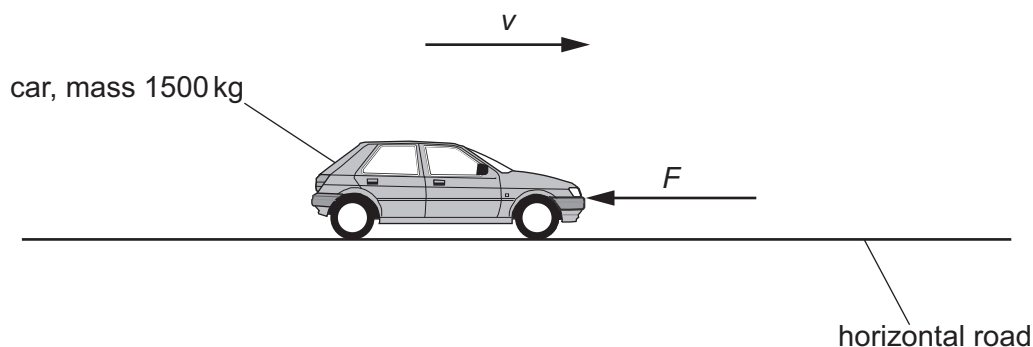
change in momentum = kg m s^{-1} [2]

- (ii)** Determine the average force on the ball during the collision with the ground.

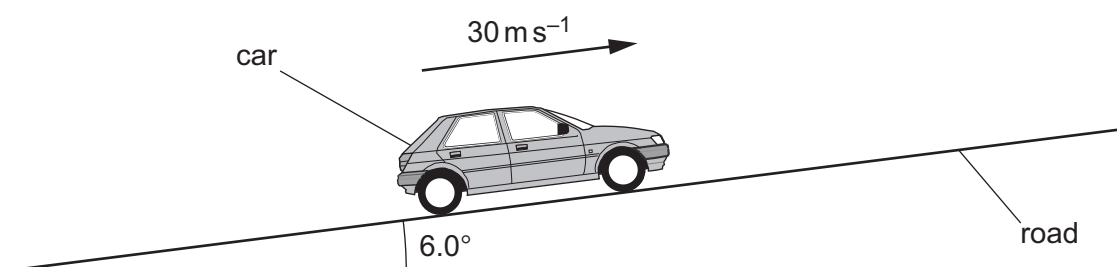
force = N [2]

13. M/J 25/P23/Q3/c(i)

- 3** A car of mass 1500 kg is travelling along a straight horizontal road at constant velocity v . The car is subject to a total resistive force F , as shown in Fig. 3.1.

**Fig. 3.1**

- (b)** The car now moves up a slope at a constant speed of 30 m s^{-1} . The slope is at an angle to the horizontal of 6.0° , as shown in Fig. 3.2.

**Fig. 3.2**

The total resistive force acting on the car is 1600 N.

- (i)** Show that the increase in gravitational potential energy of the car in a time of 1.0 s is 46 000 J.

[2]

- (ii)** Use the information in **(b)(i)** to determine the power developed by the engine to move the car up the slope.

power = _____ W [2]

- (c)** The car picks up a passenger and then continues up the slope at the same speed as in **(b)**.

State and explain the effect, if any, that the passenger has on:

- (i)** the air resistance acting on the car

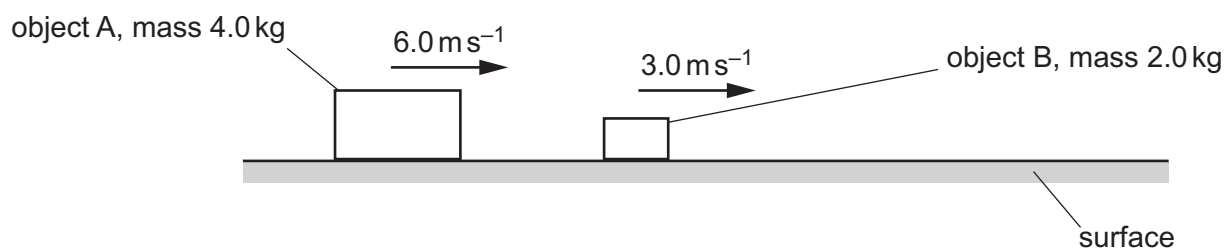
[1]

14. M/J 25/P23/Q4/a,b

- 4 (a)** State the principle of conservation of momentum.

[2]

- (b)** An object A of mass 4.0 kg travels at a velocity of 6.0 ms^{-1} to the right on a horizontal frictionless surface. It moves towards a second object B of mass 2.0 kg that is moving at a velocity of 3.0 ms^{-1} in the same direction as A, as shown in Fig. 4.1.

**Fig. 4.1**

Object A collides with object B. The two objects join and move off together with velocity v .

Calculate:

- (i)** velocity v

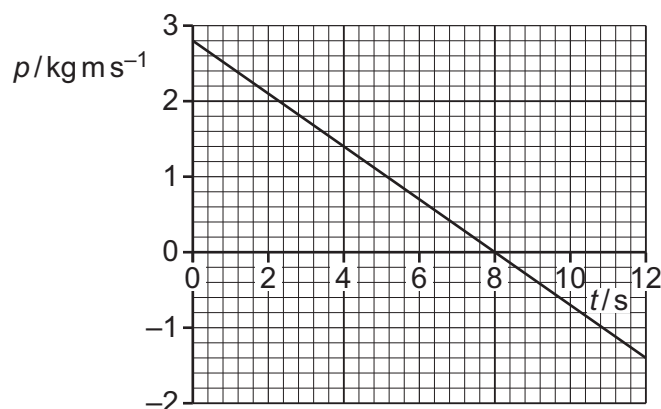
$$v = \quad \text{ms}^{-1} \quad [2]$$

- (ii)** the percentage of the total initial kinetic energy of the two objects that is transferred to other forms of energy during the collision.

$$\text{percentage} = \quad \% \quad [2]$$

15. M/J 25/P24/Q2/a,b,c,d,e

- 2 An object of constant mass moves in a straight line. The variation with time t of the momentum p of the object is shown in Fig. 2.1.

**Fig. 2.1**

- (a) Define momentum.

[1]

- (b) Calculate the change in momentum of the object from time $t = 0$ to $t = 12$ s.

change in momentum = kg ms⁻¹ [1]

- (c) Calculate the magnitude of the resultant force acting on the object.

force = N [2]

- (d) Describe the variation of the speed of the object from time $t = 0$ to $t = 8.0$ s.

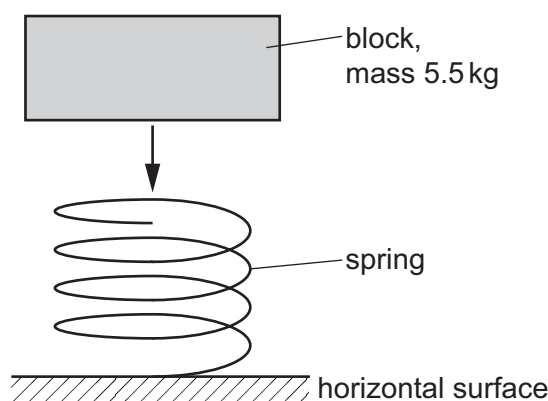
[1]

- (e) By reference to Fig. 2.1, explain why the resultant force acting on the object during the first 8.0 s of its motion cannot be due to air resistance.

[2]

16. M/J 25/P24/Q3/e

- 3 The lower end of a vertical spring is fixed to a horizontal surface, as shown in Fig. 3.1.

**Fig. 3.1**

The mass of the spring is negligible. A block of mass 5.5 kg drops vertically onto the spring and is brought to rest as the spring is compressed.

- (a) The block has kinetic energy 110 J as it makes contact with the spring.

Calculate the speed of the block as it makes contact with the spring.

speed = ms^{-1} [2]

- (b) The gravitational potential energy of the block decreases by 20 J as the spring is compressed to its maximum compression x_0 .

Show that x_0 is 0.37 m.

[2]

- (c) Assume that, as the spring compresses, all of the energy lost by the block is converted into the elastic potential energy of the spring.

Use the data from (a) and (b) to determine the maximum elastic potential energy of the spring.

Show your working.

maximum elastic potential energy = J [1]

- (d) The variation of the force F acting on the spring with the compression x of the spring is shown in Fig. 3.2.

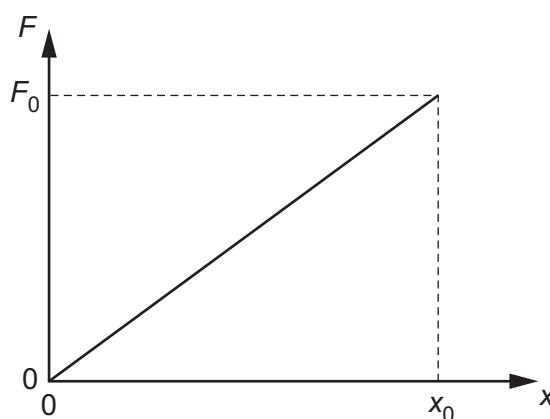


Fig. 3.2

Use the information in (b) and your answer in (c) to show that the maximum force F_0 exerted on the spring by the block is 700 N.

[2]

- (e) Use the information in (d) to determine, for the instant that the block is first brought to rest by the spring, the magnitude of:

- (i) the resultant force acting on the block

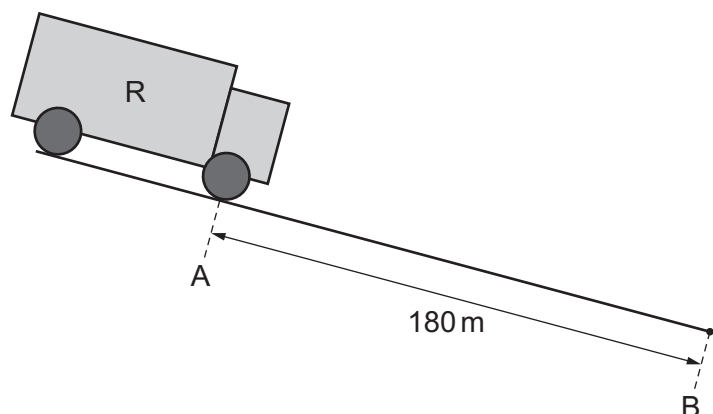
resultant force = N [2]

- (ii) the acceleration of the block.

acceleration = ms^{-2} [2]

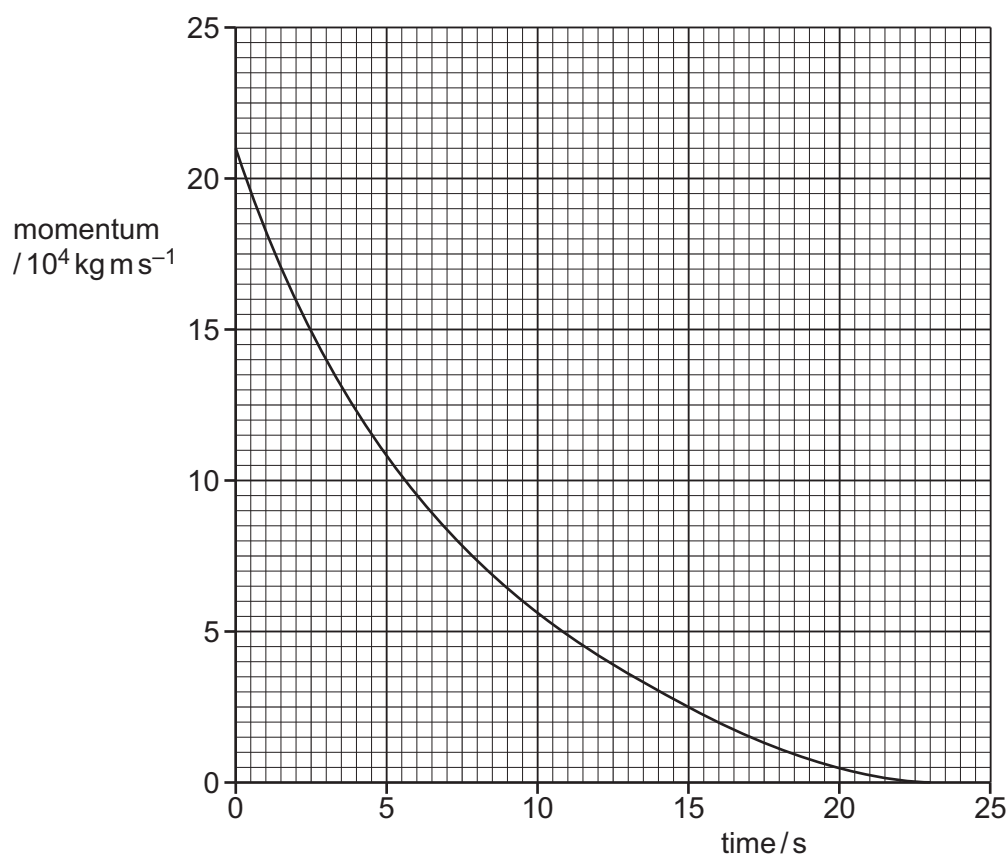
17. MAR 25/P22/Q3/b

- 3 (a) A truck R of mass 9400 kg moves with constant acceleration in a straight line down a slope, as illustrated in Fig. 3.1.

**Fig. 3.1**

At point A the speed of the truck is 13 ms^{-1} and at point B the speed of the truck is 22 ms^{-1} . A and B are a distance of 180 m apart.

- (b) A short time after passing point B truck R moves in a straight line on horizontal ground. The driver of the truck applies the brakes. Fig. 3.2 shows the variation with time of the momentum of the truck.

**Fig. 3.2**

- (i) Define force.

[1]

- (ii) Show that the average resultant force F acting on truck R between time $t = 0$ and $t = 15\text{ s}$ is $-1.2 \times 10^4\text{ N}$.

[1]

- (iii) An identical truck S has the same initial momentum as truck R. Truck S experiences a constant force equal to the force F in (b)(ii).

State and explain whether truck S will take more, less or the same amount of time to come to rest as truck R.

[3]

18. MAR 25/P22/Q7/b

- 7 An isolated stationary nucleus Q decays into nucleus R and an α -particle. The α -particle has speed $1.5 \times 10^7 \text{ ms}^{-1}$.

- (b) By considering momentum, calculate the speed of nucleus R after the decay.

speed = ms^{-1} [3]

19. O/N 24/P21/Q2/a,b(i,iii),c

- 2 (a) Define linear momentum.

[1]

- (b) A car of mass 1800 kg is moving in a straight line. Fig. 2.1 shows the variation with time t of the momentum p of the car.

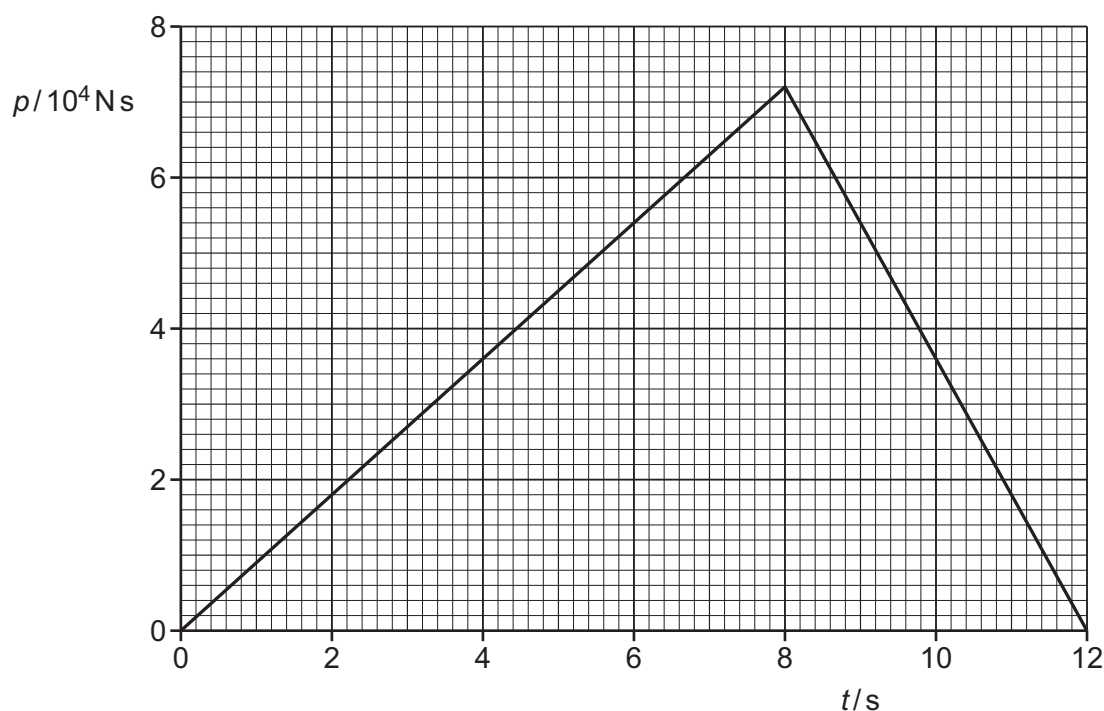


Fig. 2.1

- (i) Calculate the maximum speed reached by the car.

maximum speed = ms^{-1} [1]

- (iii) Show that the acceleration of the car at time $t = 4.0 \text{ s}$ is 5.0 ms^{-2} .

[2]

- (c) On Fig. 2.2, sketch the variation with time t of the acceleration a of the car in (b) from $t = 0$ to $t = 12.0 \text{ s}$.

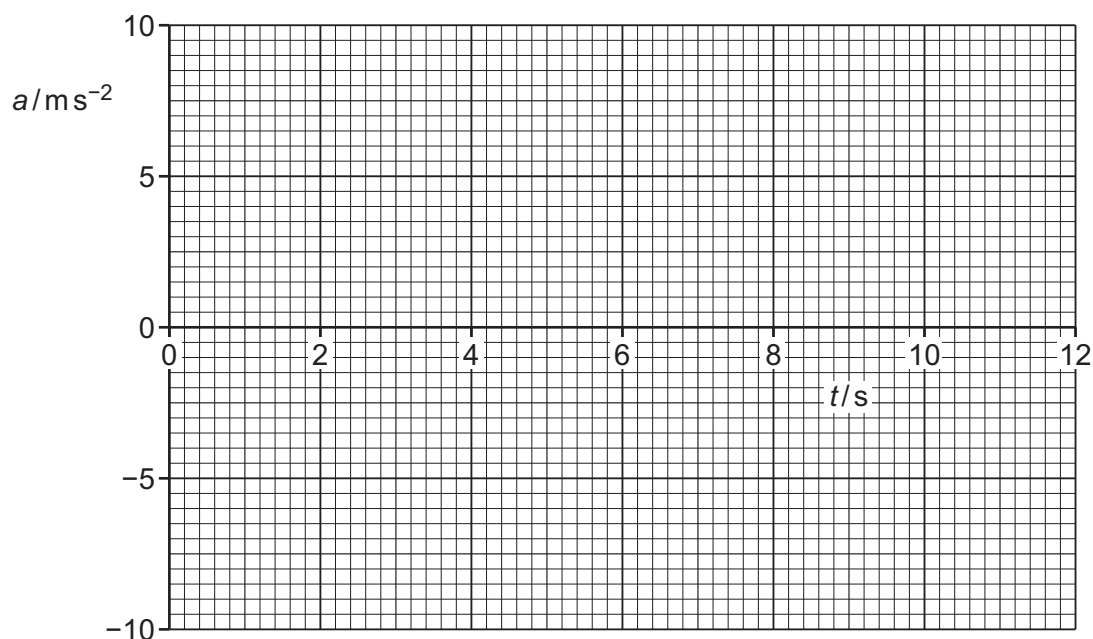


Fig. 2.2

[3]

20. O/N 24/P22/Q2/a,b

2 (a) Define momentum.

[1]

- (b) A child stands on a scooter on horizontal ground. The combined mass of the child and the scooter is 16 kg. The child starts from rest and pushes once on the ground with her foot which causes her to accelerate. The push lasts for a time of 1.1 s. The speed of the child and the scooter after the push is 0.60 ms^{-1} .

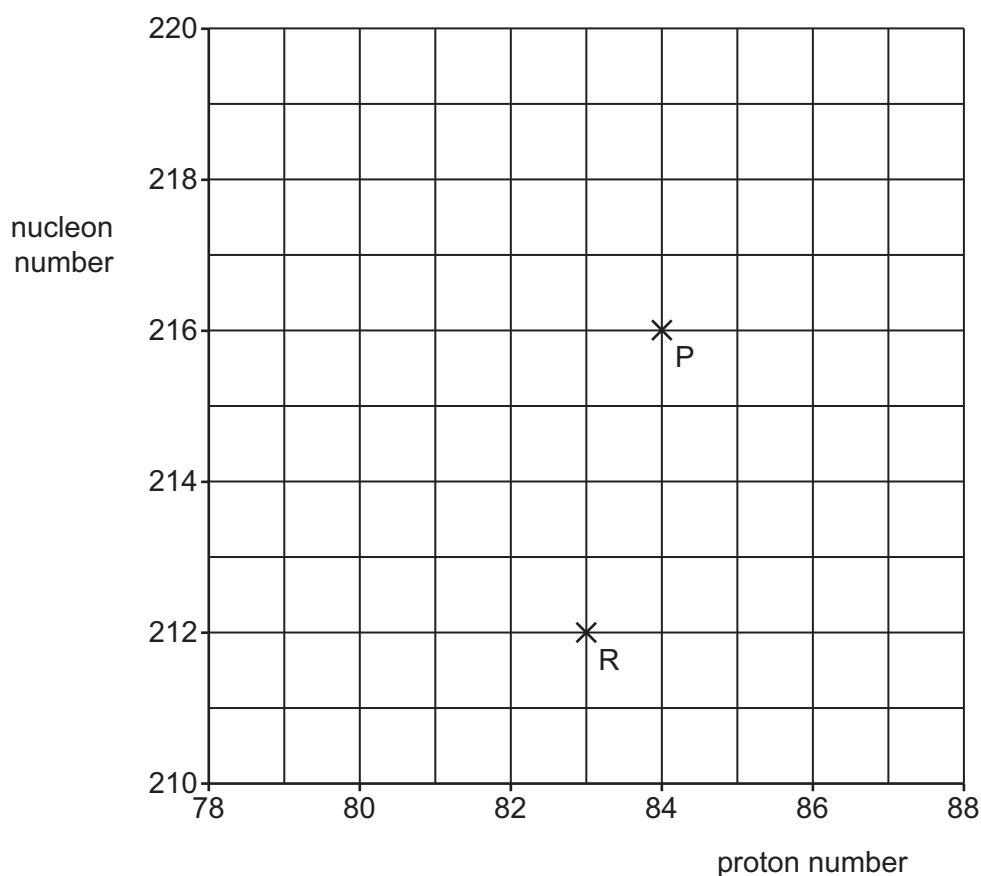
Determine the average resultant force acting horizontally on the child and the scooter during the push.

average force = N [2]

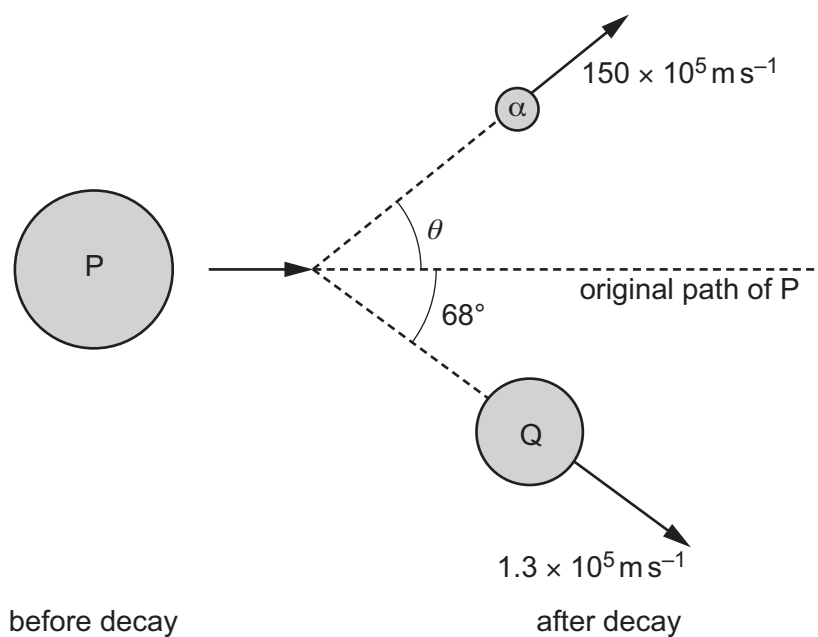
21. O/N 24/P22/Q6/c

also Unit 5 (2 marks); this unit 3 marks

- (b) Nucleus P undergoes α -decay to form nucleus Q. Nucleus Q then undergoes a further decay to form nucleus R. The proton and nucleon numbers of P and R are shown in Fig. 6.1.

**Fig. 6.1**

- (c) Before the α -decay, P is travelling at a constant velocity. After the decay, Q has a velocity of $1.3 \times 10^5 \text{ m s}^{-1}$ at an angle of 68° to the original path of P. The α -particle has a velocity of $150 \times 10^5 \text{ m s}^{-1}$ at an angle of θ to the original path of P, as shown in Fig. 6.2.

**Fig. 6.2** (not to scale)

- (i) Use the principle of conservation of momentum to determine θ .

$$\theta = \quad \quad \quad ^\circ \quad [3]$$

- (ii) Calculate the kinetic energy of the α -particle.

kinetic energy = J [2]

22. O/N 24/P23/Q2/a,b,c

- 2 (a)** State the principle of conservation of momentum.

[2]

- (b)** A ball X has mass 240 g and moves in a straight line on a horizontal frictionless surface with an initial speed of 16 m s^{-1} . The ball collides with a stationary ball Y that has mass 480 g. After the collision, ball X is stationary, as shown in Fig. 2.1.

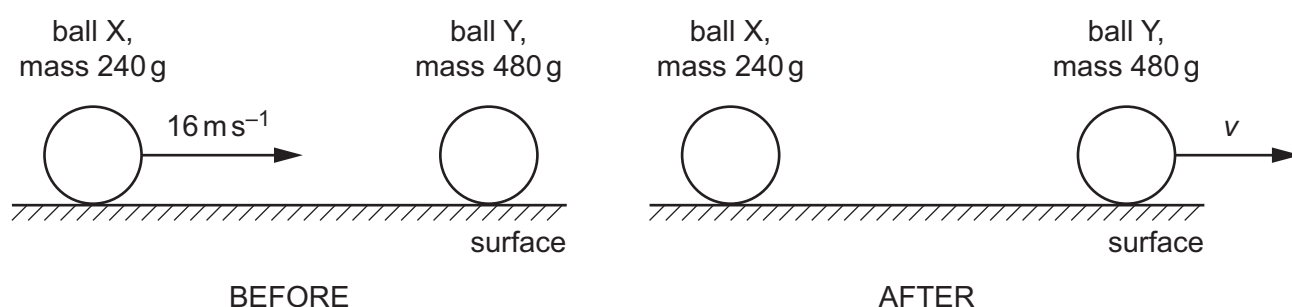


Fig. 2.1

- (i)** Show that the speed v of ball Y after the collision is 8.0 m s^{-1} .

[1]

- (ii)** Calculate the change in the total kinetic energy ΔE_K of the balls due to the collision.

$\Delta E_K =$ J [3]

- (c)** The collision in **(b)** lasts for a time of 2.0 ms. Assume that the contact force between the balls is constant during this time.

- (i)** Determine the magnitude and direction of the force exerted on ball X by ball Y during the collision.

magnitude = N

direction

[3]

- (ii)** Compare the magnitude and direction of the force exerted on ball Y by ball X during the collision with the answers in **(c)(i)**. No further calculations are required.

[2]

23. M/J 24/P21/Q1/b,c,d

- 1 The drag force F_D acting on an object falling through air is given by

$$F_D = \frac{1}{2} C \rho A v^2$$

where A is the cross-sectional area of the object,
 v is the velocity of the object in the air,
 ρ is the density of the air and
 C is a constant called the drag coefficient.

- (b) Fig. 1.1 shows a sphere falling at terminal velocity in air.

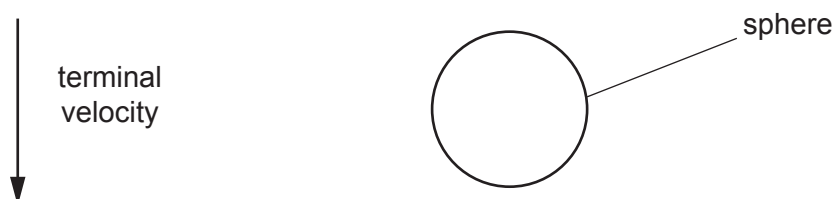


Fig. 1.1

Assume that the upthrust on the sphere is negligible.

On Fig. 1.1, draw and label arrows to show the directions of the **two** forces acting on the sphere. [2]

- (c) The mass of the sphere is 49 g.

Calculate the drag force F_D acting on the sphere.

$$F_D = \quad \quad \quad \text{N} \quad [2]$$

- (d) The sphere is falling in air at a terminal velocity of 25 in SI base units.
 The density of the air is 1.2 in SI base units.
 The diameter of the sphere is 0.060 in SI base units.

Use your answer in (c) to calculate the drag coefficient C for the sphere.

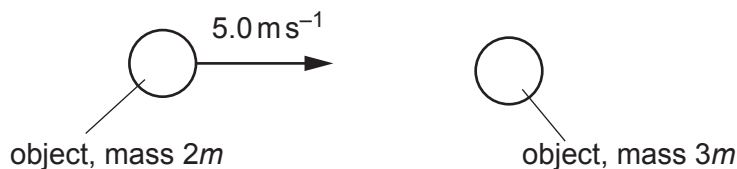
$$C = \quad \quad \quad [3]$$

24. M/J 24/P21/Q3/a,b,c

- 3 (a) State the principle of conservation of momentum.

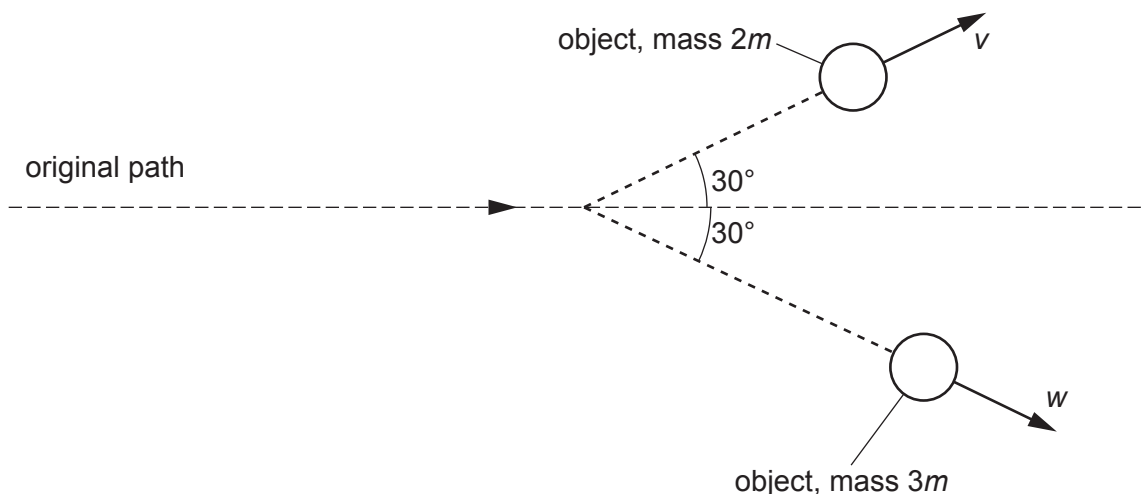
[2]

- (b) An object of mass $2m$ is travelling at a speed of 5.0 ms^{-1} in a straight line. It collides with an object of mass $3m$ which is initially stationary, as shown in Fig. 3.1.

**Fig. 3.1**

After the collision, the object of mass $2m$ moves with velocity v at an angle of 30° to its original direction of motion.

The object of mass $3m$ moves with velocity w also at an angle of 30° , as shown in Fig. 3.2.

**Fig. 3.2**

By considering the conservation of momentum in two dimensions, calculate the magnitudes of v and w .

$$v = \quad \text{ms}^{-1}$$

$$w = \quad \text{ms}^{-1}$$

[4]

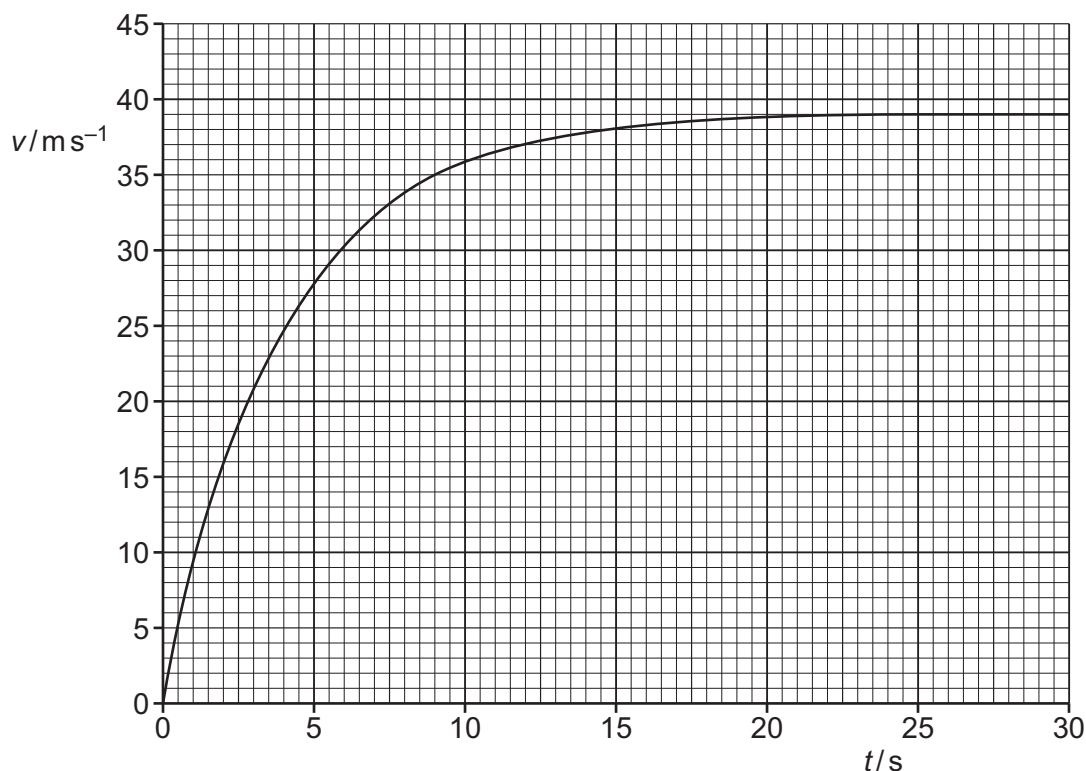
- (c) An object of mass 4.2 kg is travelling in a straight line at a speed of 6.0 ms^{-1} . The object is brought to rest in a distance of 0.050 m by a constant force.

Calculate the magnitude of this force.

$$\text{force} = \quad \text{N} \quad [3]$$

25. M/J 24/P22/Q2/a(i),b,c

- 2** A skydiver jumps from an aircraft at time $t = 0$ and falls vertically downwards. The variation with t of her velocity v is shown in Fig. 2.1.

**Fig. 2.1**

- (a) (i)** Using Fig. 2.1, state the terminal velocity of the skydiver.

terminal velocity = ms^{-1} [1]

- (b)** The mass of the skydiver and her equipment is 68 kg. The upthrust on the skydiver is negligible.

After reaching terminal velocity, the skydiver opens her parachute at time t_1 . A total drag force of 1800 N acts on the skydiver.

Determine the magnitude and direction of the acceleration of the skydiver at time t_1 .

acceleration = ms^{-2}

direction = [3]

- (c)** The parachute is fully open at time t_2 . At a later time t_3 the skydiver reaches a constant velocity of 5.7 ms^{-1} .

- (i)** Describe and explain the variation with time of the magnitude of her acceleration between time t_2 and time t_3 .

[2]

- (ii)** Calculate the change in momentum of the skydiver between time t_1 and time t_3 .

change in momentum = Ns [2]

26. M/J 24/P23/Q1/b,c,d(ii)

- 1 The drag force F_D acting on a sphere falling through a liquid is given by

$$F_D = 6\pi\eta rv$$

where r is the radius of the sphere,
 v is the speed of the sphere in the liquid and
 η is a property of the liquid called the viscosity.

- (b) The sphere has a radius of 3.0 cm and is falling vertically downwards at a terminal velocity of 2.0 m s^{-1} through the liquid. The drag force acting on the sphere is 0.096 N.

Calculate the viscosity of the liquid.

viscosity = $\text{kg m}^{-1} \text{s}^{-1}$ [2]

- (c) The sphere is shown in Fig. 1.1.

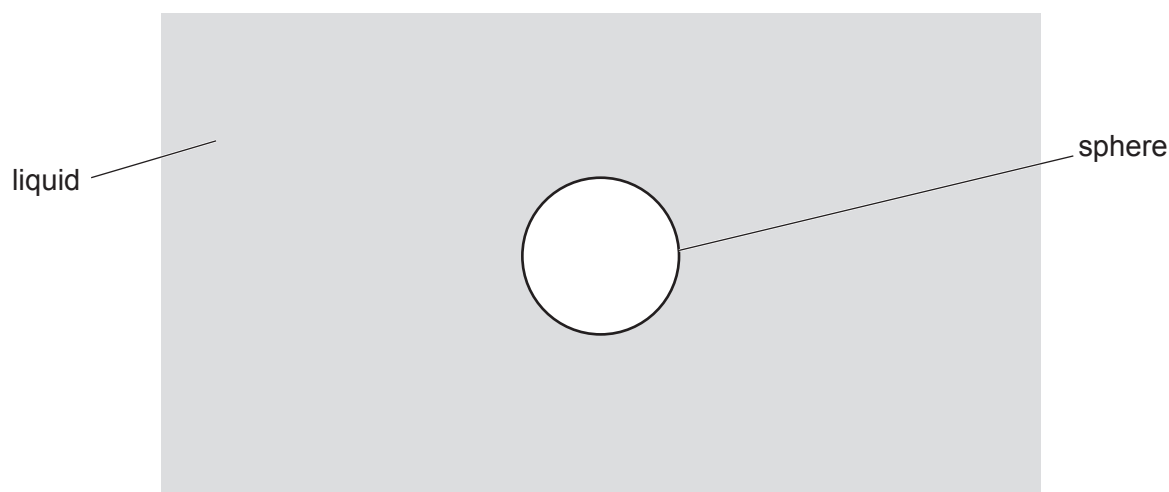


Fig. 1.1

On Fig. 1.1, draw and label arrows to represent the directions of the **three** forces acting on the sphere as it falls at terminal velocity through the liquid. [2]

- (d)

- (ii) Calculate the mass of the sphere.

mass = kg [2]

27. MAR 24/P22/Q2/c(i,iii)

(b) An Olympic diver stands on a platform above a pool of water, as shown in Fig. 2.1.

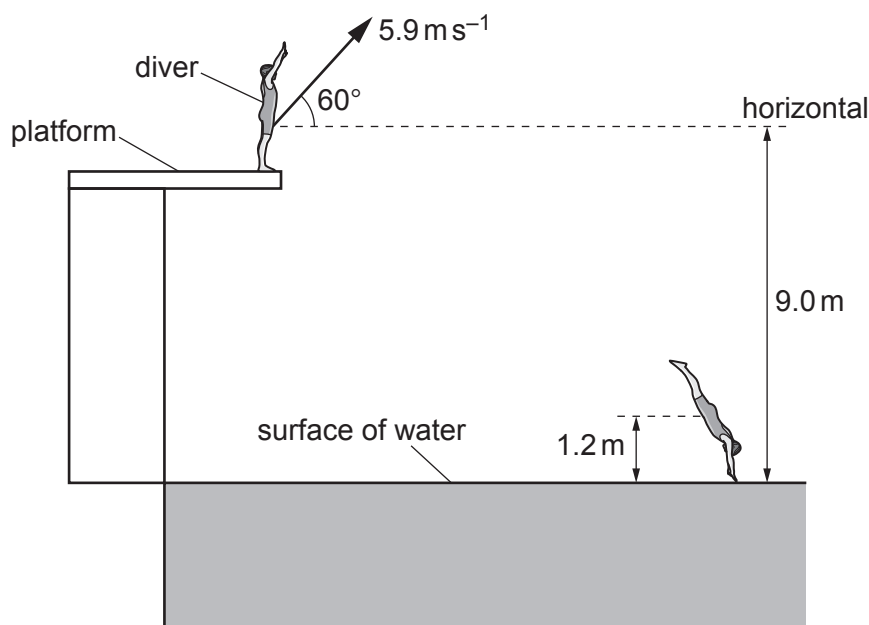


Fig. 2.1 (not to scale)

When the diver is on the platform his centre of gravity is a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is a vertical height of 1.2 m above the surface of the water.

Calculate the speed of the diver at the instant he hits the surface of the water.

speed = m s^{-1} [3]

(c) The diver in (b) enters the water and decelerates.

(i) Describe and explain the variation of the viscous drag force acting on the diver in the water as he moves downwards.

[2]

(iii) At a particular instant when the diver is entirely underwater his horizontal velocity is zero. The viscous drag force acting on him at this instant is 950 N vertically upwards. The diver has mass 78 kg.

Determine the magnitude and direction of the acceleration of the diver.

acceleration = m s^{-2}

direction

[4]

28. MAR 24/P22/Q4/b

4 A nucleus P undergoes α -decay to form nucleus Q.

(b) (i) State the principle of conservation of momentum.

[2]

(ii) Before the decay, nucleus P has a speed of $3.2 \times 10^5 \text{ ms}^{-1}$. After the decay, nucleus Q is stationary.

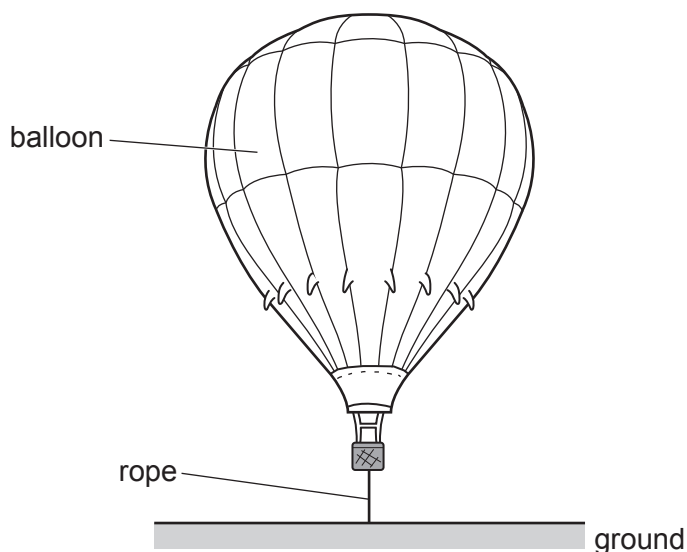
Calculate the speed of the alpha particle after the decay.

speed =

ms^{-1} [2]

29. O/N 23/P21/Q2/a(iii),b

- 2 A hot-air balloon floats just above the ground. The balloon is stationary and is held in place by a vertical rope, as shown in Fig. 2.1.

**Fig. 2.1**

The balloon has a weight W of $3.39 \times 10^4 \text{ N}$. The tension T in the rope is $4.00 \times 10^2 \text{ N}$.
Upthrust U acts on the balloon.
The density of the surrounding air is 1.23 kg m^{-3} .

(a)

- (iii) The balloon is released from the rope.

Calculate the initial acceleration of the balloon.

acceleration = ms^{-2} [3]

- (b) The balloon is stationary at a height of 500 m above the ground. A tennis ball is released from rest and falls vertically from the balloon.

A passenger in the balloon uses the equation $v^2 = u^2 + 2as$ to calculate that the ball will be travelling at a speed of approximately 100 ms^{-1} when it hits the ground.

Explain why the actual speed of the ball will be much lower than 100 ms^{-1} when it hits the ground.

[3]**30. O/N 23/P21/Q3/b(ii)**

- 3 A trolley A moves along a horizontal surface at a constant velocity towards another trolley B which is moving at a lower constant speed in the same direction. Fig. 3.1 shows the trolleys at time $t = 0$.

**Fig. 3.1**

Table 3.1 shows data for the trolleys.

Table 3.1

trolley	mass/kg	initial speed/ ms^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 3.2 shows the variation with time t of the velocity v for trolley B.

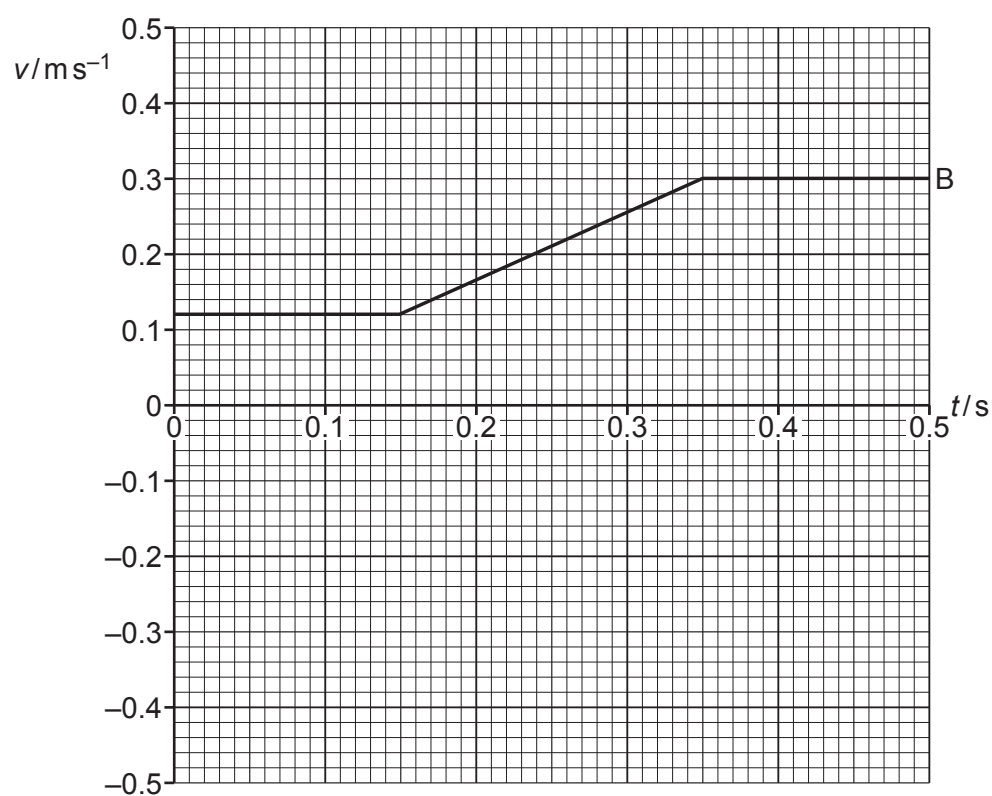


Fig. 3.2

(b) Use Table 3.1 and Fig. 3.2 to determine:

(ii) the magnitude and direction of the final velocity of trolley A.

magnitude =

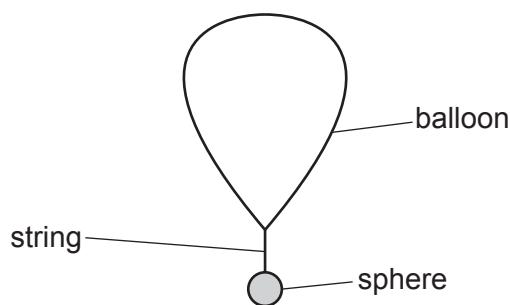
ms^{-1}

direction

[3]

31. O/N 23/P22/Q2/b,c

- 2 A high-altitude balloon is stationary in still air. A solid sphere is suspended from the balloon by a string, as shown in Fig. 2.1.

**Fig. 2.1** (not to scale)

The volume of the balloon is 7.5 m^3 . The total weight of the balloon, string and sphere is 65 N . The upthrust acting on the string and sphere is negligible.

- (b) The string breaks, releasing the sphere.

- (i) State the magnitude of the acceleration of the sphere immediately after the string breaks.

acceleration = ms^{-2} [1]

- (ii) State and explain the variation, if any, in the magnitude of the acceleration of the sphere when it is moving downwards **before** it reaches terminal (constant) velocity.

[3]

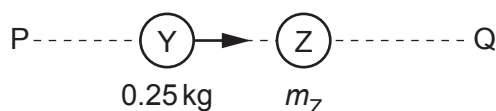
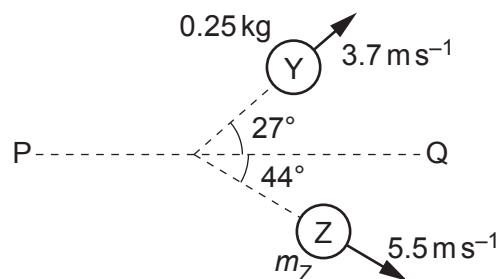
- (c) The sphere has a mass of 4.0 kg .

Calculate the total resistive force acting on the sphere at the instant when its acceleration is 1.9 ms^{-2} .

resistive force = N [2]

32. O/N 23/P22/Q4/a,b

- 4 (a) A ball Y moves along a horizontal frictionless surface and collides with a ball Z, as illustrated in the views from above in Fig. 4.1 and Fig. 4.2.

**BEFORE COLLISION****Fig. 4.1** (not to scale)**AFTER COLLISION****Fig. 4.2** (not to scale)

Ball Y has a mass of 0.25 kg and initially moves along a line PQ.
Ball Z has a mass m_Z and is initially stationary.

After the collision, ball Y has a final velocity of 3.7 m s^{-1} at an angle of 27° to line PQ and ball Z has a final velocity of 5.5 m s^{-1} at an angle of 44° to line PQ.

- (i) Calculate the component of the final momentum of ball Y in the direction perpendicular to line PQ.

component of momentum = Ns [2]

- (ii) By considering the component of the final momentum of each ball in the direction perpendicular to line PQ, calculate m_Z .

$m_Z =$ kg [1]

- (iii) During the collision, the average force exerted on Y by Z is F_Y and the average force exerted on Z by Y is F_Z .

Compare the magnitudes and directions of F_Y and F_Z . Numerical values are not required.

magnitudes:

directions:

[2]

- (b) Two blocks, A and B, move directly towards each other along a horizontal frictionless surface, as shown in the view from above in Fig. 4.3.



Fig. 4.3

The blocks collide perfectly elastically. Before the collision, block A has a speed of 4 m s^{-1} and block B has a speed of 6 m s^{-1} . After the collision, block B moves back along its original path with a speed of 2 m s^{-1} .

Calculate the speed of block A after the collision.

speed = m s^{-1} [1]

33. O/N 23/P23/Q2/b

- 2** A ball on horizontal ground is kicked towards a vertical wall. Fig. 2.1 shows the path of the ball.

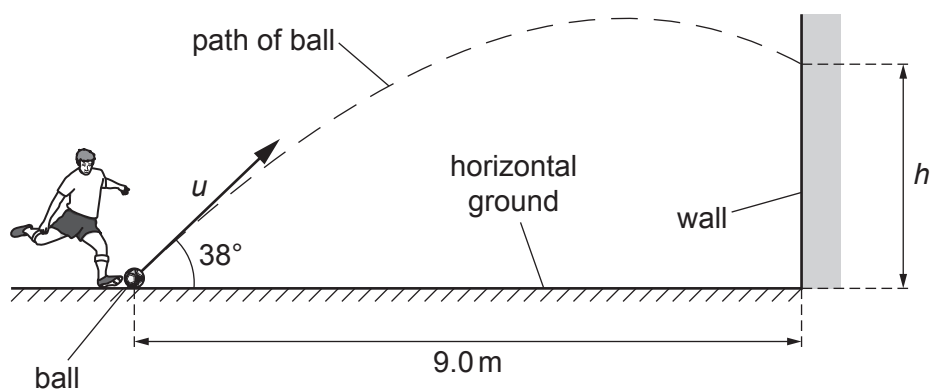


Fig. 2.1 (not to scale)

The ball has an initial velocity u at an angle of 38° to the ground. The ball travels a horizontal distance of 9.0 m before striking the wall at a height h above the ground. The horizontal component u_H of the initial velocity of the ball is 9.5 m s^{-1} .

Air resistance is negligible.

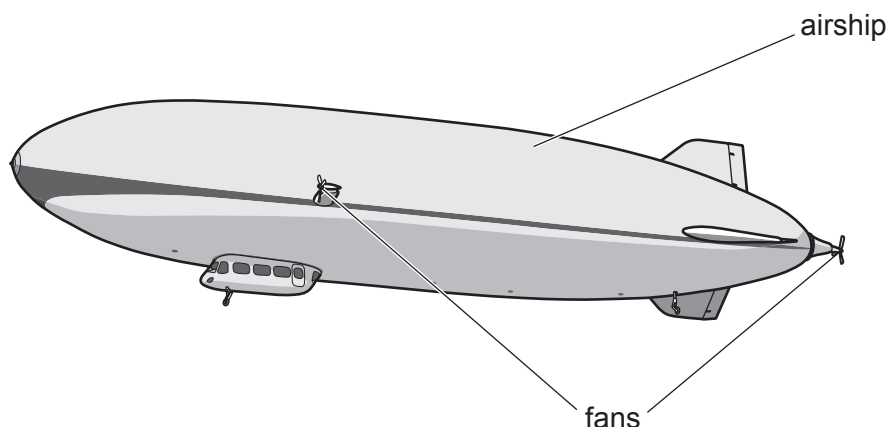
- (b)** The speed of the ball just after striking the wall is less than its speed just before striking the wall.

State what this indicates about the nature of the collision of the ball with the wall.

[1]

34. O/N 23/P23/Q3/c(i,ii)

- (b) Fig. 3.1 shows an airship in flight. The airship is propelled by identical fans that can be angled to control the motion of the airship.

**Fig. 3.1**

The upthrust on the airship is 93 000 N.
The density of the surrounding air is 1.2 kg m^{-3} .

- (i) Calculate the volume of air displaced by the airship.

volume = m^3 [1]

- (ii) When fully loaded, the weight of the airship is greater than the upthrust.
To maintain horizontal flight, the fans provide a total vertical force of $3.0 \times 10^3 \text{ N}$ upwards on the airship.

Calculate the mass of the airship.

mass = kg [2]

- (c) At a certain time, the airship in (b) is stationary. The thrust force exerted by a fan on the airship is 2800 N.

To produce this force, a mass of 64 kg of air is propelled through the blades of the fan in a time of 0.50 s. Assume that this air is initially stationary at the entrance to the fan.

Calculate:

- (i) the change in momentum Δp of the air propelled through the fan blades in this time

$\Delta p =$ kg ms^{-1} [2]

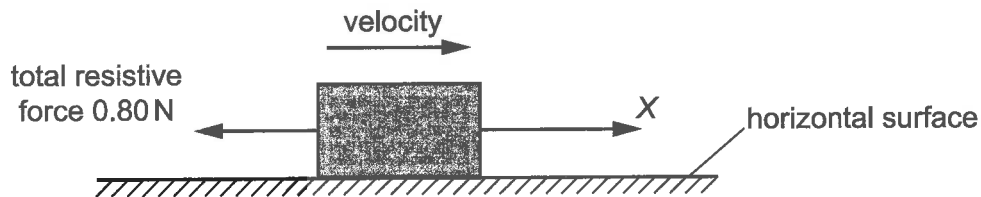
- (ii) the speed of the air as it leaves the fan

speed = ms^{-1} [2]

35. M/J 23/P22/Q3

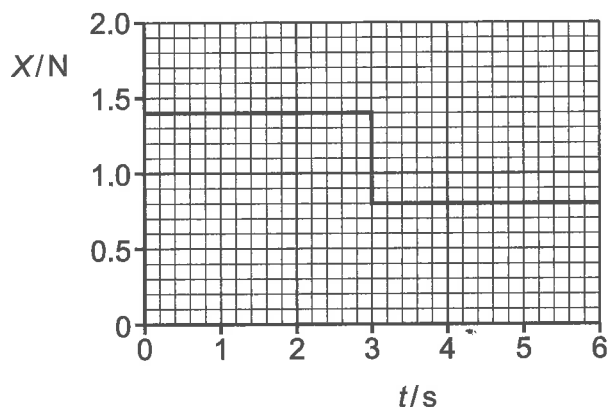
booklet (scan + OCR)

A block is pulled by a force X in a straight line along a rough horizontal surface, as shown in Fig. 3.1.

**Fig. 3.1**

Assume that the total resistive force opposing the motion of the block is 0.80 N at all speeds of the block.

The variation with time t of the magnitude of the force X is shown in Fig. 3.2.

**Fig. 3.2**

- (a) (i) Define force.

..... [1]

- (ii) Determine the change in momentum of the block from time $t = 0$ to time $t = 3.0$ s.

change in momentum = kg m s^{-1} [2]

- (b) (i) Describe and explain the motion of the block between time $t = 3.0$ s and time $t = 6.0$ s.

.....

..... [2]

- (ii) Force X produces a total power of 2.0 W when moving the block between time $t = 3.0\text{ s}$ and time $t = 6.0\text{ s}$.
Calculate the distance moved by the block during this time interval.

distance = m [3]

- (c) The block is at rest at time $t = 0$.

On Fig. 3.3, sketch a graph to show the variation of the momentum of the block with time t from $t = 0$ to $t = 6.0\text{ s}$.

Numerical values of momentum are not required.

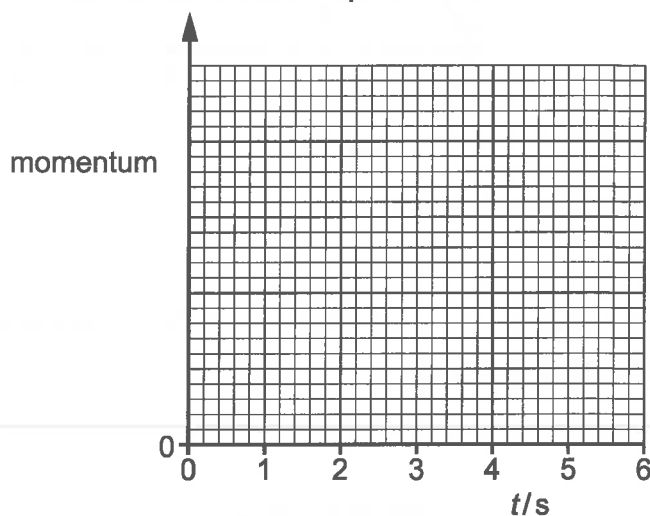


Fig. 3.3

[2]

36. M/J 23/P21/Q3

booklet (scan + OCR)

A block is pulled in a straight line along a rough horizontal surface by a varying force X , as shown in Fig. 3.1.

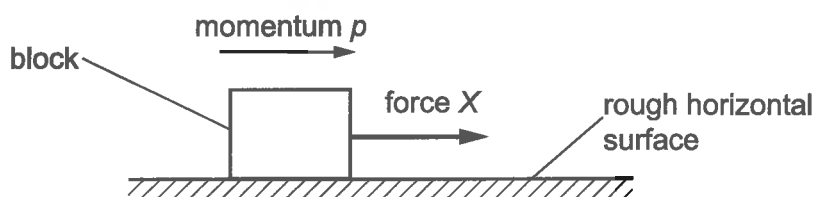


Fig. 3.1

Air resistance is negligible. Assume that the frictional force exerted on the block by the surface is constant and has magnitude 2.0 N .

The variation with time t of the momentum p of the block is shown in Fig. 3.2.

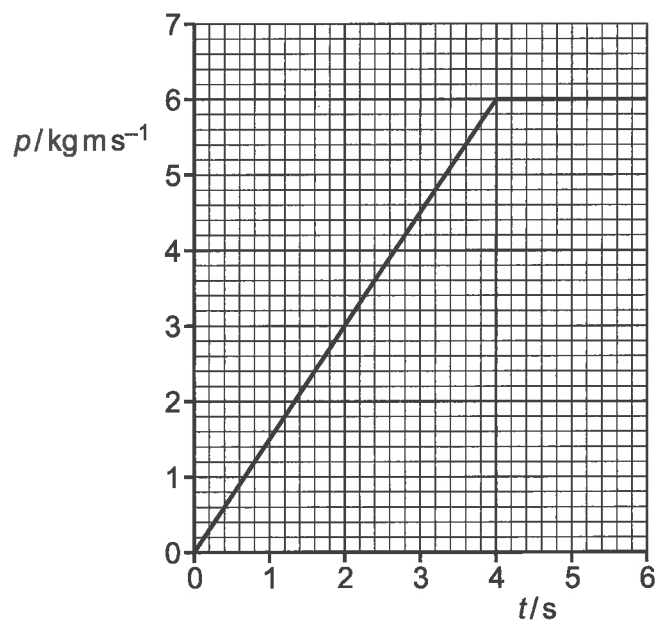


Fig. 3.2

(a) State Newton's second law of motion.

..... [1]

(b) Use Fig. 3.2 to determine, for the block at time $t = 2.0$ s, the magnitude of:

(i) the resultant force on the block

resultant force = N [1]

(ii) the force X .

$X =$ N [1]

- (c) On Fig. 3.3, sketch a graph to show the variation of force X with time t from $t = 0$ to $t = 6.0$ s.

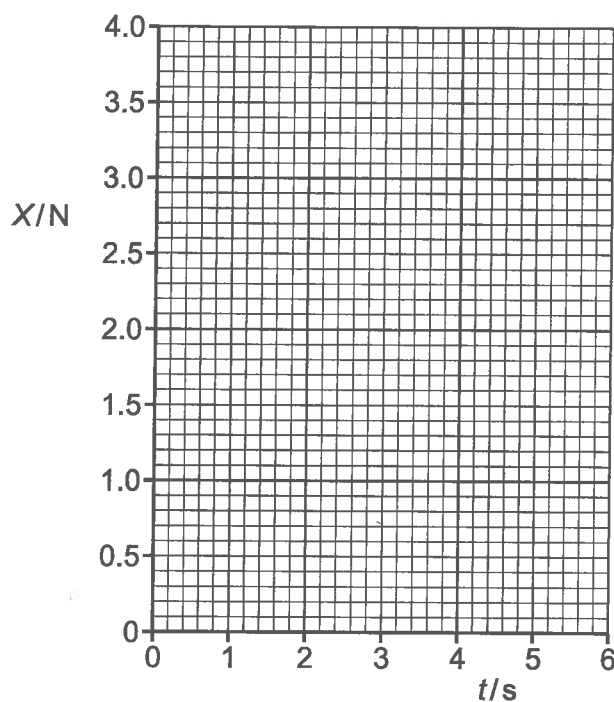


Fig. 3.3

[3]

37. M/J 23/P23/Q3

booklet (scan + OCR)

- (a) State the principle of conservation of momentum.

.....

..... [2]

- (b) A firework is initially stationary. It explodes into three fragments A, B and C that move in a horizontal plane, as shown in the view from above in Fig. 3.1.

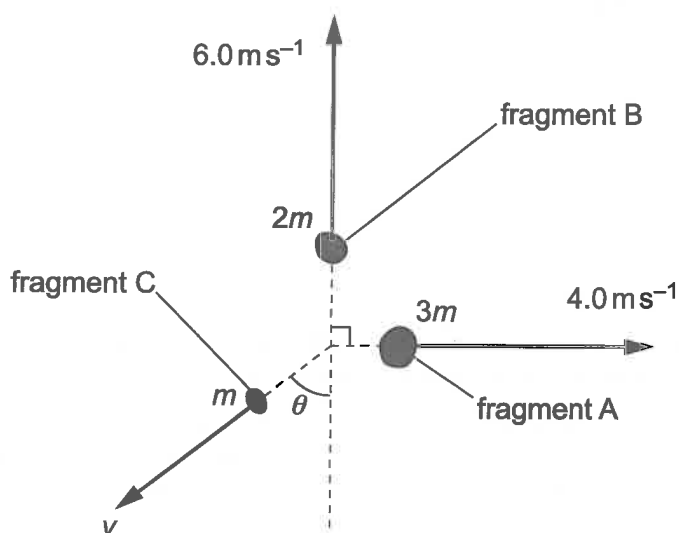


Fig. 3.1

Fragment A has a mass of $3m$ and moves away from the explosion at a speed of 4.0 ms^{-1} .

Fragment B has a mass of $2m$ and moves away from the explosion at a speed of 6.0 ms^{-1} at right angles to the direction of A.

Fragment C has a mass of m and moves away from the explosion at a speed v and at an angle θ as shown in Fig. 3.1.

Calculate:

(i) the angle θ

$$\theta = \dots\dots\dots^\circ \quad [3]$$

(ii) the speed v .

$$v = \dots\dots\dots \text{ ms}^{-1} \quad [2]$$

- (c) The firework in (b) contains a chemical that has mass 5.0 g and has chemical energy per unit mass 700 J kg^{-1} . When the firework explodes, all of the chemical energy is transferred to the kinetic energy of fragments A, B and C.

(i) Show that the total chemical energy in the firework is 3.5 J .

[1]

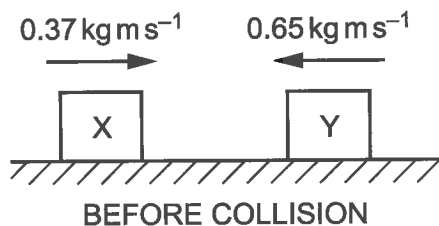
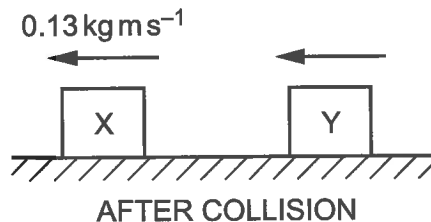
(ii) Calculate the mass m .

$$m = \dots\dots\dots \text{ kg} \quad [3]$$

38. MAR 23/P22/Q4

booklet (scan + OCR)

Two blocks slide directly towards each other along a frictionless horizontal surface, as shown in Fig. 4.1. The blocks collide and then move as shown in Fig. 4.2.

**Fig. 4.1****Fig. 4.2**

Block X initially moves to the right with a momentum of 0.37 kg m s^{-1} . Block Y initially moves to the left with a momentum of 0.65 kg m s^{-1} . After the blocks collide, block X moves to the left back along its original path with a momentum of 0.13 kg m s^{-1} . Block Y also moves to the left after the collision.

- (a) Block X has an initial kinetic energy of 0.30 J .
Calculate the mass of block X.

mass = kg [3]

- (b) Determine the magnitude of the momentum of block Y after the collision.

momentum = kg m s^{-1} [1]

- (c) Block X exerts an average force of 7.7 N on block Y during the collision.
Calculate the time that the blocks are in contact with each other.

time = s [2]

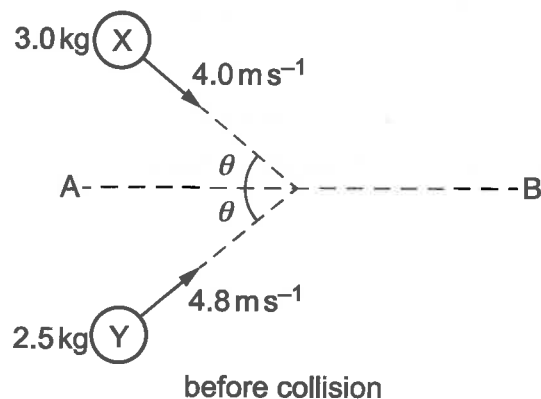
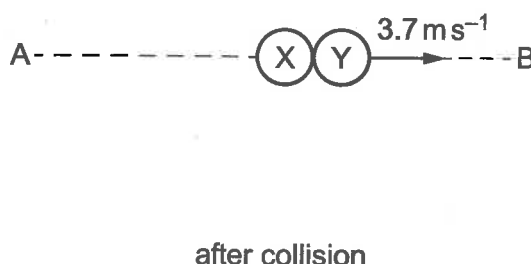
39. M/J 22/P22/Q4

booklet (scan + OCR)

(a) State the principle of conservation of momentum.

.....

..... [2]

(b) Two balls, X and Y, move along a horizontal frictionless surface, as shown from above in Fig. 4.1.**Fig. 4.1** (not to scale)**Fig. 4.2** (not to scale)

Ball X has a mass of 3.0 kg and a velocity of 4.0 ms^{-1} in a direction at angle θ to a line AB. Ball Y has a mass of 2.5 kg and a velocity of 4.8 ms^{-1} in a direction at angle θ to the line AB. The balls collide and stick together. After colliding, the balls have a velocity of 3.7 ms^{-1} along the line AB on the horizontal surface, as shown in Fig. 4.2.

(i) By considering the components of the momenta along the line AB, calculate θ .

$$\theta = \dots\dots\dots^\circ \quad [3]$$

(ii) By calculation of kinetic energies, state and explain whether the collision of the balls is inelastic or perfectly elastic.

.....

..... [2]

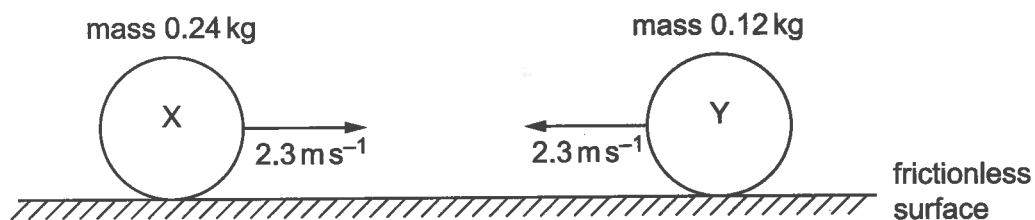
40. O/N 21/P21/Q2

booklet (scan + OCR)

(a) Define *momentum*.

[1]

(b) Two balls X and Y, of equal diameter but different masses 0.24 kg and 0.12 kg respectively, slide towards each other on a frictionless horizontal surface, as shown in Fig. 2.1.

**Fig. 2.1**

Both balls have initial speed 2.3 m s^{-1} before they collide with each other. Fig. 2.2 shows the variation with time t of the force F_Y exerted on ball Y by ball X during the collision.

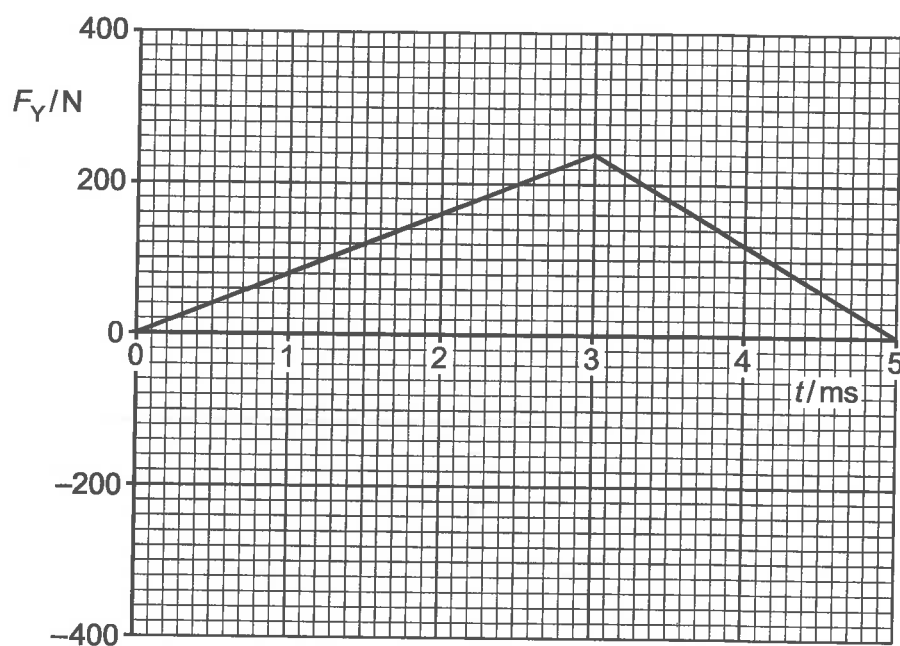


Fig. 2.2

- (i) Calculate the kinetic energy of ball X before the collision.

kinetic energy = J [3]

- (ii) The area enclosed by the lines and the time axis in Fig. 2.2 represents the change in momentum of ball Y during the collision.

Determine the magnitude of the change in momentum of ball Y.

change in momentum = N s [2]

- (iii) Calculate the magnitude of the velocity of ball Y after the collision.

velocity = ms^{-1} [2]

- (c) On Fig. 2.3, sketch the variation with time t of the force F_X exerted on ball X by ball Y during the collision in (b).

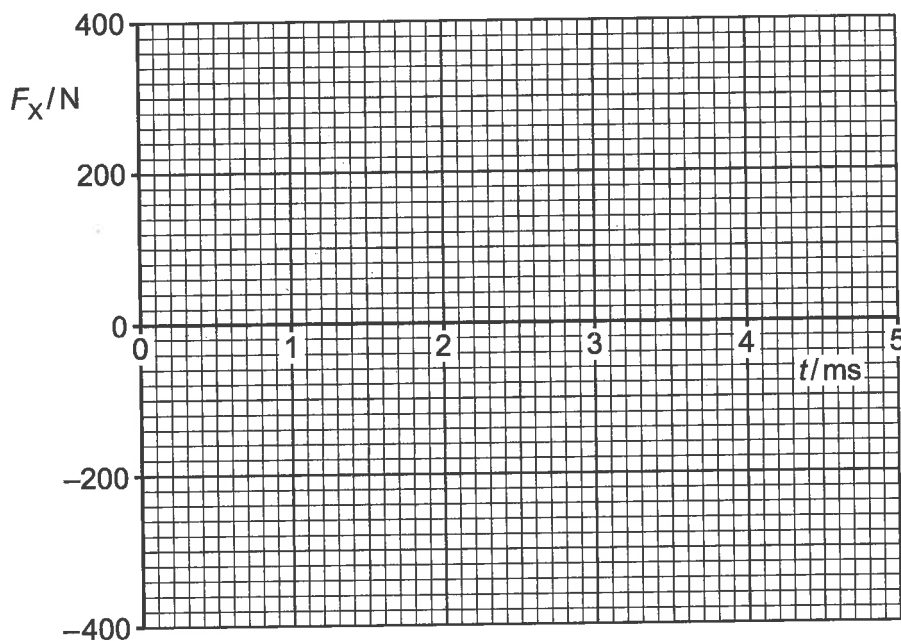


Fig. 2.3

[3]

41. M/J 21/P22/Q2

booklet (scan + OCR)

A ball is thrown vertically downwards to the ground, as illustrated in Fig. 2.1.

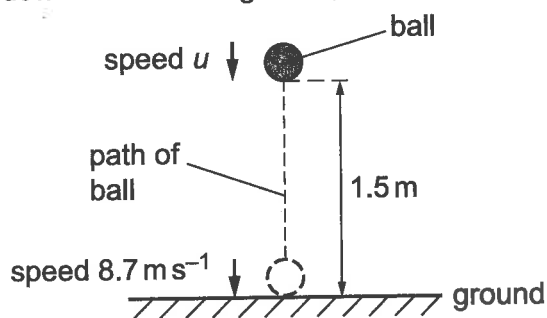


Fig. 2.1

The ball is thrown with speed u from a height of 1.5 m. The ball then hits the ground with speed 8.7 m s^{-1} . Assume that air resistance is negligible.

- (a) Calculate speed u .

$$u = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (b) State how Newton's third law applies to the collision between the ball and the ground.

.....
 [2]

- (c) The ball is in contact with the ground for a time of 0.091 s. The ball rebounds vertically and leaves the ground with speed 5.4 ms^{-1} . The mass of the ball is 0.059 kg.
- (i) Calculate the magnitude of the change in momentum of the ball during the collision.

change in momentum = N s [2]

- (ii) Determine the magnitude of the average resultant force that acts on the ball during the collision.

average resultant force = N [1]

- (iii) Use your answer in (c)(ii) to calculate the magnitude of the average force exerted by the ground on the ball during the collision.

average force = N [2]

- (d) The ball was thrown downwards at time $t = 0$ and hits the ground at time $t = T$.
On Fig. 2.2, sketch a graph to show the variation of the speed of the ball with time t from $t = 0$ to $t = T$. Numerical values are not required.

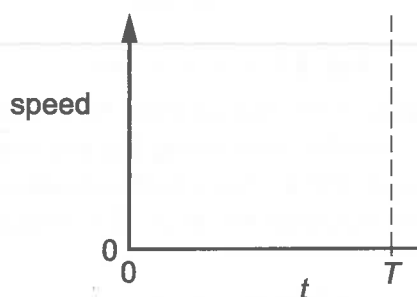


Fig. 2.2

[1]

- (e) In practice, air resistance is not negligible.
State and explain the variation, if any, with time t of the gradient of the graph in (d) when air resistance is not negligible.

.....
..... [2]

42. M/J 21/P21/Q2

booklet (scan + OCR)

A person uses a trolley to move suitcases at an airport. The total mass of the trolley and suitcases is 72 kg.

- (a) The person pushes the trolley and suitcases along a horizontal surface with a constant speed of 1.4 ms^{-1} and then releases the trolley. The released trolley moves in a straight line and comes to rest. Assume that a constant total resistive force of 18 N opposes the motion of the trolley and suitcases.

- (i) Calculate the power required to overcome the total resistive force on the trolley and suitcases when they move with a constant speed of 1.4 m s^{-1} .

power = W [2]

- (ii) Calculate the time taken for the trolley to come to rest after it is released.

time = s [3]

- (b) At another place in the airport, the trolley and suitcases are on a slope, as shown in Fig. 2.1.

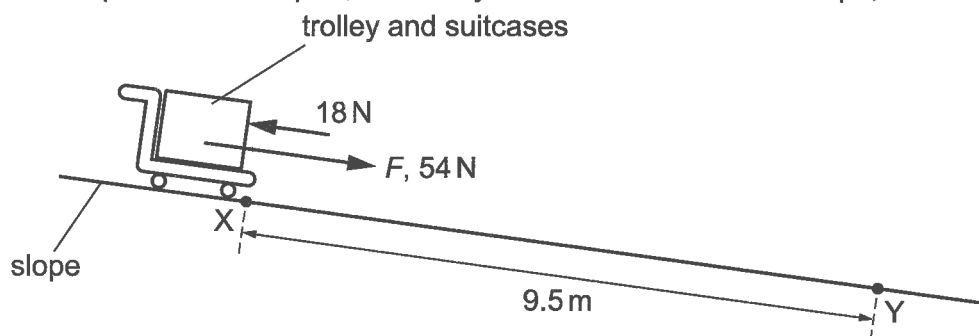


Fig. 2.1 (not to scale)

The person releases the trolley from rest at point X. The trolley moves down the slope in a straight line towards point Y. The distance along the slope between points X and Y is 9.5 m. The component F of the weight of the trolley and suitcases that acts along the slope is 54 N. Assume that a constant total resistive force of 18 N opposes the motion of the trolley and suitcases.

- (i) Calculate the speed of the trolley at point Y.

speed = ms^{-1} [3]

- (ii) Calculate the work done by F for the movement of the trolley from X to Y.

work done = J [1]

(iii) The trolley is released at point X at time $t = 0$.

On Fig. 2.2, sketch a graph to show the variation with time t of the work done by F for the movement of the trolley from X to Y.

Numerical values of the work done and t are not required.

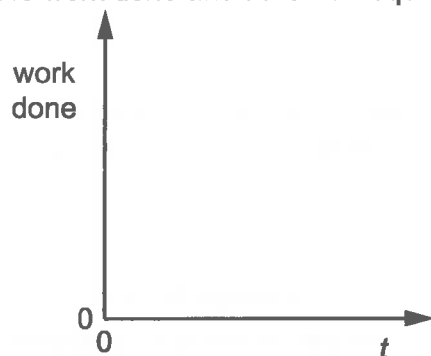


Fig. 2.2

[2]

(c) The angle of the slope in (b) is constant. The frictional forces acting on the wheels of the moving trolley are also constant.

Explain why, in practice, it is incorrect to assume that the total resistive force opposing the motion of the trolley and suitcases is constant as the trolley moves between X and Y.

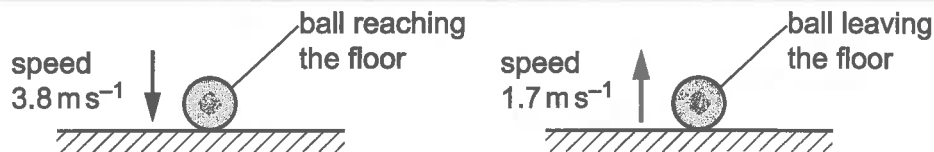
..... [1]

43. O/N 20/P21/Q3

booklet (scan + OCR)

(a) Define force.

- [1]
- (b)** A ball falls vertically downwards towards a horizontal floor and then rebounds along its original path, as illustrated in Fig. 3.1.

**Fig. 3.1**

The ball reaches the floor with speed 3.8 m s^{-1} . The ball is then in contact with the floor for a time of 0.081 s before leaving it with speed 1.7 m s^{-1} . The mass of the ball is 0.062 kg .

- (i)** Calculate the loss of kinetic energy of the ball during the collision.

loss of kinetic energy = J [2]

- (ii)** Determine the magnitude of the change in momentum of the ball during the collision.

change in momentum = N s [2]

- (iii)** Show that the magnitude of the average resultant force acting on the ball during the collision is 4.2 N .

[1]

- (iv)** Use the information in **(iii)** to calculate the magnitude of:
- the average force of the floor on the ball during the collision

average force = N

- the average force of the ball on the floor during the collision.

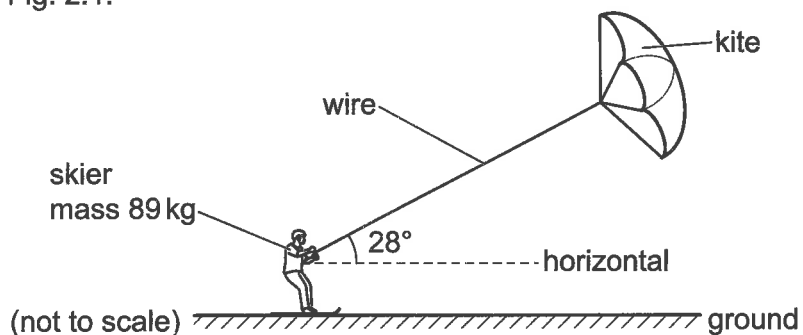
average force = N [2]

44. M/J 20/P23/Q2

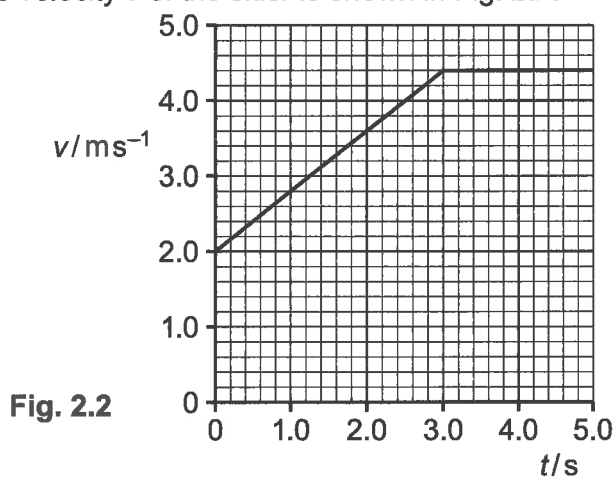
booklet (scan + OCR)

(a) State Newton's first law of motion.

[1]

(b) A skier is pulled in a straight line along horizontal ground by a wire attached to a kite, as shown in Fig. 2.1.**Fig. 2.1**

The mass of the skier is 89 kg. The wire is at an angle of 28° to the horizontal. The variation with time t of the velocity v of the skier is shown in Fig. 2.2.

**Fig. 2.2**

- (i) Use Fig. 2.2 to determine the distance moved by the skier from time $t = 0$ to $t = 5.0$ s.

distance = m [2]

- (ii) Use Fig. 2.2 to show that the acceleration a of the skier is 0.80 m s^{-2} at time $t = 2.0$ s.

[2]

- (iii) The tension in the wire at time $t = 2.0$ s is 240 N.

Calculate:

- the horizontal component of the tension force acting on the skier

horizontal component of force = N [1]

- the total resistive force R acting on the skier in the horizontal direction.

$R =$ N [2]

- (iv) The skier is now lifted upwards by a gust of wind. For a few seconds the skier moves horizontally through the air with the wire at an angle of 45° to the horizontal, as shown in Fig. 2.3.

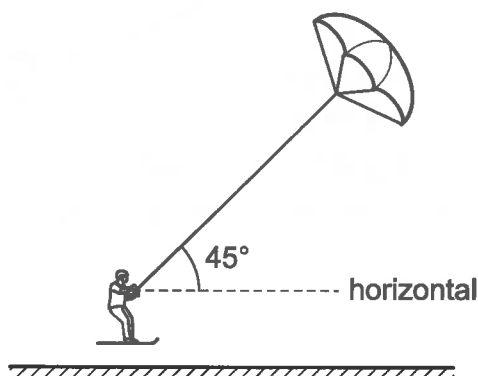


Fig. 2.3 (not to scale)

By considering the vertical components of the forces acting on the skier, determine the new tension in the wire when the skier is moving horizontally through the air.

tension = N [2]

45. O/N 19/P22/Q3

booklet (scan + OCR)

(a) State Newton's third law of motion.

.....
 [2]

(b) A block X of mass m_X slides in a straight line along a horizontal frictionless surface, as shown in Fig. 3.1.

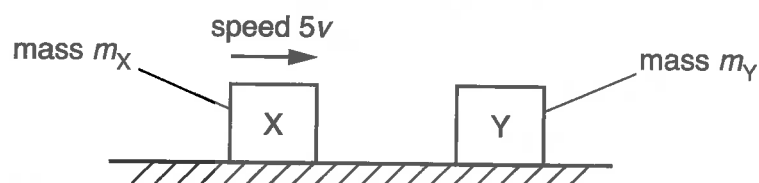


Fig. 3.1

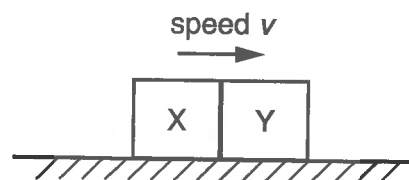


Fig. 3.2

The block X, moving with speed $5v$, collides head-on with a stationary block Y of mass m_Y . The two blocks stick together and then move with common speed v , as shown in Fig. 3.2.

(i) Use conservation of momentum to show that the ratio $\frac{m_Y}{m_X}$ is equal to 4.

[2]

(ii) Calculate the ratio

$$\frac{\text{total kinetic energy of X and Y after collision}}{\text{total kinetic energy of X and Y before collision}}.$$

ratio = [3]

(iii) State the value of the ratio in (ii) for a perfectly elastic collision.

ratio = [1]

(c) The variation with time t of the momentum of block X in (b) is shown in Fig. 3.3.

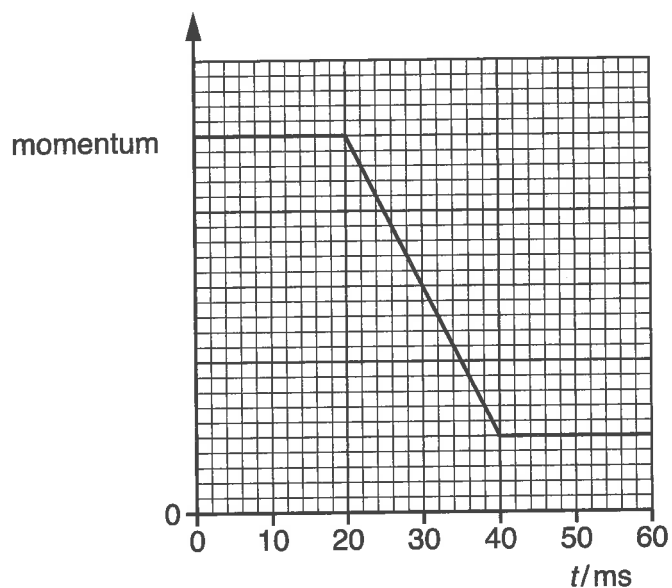


Fig. 3.3

Block X makes contact with block Y at time $t = 20$ ms.

(i) Describe, qualitatively, the magnitude and direction of the resultant force, if any, acting on block X in the time interval:

1. $t = 0$ to $t = 20$ ms

.....

2. $t = 20$ ms to $t = 40$ ms.

.....

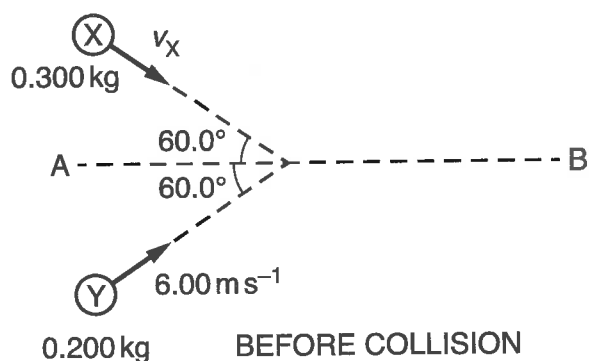
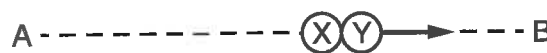
..... [3]

(ii) On Fig. 3.3, sketch the variation of the momentum of block Y with time t from $t = 0$ to $t = 60$ ms. [3]

46. O/N 19/P23/Q4/a

booklet (scan + OCR)

A ball X moves along a horizontal frictionless surface and collides with another ball Y, as illustrated in Fig. 4.1.

**Fig. 4.1** (not to scale)**Fig. 4.2** (not to scale)

Ball X has mass 0.300 kg and initial velocity v_X at an angle of 60.0° to line AB.

Ball Y has mass 0.200 kg and initial velocity 6.00 ms^{-1} at an angle of 60.0° to line AB.

The balls stick together during the collision and then travel along line AB, as illustrated in Fig. 4.2.

- (a) (i)** Calculate, to three significant figures, the component of the initial momentum of ball Y that is perpendicular to line AB.

component of momentum = kg ms^{-1} [2]

- (ii)** By considering the component of the initial momentum of each ball perpendicular to line AB, calculate, to three significant figures, v_X .

$v_X = \text{.....ms}^{-1}$ [1]

- (iii)** Show that the speed of the two balls after the collision is 2.4 ms^{-1} .

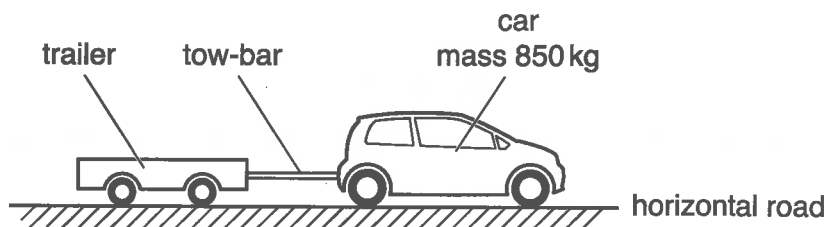
[2]

47. M/J 19/P22/Q2/a,b

booklet (scan + OCR)

(a) State Newton's second law of motion.

.....[1]

(b) A car of mass 850 kg tows a trailer in a straight line along a horizontal road, as shown in Fig. 2.1.**Fig. 2.1**

The car and the trailer are connected by a horizontal tow-bar.

The variation with time t of the velocity v of the car for a part of its journey is shown in Fig. 2.2.

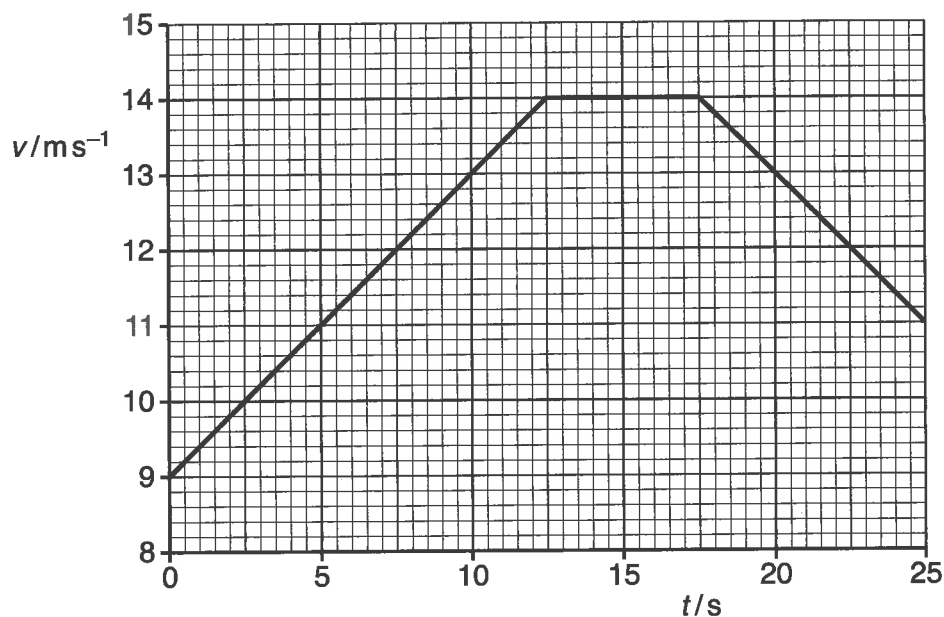


Fig. 2.2

- (i) Calculate the distance travelled by the car from time $t = 0$ to $t = 10$ s.

distance = m [2]

- (ii) At time $t = 10$ s, the resistive force acting on the car due to air resistance and friction is 510 N. The tension in the tow-bar is 440 N.

For the car at time $t = 10$ s:

1. use Fig. 2.2 to calculate the acceleration

acceleration = ms^{-2} [2]

2. use your answer to calculate the resultant force acting on the car

resultant force = N [1]

3. show that a horizontal force of 1300 N is exerted on the car by its engine

[1]

4. determine the useful output power of the engine.

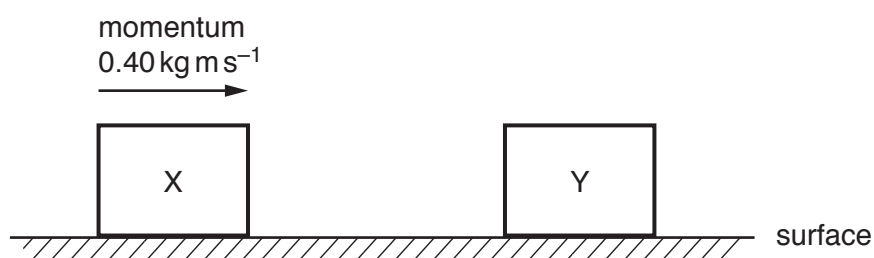
output power = W [2]

48. M/J 19/P21/Q2

booklet selection; question from the official paper

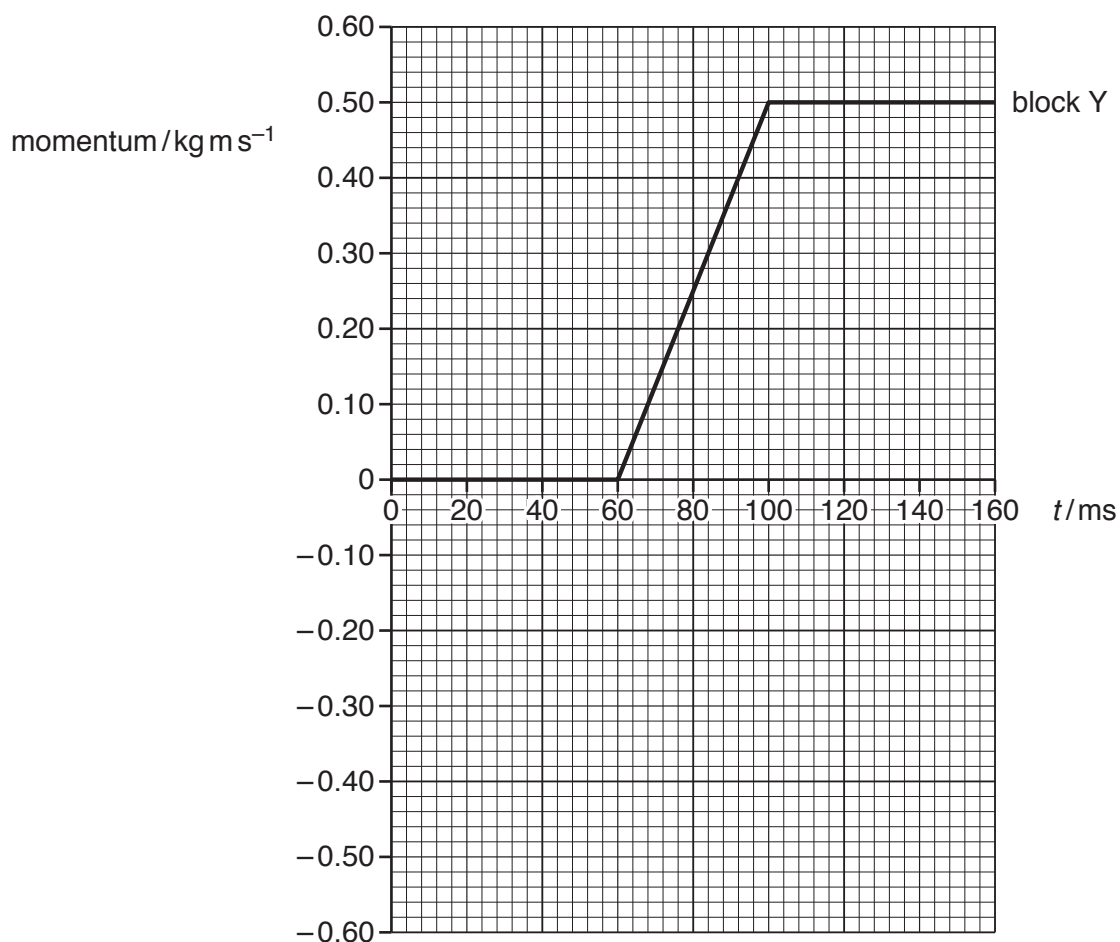
Question from the official paper: printed pages 144–145 of the scanned booklet are missing

- 2 A block X slides along a horizontal frictionless surface towards a stationary block Y, as illustrated in Fig. 2.1.

**Fig. 2.1**

There are no resistive forces acting on block X as it moves towards block Y. At time $t = 0$, block X has momentum 0.40 kg m s^{-1} . A short time later, the blocks collide and then separate.

The variation with time t of the momentum of block Y is shown in Fig. 2.2.

**Fig. 2.2**

- (a) Define *linear momentum*.

- (b) Use Fig. 2.2 to:

[1]

- (i) determine the time interval over which the blocks are in contact with each other

time interval = _____ ms [1]

- (ii) describe, without calculation, the magnitude of the acceleration of block Y from:

1. time $t = 80$ ms to $t = 100$ ms
2. time $t = 100$ ms to $t = 120$ ms.

[2]

- (c) Use Fig. 2.2 to determine the magnitude of the force exerted by block X on block Y.

force = _____ N [2]

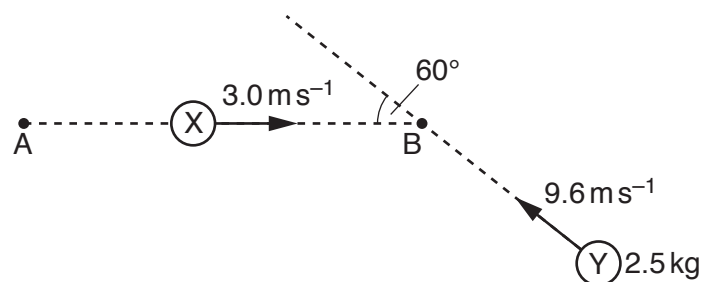
- (d) On Fig. 2.2, sketch the variation of the momentum of block X with time t from $t = 0$ to $t = 160$ ms. [3]

49. MAR 19/P22/Q3

booklet selection; question and answer from the official paper

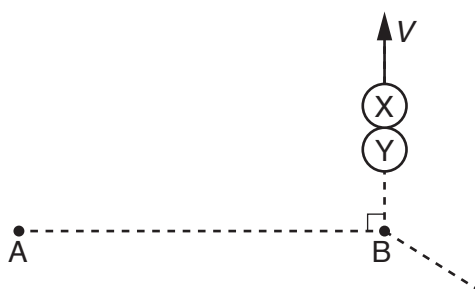
The scanned booklet lost this question (missing pages); only its answer survives

- 3** Two balls, X and Y, move along a horizontal frictionless surface, as illustrated in Fig. 3.1.

**Fig. 3.1** (not to scale)

Ball X has an initial velocity of 3.0 m s^{-1} in a direction along line AB. Ball Y has a mass of 2.5 kg and an initial velocity of 9.6 m s^{-1} in a direction at an angle of 60° to line AB.

The two balls collide at point B. The balls stick together and then travel along the horizontal surface in a direction at right-angles to the line AB, as shown in Fig. 3.2.

**Fig. 3.2**

- (a) By considering the components of momentum in the direction from A to B, show that ball X has a mass of 4.0 kg .

[2]

- (b) Calculate the common speed V of the two balls after the collision.

$V =$ m s^{-1} [2]

- (c) Determine the difference between the initial kinetic energy of ball X and the initial kinetic energy of ball Y.

difference in kinetic energy = J [2]

50. O/N 18/P22/Q3

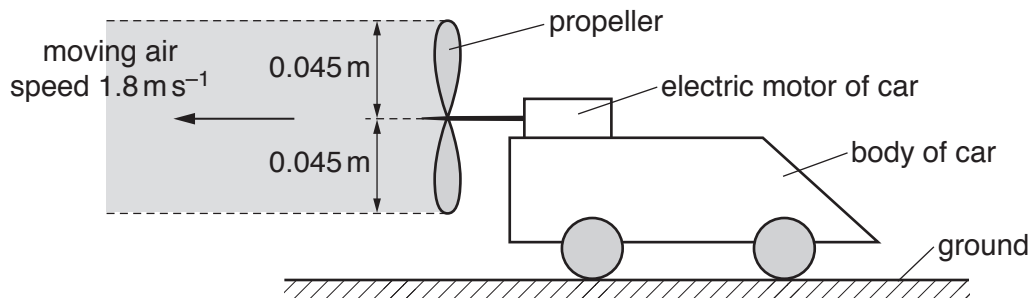
booklet selection; question and answer from the official paper

The scanned booklet lost this question (missing pages); only its answer survives

- 3 (a)** State the principle of conservation of momentum.

[2]

- (b)** The propulsion system of a toy car consists of a propeller attached to an electric motor, as illustrated in Fig. 3.1.

**Fig. 3.1**

The car is on horizontal ground and is initially held at rest by its brakes. When the motor is switched on, it rotates the propeller so that air is propelled horizontally to the left. The density of the air is 1.3 kg m^{-3} .

Assume that the air moves with a speed of 1.8 m s^{-1} in a uniform cylinder of radius 0.045 m . Also assume that the air to the right of the propeller is stationary.

- (i)** Show that, in a time interval of 2.0 s , the mass of air propelled to the left is 0.030 kg .

[2]

- (ii)** Calculate

1. the increase in the momentum of the mass of air in **(b)(i)**,

increase in momentum = Ns

2. the force exerted on this mass of air by the propeller.

force = N
[3]

- (iii)** Explain how Newton's third law applies to the movement of the air by the propeller.

[2]

- (iv)** The total mass of the car is 0.20 kg . The brakes of the car are released and the car begins to move with an initial acceleration of 0.075 m s^{-2} .

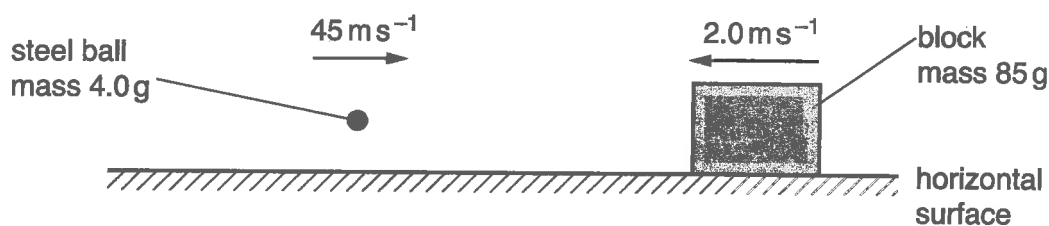
Determine the initial frictional force acting on the car.

frictional force = N [2]

51. O/N 18/P21/Q2

booklet (scan + OCR)

A wooden block moves along a horizontal frictionless surface, as shown in Fig. 2.1.

**Fig. 2.1**

The block has mass 85 g and moves to the left with a velocity of 2.0 m s^{-1} . A steel ball of mass 4.0 g is fired to the right. The steel ball, moving horizontally with a speed of 45 m s^{-1} , collides with the block and remains embedded in it. After the collision the block and steel ball both have speed v .

(a) Calculate v .

$$v = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

(b) (i) For the block and ball, state

1. the relative speed of approach before collision,

$$\text{relative speed of approach} = \dots\dots\dots \text{ m s}^{-1}$$

2. the relative speed of separation after collision.

$$\text{relative speed of separation} = \dots\dots\dots \text{ m s}^{-1} \quad [1]$$

(ii) Use your answers in **(i)** to state and explain whether the collision is elastic or inelastic.

..... [1]

(c) Use Newton's third law to explain the relationship between the rate of change of momentum of the ball and the rate of change of momentum of the block during the collision.

..... [2]

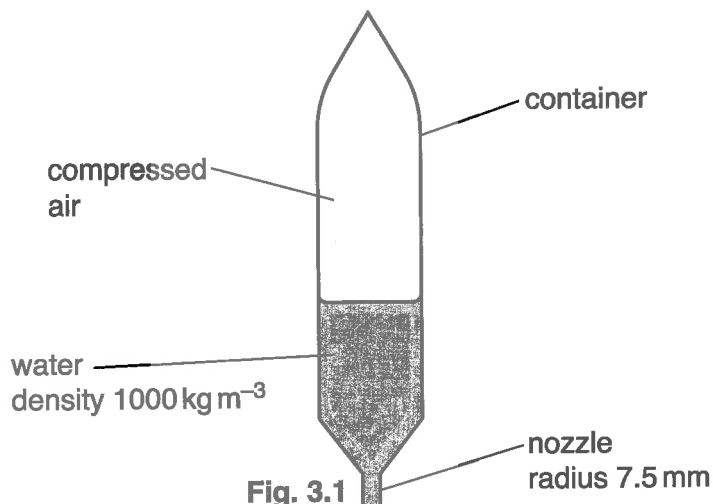
52. O/N 18/P23/Q3

booklet (scan + OCR)

(a) State Newton's second law of motion.

..... [1]

(b) A toy rocket consists of a container of water and compressed air, as shown in Fig. 3.1.



Water is pushed vertically downwards through a nozzle by the compressed air. The rocket moves vertically upwards.

The nozzle has a circular cross-section of radius 7.5 mm. The density of the water is 1000 kg m^{-3} . Assume that the water leaving the nozzle has the shape of a cylinder of radius 7.5 mm and has a constant speed of 13 ms^{-1} relative to the rocket.

- (i)** Show that the mass of water leaving the nozzle in the first 0.20 s after the rocket launch is 0.46 kg.

[2]

(ii) Calculate

1. the change in the momentum of the mass of water in **(b)(i)** due to leaving the nozzle,

change in momentum = N s

2. the force exerted on this mass of water by the rocket.

force = N

[3]

(iii) State and explain how Newton's third law applies to the movement of the rocket by the water.

.....

..... [2]

- (iv) The container has a mass of 0.40 kg. The initial mass of water before the rocket is launched is 0.70 kg. The mass of the compressed air in the rocket is negligible. Assume that the resistive force on the rocket due to its motion is negligible. For the rocket at a time of 0.20 s after launching,
1. show that its total mass is 0.64 kg,

2. calculate its acceleration.

acceleration = ms^{-2} [3]

53. M/J 18/P22/Q3

booklet (scan + OCR)

A child on a sledge slides down a steep hill and then travels in a straight line up an ice-covered slope, as illustrated in Fig. 3.1.

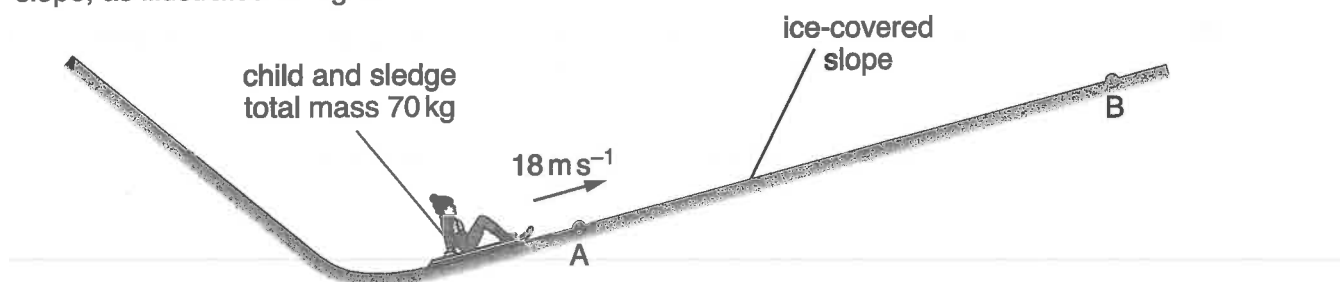


Fig. 3.1 (not to scale)

The sledge passes point A with speed 18 ms^{-1} at time $t = 0$ and then comes to rest at point B. The child applies a brake to the sledge at point B. The brake does not keep the sledge stationary and it immediately slides back down the slope towards A.

The variation with time t of the velocity v of the sledge from $t = 0$ to $t = 24 \text{ s}$ is shown in Fig. 3.2.

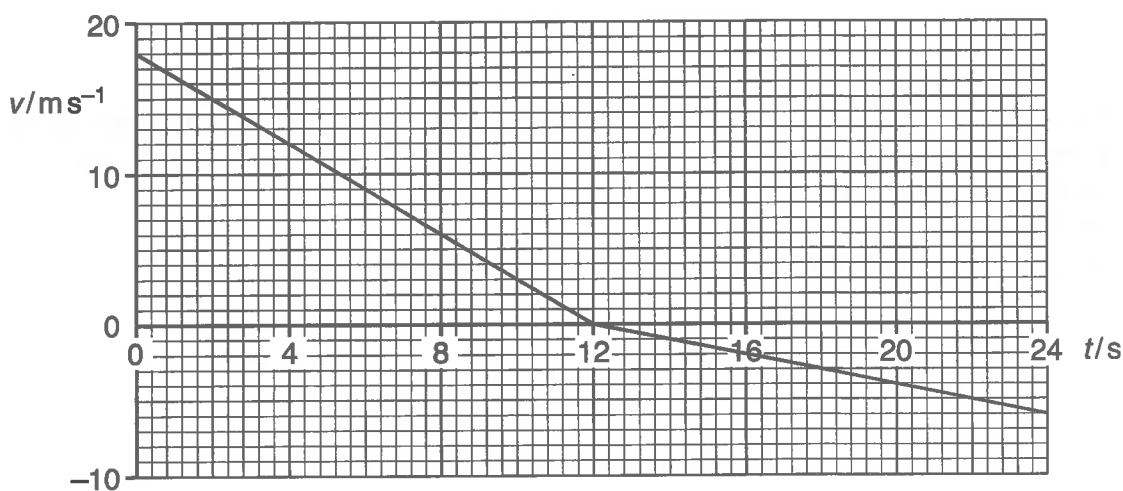


Fig. 3.2

- (a) State the time taken for the sledge to travel from A to B.

time = s [1]

- (b) Determine the displacement of the sledge up the slope from point A at time $t = 24$ s.

displacement = m [3]

- (c) Show that the acceleration of the sledge as it moves from B back towards A is 0.50 m s^{-2} .

[2]

- (d) The child and sledge have a total mass of 70 kg. The component of the total weight of the child and sledge that acts down the slope is 80 N.

Determine

- (i) the frictional force on the sledge as it moves from B towards A,

frictional force = N [2]

- (ii) the angle θ of the slope to the horizontal.

$\theta =$ $^{\circ}$ [2]

- (e) The child on the sledge blows a whistle between $t = 4.0$ s and $t = 8.0$ s. The whistle emits sound of frequency 900 Hz. The speed of the sound in the air is 340 m s^{-1} . A man standing at point A hears the sound.

Use Fig. 3.2 to

- (i) determine the initial frequency of the sound heard by the man,

initial frequency = Hz [2]

- (ii) describe and explain qualitatively the variation, if any, in the frequency of the sound heard by the man.

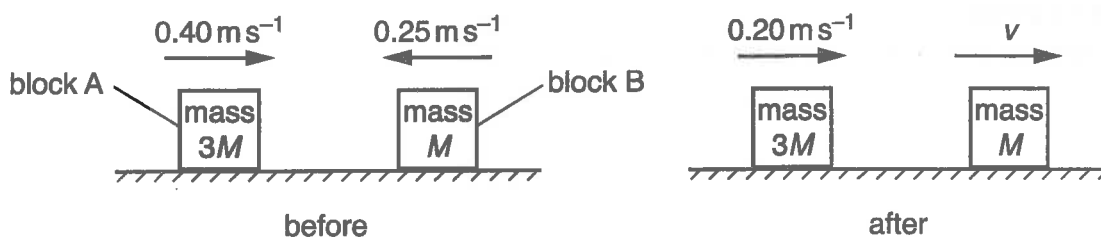
..... [1]

54. M/J 18/P21/Q3

booklet (scan + OCR)

(a) State what is meant by the *mass* of a body.

[1]

(b) Two blocks travel directly towards each other along a horizontal, frictionless surface. The blocks collide, as illustrated in Fig. 3.1.**Fig. 3.1**Block A has mass $3M$ and block B has mass M .Before the collision, block A moves to the right with speed 0.40 ms^{-1} and block B moves to the left with speed 0.25 ms^{-1} .After the collision, block A moves to the right with speed 0.20 ms^{-1} and block B moves to the right with speed v .**(i)** Use Newton's third law to explain why, during the collision, the change in momentum of block A is equal and opposite to the change in momentum of block B.

[2]

(ii) Determine speed v .

$$v = \dots \text{ ms}^{-1} \quad [3]$$

(iii) Calculate, for the blocks,
1. the relative speed of approach,

$$\text{relative speed of approach} = \dots \text{ ms}^{-1}$$

2. the relative speed of separation.

$$\text{relative speed of separation} = \dots \text{ ms}^{-1} \quad [2]$$

(iv) Use your answers in **(b)(iii)** to state and explain whether the collision is elastic or inelastic.

[1]

55. O/N 17/P23/Q3

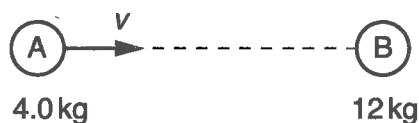
booklet (scan + OCR)

- (a)**
- State the principle of conservation of momentum.

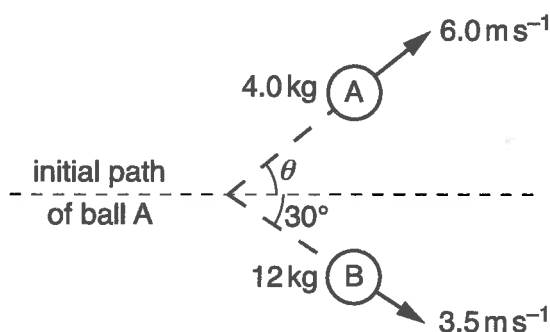
.....

..... [2]

- (b)**
- Ball A moves with speed
- v
- along a horizontal frictionless surface towards a stationary ball B, as shown in Fig. 3.1.



before collision

Fig. 3.1

after collision

Fig. 3.2 (not to scale)

Ball A has mass 4.0 kg and ball B has mass 12 kg .

The balls collide and then move apart as shown in Fig. 3.2.

Ball A has velocity 6.0 ms^{-1} at an angle of θ to the direction of its initial path.

Ball B has velocity 3.5 ms^{-1} at an angle of 30° to the direction of the initial path of ball A.

- (i)**
- By considering the components of momentum at right-angles to the direction of the initial path of ball A, calculate
- θ
- .

$$\theta = \dots\dots\dots^\circ \quad [3]$$

- (ii)**
- Use your answer in
- (i)**
- to show that the initial speed
- v
- of ball A is
- 12 ms^{-1}
- .
-
- Explain your working.

[2]

- (iii)**
- By calculation of kinetic energies, state and explain whether the collision is elastic or inelastic.

.....

.....

..... [3]

56. M/J 17/P22/Q4

booklet (scan + OCR)

(a) State Newton's first law of motion.

..... [1]

- (b)** An object A of mass 100 g is moving in a straight line with a velocity of 0.60 m s^{-1} to the right. An object B of mass 200 g is moving in the same straight line as object A with a velocity of 0.80 m s^{-1} to the left, as shown in Fig. 4.1.

**Fig. 4.1**

Objects A and B collide. Object A then moves with a velocity of 0.40 m s^{-1} to the left.

- (i)** Calculate the magnitude of the velocity of B after the collision.

magnitude of velocity = m s^{-1} [2]

- (ii)** The collision between A and B is inelastic,

Explain how the collision is inelastic and still obeys the law of conservation of energy.

..... [1]

57. M/J 17/P21/Q2

booklet (scan + OCR)

(a) State the two conditions for a system to be in equilibrium.

1.

2. [2]

(b) A paraglider P of mass 95 kg is pulled by a wire attached to a boat, as shown in Fig. 2.1.

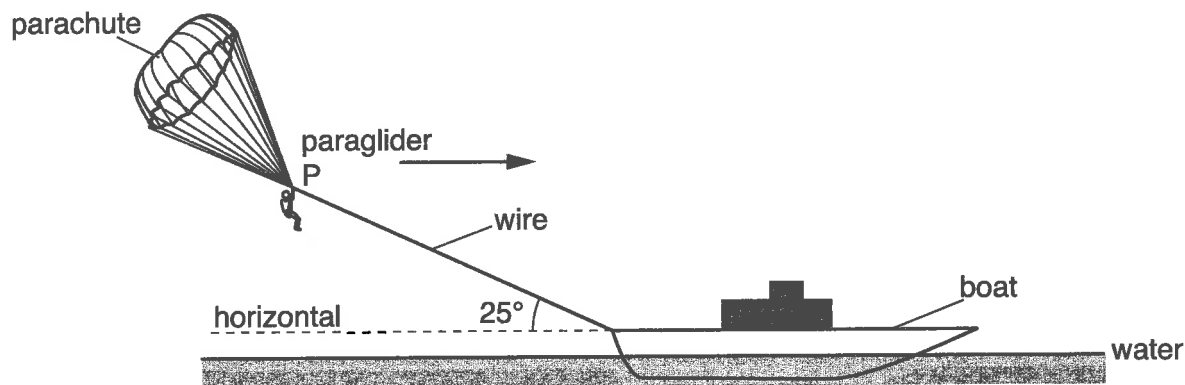


Fig. 2.1

The wire makes an angle of 25° with the horizontal water surface. P moves in a straight line parallel to the surface of the water.

The variation with time t of the velocity v of P is shown in Fig. 2.2.

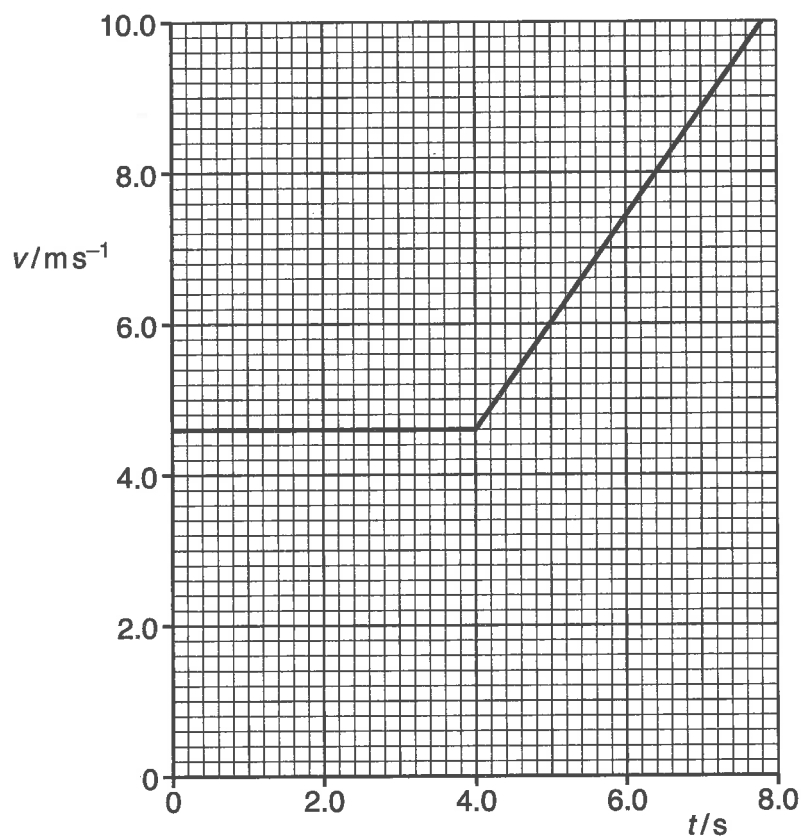


Fig. 2.2

(i) Show that the acceleration of P is 1.4 ms^{-2} at time $t = 5.0 \text{ s}$.

(ii) Calculate the total distance moved by P from time $t = 0$ to $t = 7.0$ s.

distance =m [2]

(iii) Calculate the change in kinetic energy of P from time $t = 0$ to $t = 7.0$ s.

change in kinetic energy =J [2]

(iv) The tension in the wire at time $t = 5.0$ s is 280 N.

Calculate, for the horizontal motion,

1. the vertical lift force F supporting P,

$F = \dots\dots\dots$ N [3]

2. the force R due to air resistance acting on P in the horizontal direction.

$R = \dots\dots\dots$ N [3]

58. M/J 17/P23/Q2

booklet (scan + OCR)

(a) State Newton's second law of motion.

..... [1]

- (b) A constant resultant force F acts on an object A. The variation with time t of the velocity v for the motion of A is shown in Fig. 2.1.

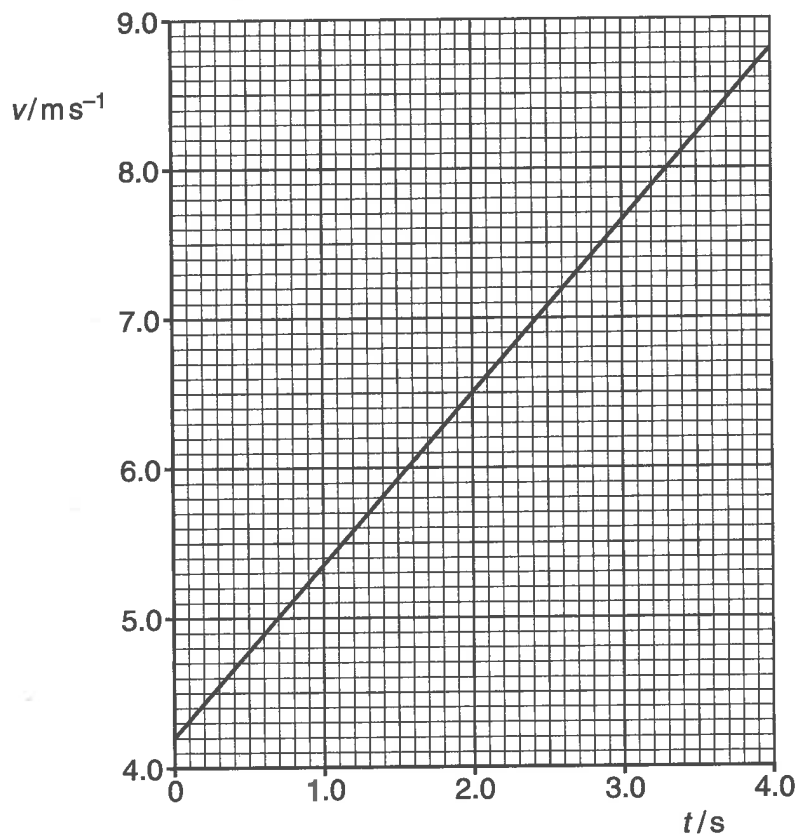


Fig. 2.1

The mass of A is 840 g.

Calculate, for the time $t = 0$ to $t = 4.0$ s,

- (i) the change in momentum of A,

change in momentum = kg m s⁻¹ [2]

- (ii) the force F .

$F =$ N [1]

- (c) The force F is removed at $t = 4.0$ s. Object A continues at constant velocity before colliding with an object B, as illustrated in Fig. 2.2.



Fig. 2.2

Object B is initially at rest. The mass of B is 730 g.

The objects A and B join together and have a velocity of 4.7 m s^{-1} .

- (i) By calculation, show that the changes in momentum of A and of B during the collision are equal and opposite.

[2]

- (ii) Explain how the answers obtained in (i) support Newton's third law.

.....

..... [2]

- (iii) By reference to the speeds of A and B, explain whether the collision is elastic.

..... [1]

59. O/N 16/P22/Q2

booklet (scan + OCR)

A ball of mass 0.030 kg moves along a curved track, as shown in Fig. 2.1.

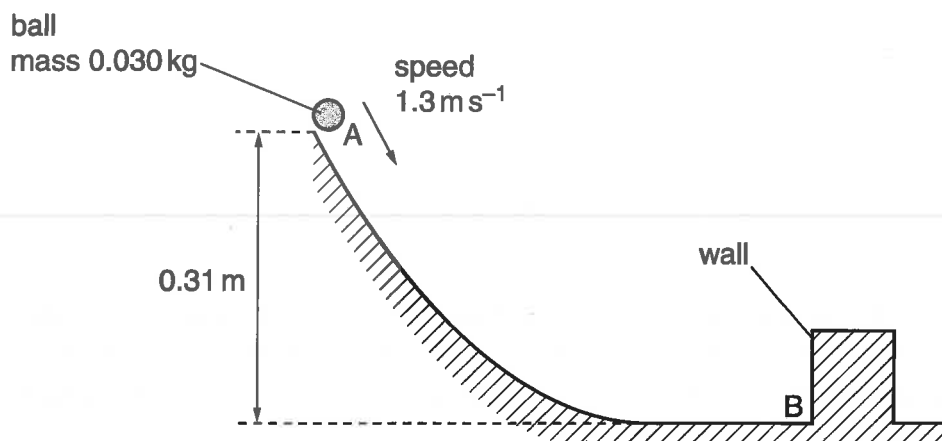


Fig. 2.1

The speed of the ball is 1.3 m s^{-1} when it is at point A at a height of 0.31 m .

The ball moves down the track and collides with a vertical wall at point B. The ball then rebounds back up the track. It may be assumed that frictional forces are negligible.

- (a) Calculate the change in gravitational potential energy of the ball in moving from point A to point B.

change in gravitational potential energy = J [2]

(b) Show that the ball hits the wall at B with a speed of 2.8 ms^{-1} .

[3]

(c) The change in momentum of the ball due to the collision with the wall is 0.096 kg ms^{-1} . The ball is in contact with the wall for a time of 20 ms.

Determine, for the ball colliding with the wall,

(i) the speed immediately after the collision,

speed = ms^{-1} [2]

(ii) the magnitude of the average force on the ball.

force = N [2]

(d) State and explain whether the collision is elastic or inelastic.

..... [1]

(e) In practice, frictional effects are significant so that the actual increase in kinetic energy of the ball in moving from A to B is 76 mJ. The length of the track between A and B is 0.60 m.

Use your answer in **(a)** to determine the average frictional force acting on the ball as it moves from A to B.

frictional force = N [2]

60. M/J 16/P22/Q1

booklet (scan + OCR)

(a) Define *acceleration*.

..... [1]

- (b) A man travels on a toboggan down a slope covered with snow from point A to point B and then to point C. The path is illustrated in Fig. 1.1.

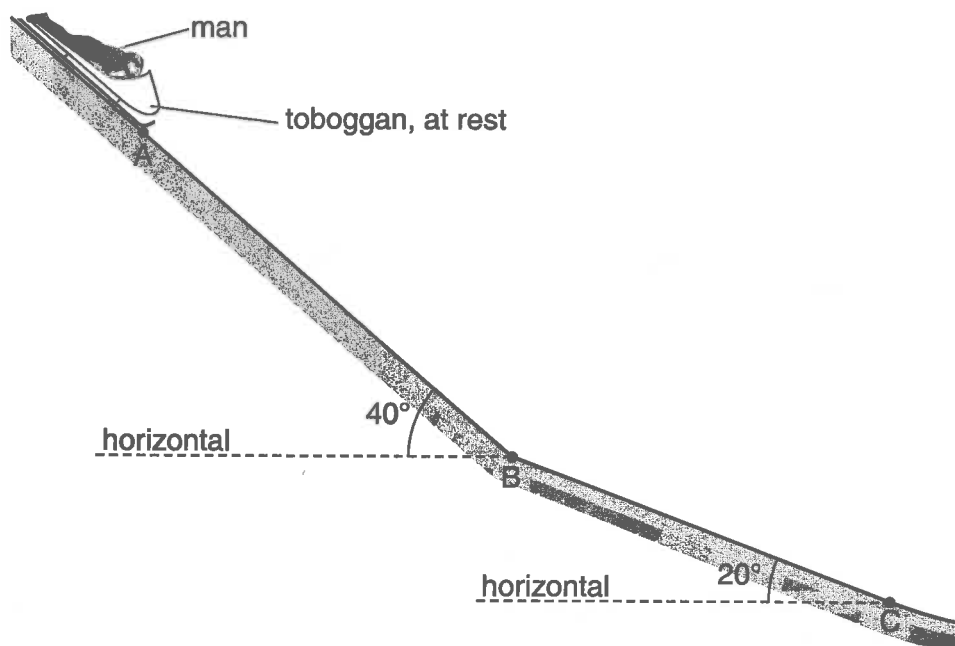


Fig. 1.1 (not to scale)

The slope AB makes an angle of 40° with the horizontal and the slope BC makes an angle of 20° with the horizontal. Friction is not negligible.

The man and toboggan have a combined mass of 95 kg.

The man starts from rest at A and has constant acceleration between A and B. The man takes 19 s to reach B. His speed is 36 m s^{-1} at B.

- (i) Calculate the acceleration from A to B.

acceleration = ms^{-2} [2]

- (ii) Show that the distance moved from A to B is 340 m.

[1]

- (iii) For the man and toboggan moving from A to B, calculate
1. the change in kinetic energy,

change in kinetic energy = J [2]

2. the change in potential energy.

change in potential energy = J [2]

- (iv) Use your answers in (iii) to determine the average frictional force that acts on the toboggan between A and B.

frictional force = N [2]

- (v) A parachute opens on the toboggan as it passes point B. There is a constant deceleration of 3.0 m s^{-2} from B to C.
Calculate the frictional force that produces this deceleration between B and C.

frictional force = N [2]

61. M/J 16/P21/Q3

booklet (scan + OCR)

A ball of mass 150 g is at rest on a horizontal floor, as shown in Fig. 3.1.

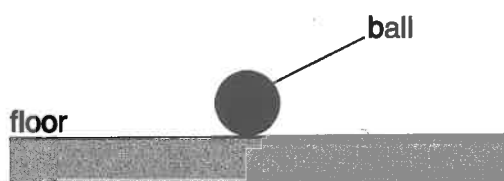


Fig. 3.1

- (a) (i) Calculate the magnitude of the normal contact force from the floor acting on the ball.

force = N [1]

- (ii) Explain your working in (i).

..... [1]

- (b) The ball is now lifted above the floor and dropped so that it falls vertically, as illustrated in Fig. 3.2.

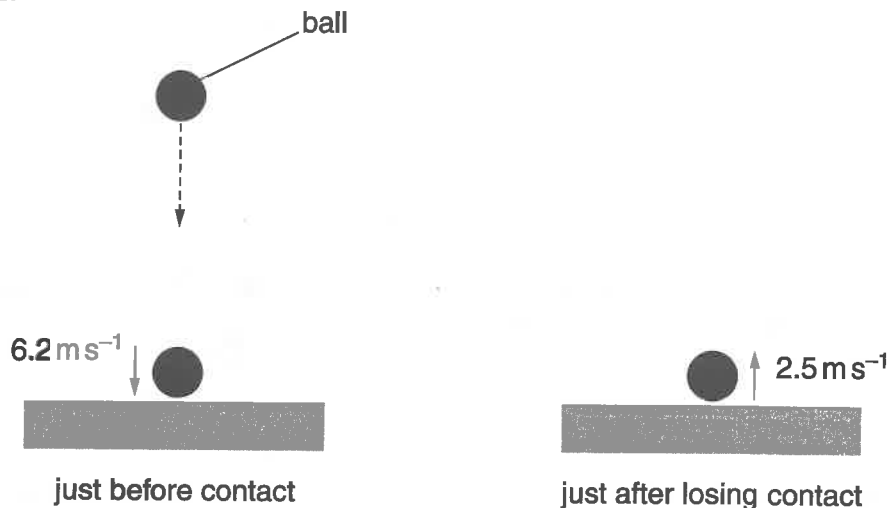


Fig. 3.2

Just before contact with the floor, the ball has velocity 6.2 ms^{-1} downwards. The ball bounces from the floor and its velocity just after losing contact with the floor is 2.5 ms^{-1} upwards. The ball is in contact with the floor for 0.12 s .

- (i) State Newton's second law of motion.

..... [1]

- (ii) Calculate the average resultant force on the ball when it is in contact with the floor.

magnitude of force = N

direction of force [3]

- (iii) State and explain whether linear momentum is conserved during the collision of the ball with the floor.

..... [2]

62. M/J 16/P23/Q5

booklet (scan + OCR)

- (a) State the law of conservation of momentum.

..... [2]

(b) Two particles A and B collide elastically, as illustrated in Fig. 5.1.

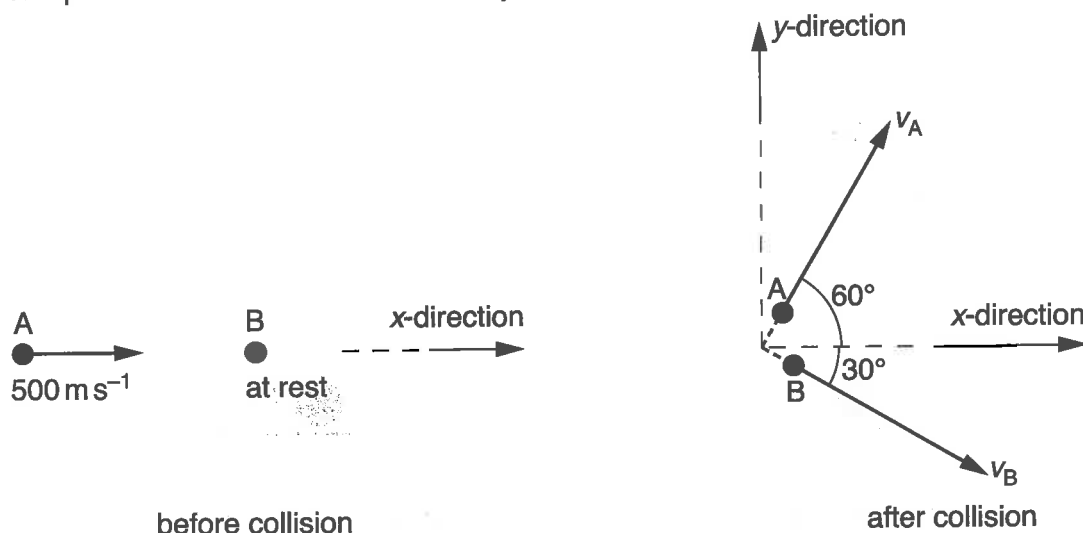


Fig. 5.1

The initial velocity of A is 500 m s^{-1} in the x-direction and B is at rest.

The velocity of A after the collision is v_A at 60° to the x-direction. The velocity of B after the collision is v_B at 30° to the x-direction.

The mass m of each particle is $1.67 \times 10^{-27} \text{ kg}$.

(i) Explain what is meant by the particles colliding *elastically*.

..... [1]

(ii) Calculate the total initial momentum of A and B.

momentum =Ns [1]

(iii) State an expression in terms of m , v_A and v_B for the total momentum of A and B after the collision

1. in the x-direction,

.....

2. in the y-direction.

..... [2]

(iv) Calculate the magnitudes of the velocities v_A and v_B after the collision.

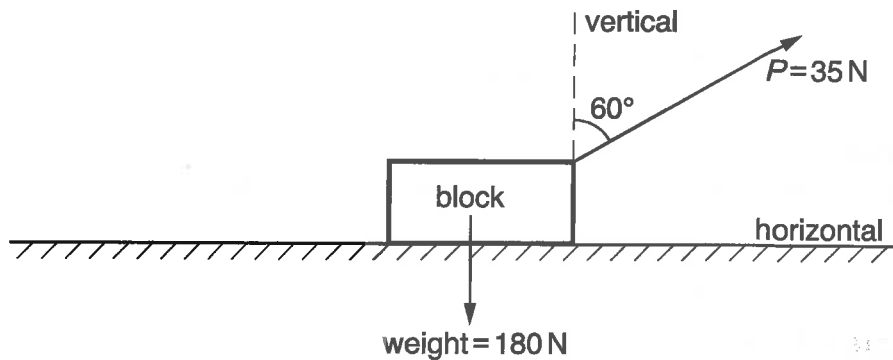
$v_A = \dots \text{ m s}^{-1}$

$v_B = \dots \text{ m s}^{-1}$ [3]

63. O/N 15/P23/Q4

booklet (scan + OCR)

A block is pulled on a horizontal surface by a force P as shown in Fig. 4.1.

**Fig. 4.1**

The weight of the block is 180 N. The force P is 35 N at 60° to the vertical.

The block moves a distance of 20 m at constant velocity.

(a) Calculate

- (i)** the vertical force that the surface applies to the block (normal reaction force),

force = N [2]

- (ii)** the work done by force P .

work done = J [2]

- (b) (i)** Explain why the block continues to move at constant velocity although work is done on the block by force P .

..... [1]

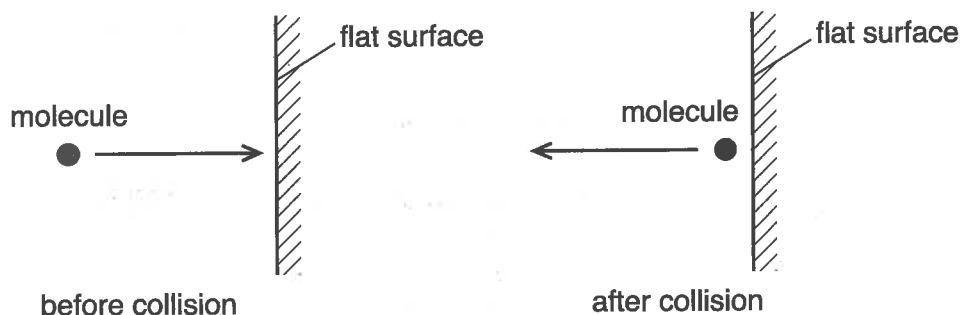
- (ii)** Explain, in terms of the forces acting, why the block remains in equilibrium.

.....
 [2]

64. M/J 15/P22/Q4

booklet (scan + OCR)

- (a) A gas molecule has a mass of $6.64 \times 10^{-27} \text{ kg}$ and a speed of 1250 ms^{-1} . The molecule collides normally with a flat surface and rebounds with the same speed, as shown in Fig. 4.1.

**Fig. 4.1**

Calculate the change in momentum of the molecule.

change in momentum = N s [2]

- (b) (i) Use the kinetic model to explain the pressure exerted by gases.

.....
.....
..... [3]

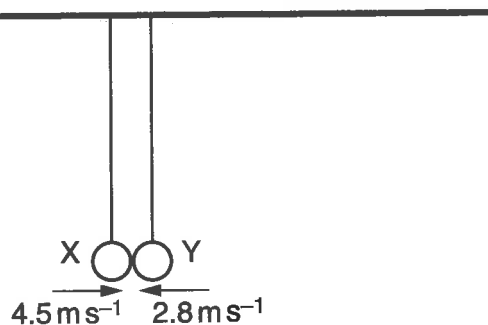
- (ii) Explain the effect of an increase in density, at constant temperature, on the pressure of a gas.

..... [1]

65. M/J 15/P21/Q3

booklet (scan + OCR)

Two balls X and Y are supported by long strings, as shown in Fig. 3.1.

**Fig. 3.1**

The balls are each pulled back and pushed towards each other. When the balls collide at the position shown in Fig. 3.1, the strings are vertical. The balls rebound in opposite directions. Fig. 3.2 shows data for X and Y during this collision.

ball	mass	velocity just before collision/ ms^{-1}	velocity just after collision/ ms^{-1}
X	50 g	+4.5	-1.8
Y	M	-2.8	+1.4

Fig. 3.2

The positive direction is horizontal and to the right.

(a) Use the conservation of linear momentum to determine the mass M of Y.

$$M = \dots\dots\dots \text{ g [3]}$$

(b) State and explain whether the collision is elastic.

..... [1]

(c) Use Newton's second and third laws to explain why the magnitude of the change in momentum of each ball is the same.

.....

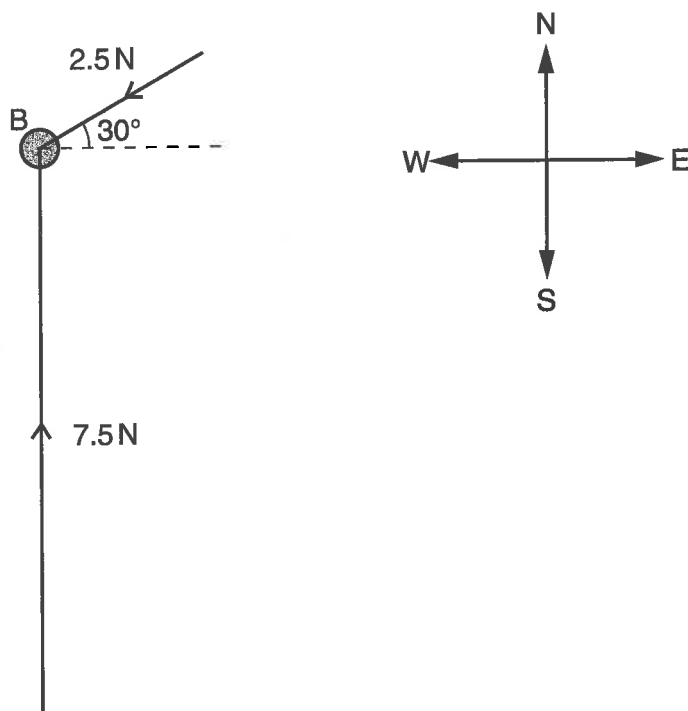
.....

..... [3]

66. O/N 14/P22/Q1/c

booklet (scan + OCR)

- (a) An object B is on a horizontal surface. Two forces act on B in this horizontal plane. A vector diagram for these forces is shown to scale in Fig. 1.1.

**Fig. 1.1**

A force of 7.5 N towards north and a force of 2.5 N from 30° north of east act on B.
The mass of B is 750 g.

- (i) On Fig. 1.1, draw an arrow to show the approximate direction of the resultant of these two forces. [1]
- (ii) 1. Show that the magnitude of the resultant force on B is 6.6 N.

[1]

2. Calculate the magnitude of the acceleration of B produced by this resultant force.

magnitude = ms^{-2} [2]

- (iii) Determine the angle between the direction of the acceleration and the direction of the 7.5 N force.

angle = ° [1]

Answers Section

1. M/J 26/P22/Q6/b(i,ii)

6(b)(i)	$(p =) mv$	C1	Allow any mass $\times$ any velocity e.g. $4(u) \times 5.60 \times 10^6$ or $1.66 \times 10^{-27} \times 5.60 \times 10^6$ or $1.67 \times 10^{-27} \times 5.60 \times 10^6$
	$= 4 \times 1.66 \times 10^{-27} \times 5.60 \times 10^6 = 3.72 \times 10^{-20} \text{ (kg m s}^{-1}\text{)}$	A1	As this is a 'show that' question the full working and answer must be seen Not 1.67×10^{-27} or 1.6×10^{-27} for A1 mark
6(b)(ii)	$4(u) \times 5.60 \times 10^6 = 197(u)v - 4(u) \times 5.38 \times 10^6$ $3.72 \times 10^{-20} = (197 \times 1.66 \times 10^{-27}v) - (4 \times 1.66 \times 10^{-27} \times 5.38 \times 10^6)$	C1	If u is present in equation it must be present in all terms Allow 1.67 instead of 1.66 as gives correct answer to 2SF
	speed = $2.2 \times 10^5 \text{ m s}^{-1}$	A1	<u>Special case:</u> $4(u) \times 5.60 \times 10^6 = 197(u)v + (4(u) \times 5.38 \times 10^6)$ leading to 4500 m s^{-1} credit 1 / 2 working must be seen <u>Alternative method</u> if candidates (correctly) assume elastic collision and use speed of approach = speed of separation speed = $5.60 \times 10^6 - 5.38 \times 10^6$ (C1) = 2.2×10^5 (A1)

2. M/J 26/P23/Q1/a,b(ii),c(i,iii)

1(a)	product of mass and velocity	B1	not speed for velocity
1(b)(ii)	0.15×1.6 or 0.15×0.8	C1	No tolerance. ECF from (b)(i) for v read-offs
	change in momentum = $0.15 \times (-0.8 - 1.6)$ = $(-)0.36 \text{ kg m s}^{-1}$	A1	Ignore sign
	OR $F = 0.15 \times (-)80$ = $(-)12$	(C1)	ECF a from (b)(i)
	Change in momentum = $(-)12 \times (0.12 - 0.09)$ = $(-)0.36 \text{ kg m s}^{-1}$	(A1)	
1(c)(i)	relative speed of approach = relative speed of separation	B1	Allow 'relative speed before collision = relative speed after collision' Allow $u_1 - u_2 = v_2 - v_1$ not relative <u>velocity</u> for speed, but allow relative velocity of approach = - relative velocity of separation
	(speed of separation = 1.6 m s^{-1} so) $v_Y - (-0.80) = 1.6$, so $v_Y = 0.80 \text{ m s}^{-1}$	B1	Allow $v_Y + 0.80 = 1.6$ Allow $0.80 + 0.80 = 1.6$ BOD Ignore $v_Y = 1.6 / 2$
1(c)(iii)	$(0.15 \times 1.6) = (0.15 \times (-0.80)) + 0.80m$ $0.36 = 0.8m$	C1	ECF from (b)(i) for v read-offs
	OR $(\frac{1}{2} \times 0.15 \times 1.6^2) = (\frac{1}{2} \times 0.15 \times 0.80^2) + (\frac{1}{2} \times m \times 0.80^2)$ $0.192 = 0.048 + 0.32m$	(C1)	Allow with the $\frac{1}{2}$ factor missing from all three terms.
	OR $m = 0.15 \times 80 / [0.80 / (0.12 - 0.09)]$ $m = 12 / 26.7$	(C1)	
	mass = 0.45 kg	A1	

3. M/J 26/P24/Q1/b(iii),c(i)

1(b)(iii)	$F = ma$ $= 1400 \times 3.2$	C1	ecf a from 1(b)(ii)
	$= 4500 \text{ N}$	A1	Allow $\pm 4500 \text{ N}$
	OR $F = W/s$ $= 1.6 \times 10^5 / 35$	(C1)	Allow $E/E_K/KE$ for W . Allow d or x for s
	$= 4600 \text{ N}$	(A1)	Allow $\pm 4600 \text{ N}$ ecf E_K from 1(b)(i) Allow 4500 or 4600 for either method.
1(c)(i)	This will remain the same as ($F = ma$ and) the force and the mass are the same	A1	Allow no change / constant / equal for 'the same' allow F and m as symbols

4. M/J 26/P24/Q4/a,b

4(a)	force	increases	stays the same	decreases	B1	Allow any other consistent ways of indicating choice provided no other marks seen. One tick on each line only.
	W		✓		B1	
	D	✓			B1	
	U		✓			
4(b)	$W = D + U$				A1	Allow any subject.

5. MAR 26/P22/Q1/a

1(a)	rate of change of momentum	B1
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6. MAR 26/P22/Q2/a,b

2(a)	<u>sum</u> / <u>total</u> momentum (of a system of bodies) is constant or <u>sum</u> / <u>total</u> momentum before = <u>sum</u> / <u>total</u> momentum after	M1
	for an isolated system / no (resultant) external force	A1
2(b)(i)	23×3.4 or $(23 + 160)v$ $23 \times 3.4 + (160 \times 0) = (23 + 160)v$	C1
	$v = 0.43 \text{ ms}^{-1}$	A1
2(b)(ii)	$E_{(K)} = \frac{1}{2} mv^2$	C1
	$E_2/E_1 = \frac{1}{2} (160+23) \times 0.43^2 / \frac{1}{2} (23) \times 3.4^2$ $= 16.9 / 132.9$	C1
	$E_2/E_1 = 0.13$	A1
2(b)(iii)	The <u>total</u> kinetic energy after the collision is less than the <u>total</u> kinetic energy before the collision, so inelastic	B1
	OR The relative speed of approach is more than the relative speed of separation, so inelastic	(B1)
	OR $E_2 < E_1$ so inelastic	(B1)

7. O/N 25/P21/Q2/b(i),c(ii)

2(b)(i)	arrow vertically downwards labelled 'weight'	B1
	arrow vertically upwards labelled 'upthrust'	B1
	arrow vertically upwards labelled '(viscous) drag'	B1
2(c)(ii)	weight = drag + upthrust	C1
	$(2.4 \times 10^{-3} \times 9.81) = (2.8 \times 10^{-3}) + (6\pi \times 4.7 \times 4.2 \times 10^{-3} \times v)$	C1
	$v = 0.056 \text{ m s}^{-1}$	A1

8. O/N 25/P22/Q1/b(iii)

1(b)(iii)	$(\Sigma)F = 33 - 19$ $(= 14 \text{ N})$	C1
	$m = 19 / 9.81$ $(= 1.94 \text{ kg})$	C1
	$a = (33 - 19) / (19 / 9.81)$ $= 7.2 \text{ m s}^{-2}$	A1

9. O/N 25/P22/Q2/b,c

2(b)	line with a positive gradient from $(t_1, 0)$ to t_2	B1
	line with increasing positive gradient from t_1 to t_2	B1
2(c)(i)	<u>sum</u> / <u>total</u> momentum before = <u>sum</u> / <u>total</u> momentum after or <u>sum</u> / <u>total</u> momentum (of a system of objects) is constant	M1
	if no (resultant) external force / for an isolated system	A1
2(c)(ii)	$(\Delta)p = F(\Delta)t$	C1
	$m = Ft / (\Delta)v$ $(m =) 5500 \times 0.36 / 8.5 = 230 \text{ (kg)}$	A1
	or	
	$F = ma$ and $a = (\Delta)v / (\Delta)t$	(C1)
	$(m =) 5500 / (8.5 / 0.36) = 230 \text{ (kg)}$	(A1)
2(c)(iii)	$(\Delta p =) 5500 \times 0.36 (= 1980 \text{ N s})$ or $(\Delta p =) 230 \times 8.5 (= 1955 \text{ N s})$	C1
	$1980 = (2.5 \times 10^3 - 230)\Delta v$ or $1955 = (2.5 \times 10^3 - 230)\Delta v$	C1
	$\Delta v = 0.87 \text{ m s}^{-1}$ or 0.86 m s^{-1}	A1
	or	
	$a = 5500 / (2.5 \times 10^3 - 230)$ $(= 2.42 \text{ m s}^{-2})$	(C1)
	$\Delta v = 2.42 \times 0.36$	(C1)
	$\Delta v = 0.87 \text{ m s}^{-1}$ or 0.86 m s^{-1}	(A1)

10. O/N 25/P24/Q1/b

1(b)(i)	product of mass and velocity	B1
1(b)(ii)	$F = \Delta p / t$	C1
	$= 13 / 3.2$	A1
	$= 4.1 \text{ N}$	
1(b)(iii)	$m = \Delta p / v$	A1
	$= 13 / (2 \times 16)$	
	$= 0.41 \text{ kg}$	
	or	
	$m = F / g$	
	$= 4.06 / 9.81$	
	$= 0.41 \text{ kg}$	

11. M/J 25/P21/Q3/c(i)

3(c)(i)	Air resistance is the same, as the speed is the same	B1
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12. M/J 25/P22/Q4/a,b

4(a)	The (total) <u>kinetic</u> energy changes / decreases so (the collision is) inelastic	B1
	OR	(B1)
	(relative) speed of approach not equal to / greater than (relative) speed of separation so (collision is) inelastic	
4(b)(i)	$p = mv$ or 0.25×3.6 or 0.25×5.2	C1
	$\Delta p = 0.25 \times (3.6 + 5.2)$	
	$= 2.2 \text{ kg m s}^{-1}$	A1
4(b)(ii)	$F = \Delta p / (\Delta)t$	C1
	$= 2.2 / 0.18$	
	$= 12 \text{ N}$	A1
	OR	(C1)
	$F = ma$ and $a = (v-u) / t$	
	$= m\Delta v / t$	
	$= 0.25 \times (3.6 + 5.2) / 0.18$	
	$= 12 \text{ N}$	(A1)

13. M/J 25/P23/Q3/c(i)

3(c)(i)	Air resistance is the same, as the speed is the same	B1
---------	--	-----------

14. M/J 25/P23/Q4/a,b

4(a)	<u>sum / total</u> momentum (of a system of bodies) is constant or <u>sum / total</u> momentum before = <u>sum / total</u> momentum after	M1
	for an isolated system / no (resultant) <u>external</u> force	A1
4(b)(i)	$p = mv$ or 4.0×6.0 or 2.0×3.0	C1
	$(4.0 \times 6.0) + (2.0 \times 3.0) = 6.0 \times v$	
	so $v = 5.0 \text{ m s}^{-1}$	A1

4(b)(ii)	KE before = $(0.5 \times 4.0 \times 6.0^2) + (0.5 \times 2.0 \times 3.0^2)$ = 81 J	C1
	KE after = $(0.5 \times 6.0 \times 5.0^2)$ = 75 J	(C1)
	percentage transferred = $[(81 - 75) / 81] \times 100$ = 7%	A1

15. M/J 25/P24/Q2/a,b,c,d,e

2(a)	product of mass and velocity	B1
2(b)	change in momentum = $(-1.4) - (+2.8)$ = $(-) 4.2 \text{ kg m s}^{-1}$	A1
2(c)	$F = \Delta p / (\Delta t)$ or $F = \text{gradient}$ = $4.2 / 12$	C1
	= 0.35 N	A1
2(d)	constant / uniform (rate of) decrease (of speed to zero).	B1
2(e)	- The (resultant) force is constant / does not decrease - air resistance would vary / decrease / not constant - the force is not zero when speed / velocity is zero / at 8 s Any two of the above 3 marking points (1 mark each, max 2)	B2

16. M/J 25/P24/Q3/e

3(b)	$(\Delta)E_{(P)} / 20 = mg(\Delta)h$ OR $(\Delta)E_{(P)} / 20 = mgx_0$	C1
	$(x_0 =) 20 / 5.5 \times 9.81 = 0.37 \text{ (m)}$ Allow $(\Delta)h$ for x_0	A1
3(c)	$[\max (E_{(P)})] = 110 + 20 = 130 \text{ J}$	A1
3(d)	$(\max) E_{(P)} / 130 = \frac{1}{2} F_{(0)} x_{(0)}$	C1
	$(F_0 =) 2 \times 130 / 0.37 = 700 \text{ (N)}$	A1
	or $(\max) E_{(P)} / 130 = 1 / 2 k x_{(0)}^2$ $(F_0 =) k x_{(0)}$	(C1)
	$(F_0 =) 2 \times 130 \times 0.37 / (0.37)^2 = 700 \text{ (N)}$	(A1)
3(e)(i)	(weight) = 5.5×9.81	C1
	resultant force = $700 - (5.5 \times 9.81)$ = 650 N	A1
3(e)(ii)	$F = ma$	C1
	$a = 650 / 5.5$ or $(700 / 5.5) - 9.81$ $a = 120 \text{ m s}^{-2}$	A1

17. MAR 25/P22/Q3/b

3(b)(i)	rate of change of momentum	B1
3(b)(ii)	(Force =) $(2.5 \times 10^4 - 21 \times 10^4) / 15 = -1.2 \times 10^4 \text{ (N)}$	A1

3(b)(iii)	The change of momentum (for S and R to come to rest) is the same	B1
	(Average) force on R (to come to rest) is $(-21 \times 10^4 / 23 =) 0.91 \times 10^4 \text{ N}$ or (Average) force on R (to come to rest) is less than the force on S / less than F	B1
	(S will come to rest in) less time (than R).	B1
	OR The change of momentum (for S and R to come to rest) is the same	(B1)
	Time for S to come to rest is $(-21 \times 10^4 / 1.2 \times 10^4 =) 17.5 \text{ s}$ (and time for R to come to rest is 23 s)	(B1)
	(S comes to rest in) less time (than R).	(B1)

18. MAR 25/P22/Q7/b

7(b)	$4(u) \times 1.5 \times 10^7$ or $222(u) \times v$	C1
	$v = 4(u) \times 1.5 \times 10^7 / 222(u)$	C1
	$= 2.7 \times 10^5 \text{ m s}^{-1}$	A1

19. O/N 24/P21/Q2/a,b(i,iii),c

2(a)	product of mass and velocity	B1
2(b)(i)	maximum speed $= 7.2 \times 10^4 / 1800$ $= 40 \text{ m s}^{-1}$	A1
2(b)(iii)	$a = \text{gradient of line / mass}$ or $a = (v - u) / t$ or $a = \Delta v / t$	C1
	$a = \text{e.g. } (3.6 \times 10^4) / (4.0 \times 1800) = 5.0 \text{ m s}^{-2}$ or $a = \text{e.g. } 20 (-0) / 4 = 5.0 \text{ m s}^{-2}$ or $F = \text{e.g. } 3.6 \times 10^4 / 4.0 = 9.0 \times 10^3$ and $a = 9.0 \times 10^3 / 1800 = 5.0 \text{ m s}^{-2}$	A1
2(c)	stepped shape, showing one constant value up to 8.0 s then stepping to a different constant value from 8.0 s with no time delay	B1
	horizontal straight line from (0, 5.0) to (8.0, 5.0)	B1
	horizontal straight line from (8.0, -10.0) to (12.0, -10.0)	B1

20. O/N 24/P22/Q2/a,b

2(a)	product of mass and velocity	B1
2(b)	$F = m\Delta v / \Delta t$	C1
	$= 16 \times 0.60 / 1.1$	
	$= 8.7 \text{ N}$	A1

21. O/N 24/P22/Q6/c

6(c)(i)	$212(u) \times 1.3 (\times 10^5) \sin 68^\circ$ or $4(u) \times 150 (\times 10^5) \sin \theta$	C1
	$4(u) \times 150 (\times 10^5) \times \sin \theta = 212(u) \times 1.3 (\times 10^5) \times \sin 68^\circ$	C1
	$\sin \theta = 0.426$	A1
	$\theta = 25^\circ$	

6(c)(ii)	$E = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 4 \text{ u} \times (150 \times 10^5)^2$	A1
	$= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (150 \times 10^5)^2$	
	$= 7.5 \times 10^{-13} \text{ J}$	

22. O/N 24/P23/Q2/a,b,c

2(a)	<u>sum / total</u> momentum (of a system of bodies) is constant or <u>sum / total</u> momentum before = <u>sum / total</u> momentum after	M1
	for an isolated system / no (resultant) <u>external</u> force	A1
2(b)(i)	$240 \times 16 = 480v$ and so $v = 8.0 \text{ m s}^{-1}$ or (initial momentum =) $240 \times 16 (= 3840 \text{ g m s}^{-1})$ and $v = 3840 / 480 = 8.0 \text{ m s}^{-1}$	A1
2(b)(ii)	$(E_k =) \frac{1}{2}mv^2$	C1
	$\Delta E_k = \frac{1}{2}[(0.24 \times 16^2) - (0.48 \times 8.0^2)]$	C1
	$= 15 \text{ J}$	A1
2(c)(i)	$F = (0.24 \times 16) / (2.0 \times 10^{-3})$ or $F = (0.48 \times 8) / (2.0 \times 10^{-3})$	C1
	$= 1900 \text{ N}$	A1
	direction: to the left	B1
2(c)(ii)	equal (magnitude)	B1
	opposite (direction)	B1

23. M/J 24/P21/Q1/b,c,d

1(b)	one arrow vertically downward labelled weight to within 10° of the vertical	B1
	one arrow vertically upwards labelled drag / drag force / F_D / air resistance / viscous force to within 10° of the vertical	B1
1(c)	(at terminal velocity) $F_D = mg$	C1
	$F_D = 0.049 \times 9.81$	A1
	$= 0.48 \text{ N}$	
1(d)	area $= \pi \times (0.060 / 2)^2$	C1
	$0.48 = \frac{1}{2} \times C \times 1.2 \times \pi \times (0.060 / 2)^2 \times 25^2$	C1
	$C = 0.45$	A1

24. M/J 24/P21/Q3/a,b,c

3(a)	<u>sum / total</u> momentum before (a collision) = <u>sum / total</u> momentum after (a collision) or <u>sum / total</u> momentum (of a system) is constant	M1
	if no (resultant) external force (acts) / for an isolated system	A1

3(b)	along direction of motion: $10m = 2mv \cos 30^\circ + 3mw \cos 30^\circ$	C1
	perpendicular to direction of motion: $2mv \cos 60^\circ = 3mw \cos 60^\circ$ $(v = 3w / 2)$	C1
	$v = 2.9 \text{ m s}^{-1}$	A1
	$w = 1.9 \text{ m s}^{-1}$	A1
3(c)	$E_K = \frac{1}{2} \times m \times v^2$ $(= \frac{1}{2} \times 4.2 \times 6.0^2)$ $(= 76 \text{ J})$	C1
	force = work done / distance	C1
	force = $76 / 0.050$ $= 1500 \text{ N}$	A1
3(c)	or	
	$a = (-)u^2 / 2s$ $= (-)6.0^2 / (2 \times 0.050)$ $(= (-)360 \text{ m s}^{-2})$	(C1)
	$F = ma$	(C1)
	$F = 4.2 \times 360$ $= 1500 \text{ N}$	(A1)
	or	
	$a = (-)u^2 / 2s$ $= (-)6.0^2 / (2 \times 0.050)$ $(= (-)360 \text{ m s}^{-2})$	(C1)
	$t = -u / a$ $= -6.0 / -360 (= 0.017 \text{ s})$ $F = \Delta p / t$	(C1)
	$F = (0 - 4.2 \times 6) / 0.017$ $= 1500 \text{ N}$	(A1)

25. M/J 24/P22/Q2/a(i),b,c

2(a)(i)	39 m s^{-1}	A1
2(b)	$(\Sigma)F = 68 \times 9.81 - 1800$ $(= -1133 \text{ N})$	C1
	$a = (\Sigma)F / m = -1133 / 68$ $= (-)17 \text{ m s}^{-2}$	A1
	upwards	B1
2c(i)	drag force decreases (as speed decreases)	B1
	(as speed decreases) resultant force decreases so (magnitude of) acceleration decreases (to zero)	B1

2(c)(ii)	$(\Delta)p = m\Delta v$ or $(\Delta)p = m(v - u)$	C1
	$= 68 \times (5.7 - 39)$	A1
	$= (-)2300 \text{ N s}$	

26. M/J 24/P23/Q1/b,c,d(ii)

1(b)	viscosity $= 0.096 / (6 \times \pi \times 0.03 \times 2.0)$	C1
	$= 0.085 \text{ kg m}^{-1} \text{ s}^{-1}$	A1
1(c)	one arrow vertically downwards labelled weight / W	B1
	arrow(s) vertically upwards labelled U / upthrust and drag / F_D / viscous force	B1
1(d)(ii)	weight $= 1.0 + 0.096 (= 1.096 \text{ N})$	C1
	$m = 1.096 / 9.81$	A1
	$= 0.11 \text{ kg}$	

27. MAR 24/P22/Q2/c(i,iii)

2(c)(i)	(As the diver moves down their) speed decreases	B1
	(So) viscous force / drag (force) decreases	B1
2c(iii)	resultant force $= 740 + 950 - (78 \times 9.81)$ $= 925$	C1
	acceleration $= F / m$	C1
	$= 925 / 78$ $= 12 \text{ m s}^{-2}$	A1
	(vertically) upwards	B1

28. MAR 24/P22/Q4/b

4(b)(i)	sum / total momentum (of a system of bodies) is constant or sum / total momentum before = sum / total momentum after	M1
	for an isolated system / no (resultant) external force	A1
4(b)(ii)	$p_a = p_P - p_Q$ $4(u)v = 215(u) \times 3.2 \times 10^5 (-0)$ $v = 215(u) \times 3.2 \times 10^5 / 4(u)$	C1
	$v = 1.7 \times 10^7 \text{ m s}^{-1}$	A1

29. O/N 23/P21/Q2/a(iii),b

2(a)(iii)	$m = W / g$ or $a = F / m$	C1
	$a = (4.00 \times 10^2) / [(3.39 \times 10^4) / 9.81]$	C1
	$a = 0.12 \text{ m s}^{-2}$	A1
2(b)	there is air resistance (which increases with speed)	B1
	(average) resultant force is less (than weight)	B1
	(average) acceleration is less (than $g / 9.81$, so speed is less than 100 m s^{-1})	B1

30. O/N 23/P21/Q3/b(ii)

3(b)(ii)	$(0.25 \times 0.48) + (0.75 \times 0.12) = (0.25 v) + (0.75 \times 0.30)$ or $(0.48 - 0.12) = (0.30 - v)$ or $(\frac{1}{2} \times 0.25 \times 0.48^2) + (\frac{1}{2} \times 0.75 \times 0.12^2) = (\frac{1}{2} \times 0.25 \times v^2) + (\frac{1}{2} \times 0.75 \times 0.30^2)$	C1
	$v = (-)0.060 \text{ m s}^{-1}$	A1
	direction: to the left / from the right / opposite to (its) initial velocity / opposite to (initial / final) velocity of B	B1

31. O/N 23/P22/Q2/b,c

2(b)(i)	acceleration = 9.8 m s^{-2}	A1
2(b)(ii)	air resistance (acting on sphere) increases	B1
	resultant force (on sphere) decreases	B1
	(magnitude of) acceleration decreases	B1
2(c)	$F = ma$ $= 4.0 \times 1.9$ $(= 7.6 \text{ N})$	C1
	resistive force = $(4.0 \times 9.81) - (4.0 \times 1.9)$ $= 32 \text{ N}$	A1

32. O/N 23/P22/Q4/a,b

4(a)(i)	component of momentum = $0.25 \times 3.7 \times \sin 27^\circ$	C1
	$= 0.42 \text{ N s}$	A1
4(a)(ii)	$m_{(Z)} \times 5.5 \times \sin 44^\circ = 0.42$ or $m_{(Z)} \times 5.5 \times \sin 44^\circ = 0.25 \times 3.7 \times \sin 27^\circ$ $m_Z = 0.11 \text{ kg}$	A1
4(a)(iii)	magnitudes: equal	B1
	directions: opposite	B1
4(b)	$4 + 6 = 2 + v$ $v = 8 \text{ m s}^{-1}$	A1

33. O/N 23/P23/Q2/b

2(b)	(collision is) inelastic	B1
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34. O/N 23/P23/Q3/c(i,ii)

3(c)(i)	$(\Delta p) = F(\Delta t)$ or $F = \Delta p / (\Delta t)$ or $F = \text{rate of change of momentum}$ or $F = m(v - u) / t$	C1
	$\Delta p = 2800 \times 0.50$	A1
	$= 1400 \text{ kg m s}^{-1}$	
3(c)(ii)	$(\Delta p =) mv$	C1
	$v = 1400 / 64$	A1
	$= 22 \text{ m s}^{-1}$	

35. M/J 23/P22/Q3

- (a) (i) rate of change of momentum [1]
 (ii) change in momentum = $(1.4 - 0.80) \times 3.0$
 $= 1.8 \text{ kg m s}^{-1}$ [2]
- (b) (i) resultant force (on block) is zero
 (so) velocity is constant [2]
 (ii) $P = Fv$ or $P = Fs/t$
 $v = 2.0 / 0.80 (= 2.5 \text{ m s}^{-1})$
 distance = 2.5×3.0
 $= 7.5 \text{ m}$
 or
 $P = W/t$ or $P = Fs/t$
 $W = 2.0 \times 3.0 (= 6.0 \text{ J})$
 distance = $6.0 / 0.80$
 $= 7.5 \text{ m}$ [3]
- (c) 0 to 3.0 s: upward sloping straight line from the origin.
 3.0 to 6.0 s: horizontal line at non-zero value of momentum with no 'step change' in momentum at 3.0s [2]

36. M/J 23/P21/Q3

- (a) (resultant) force (on an object) is proportional to / equal to the rate of change of momentum [1]
- (b) (i) resultant force = e.g. $6.0 / 4.0$
 $= 1.5 \text{ N}$ [1]
 (ii) force X = $1.5 + 2.0$
 $= 3.5 \text{ N}$ [1]
- (c) from $t = 0$ to $t = 4.0 \text{ s}$: horizontal line at any non-zero value of X
 from $t = 0$ to $t = 4.0 \text{ s}$: horizontal line at $X = 3.5 \text{ N}$
 from $t = 4.0 \text{ s}$ to $t = 6.0 \text{ s}$: horizontal line at $X = 2.0 \text{ N}$ [3]

37. M/J 23/P23/Q3

- (a) sum / total momentum before = sum / total momentum after
 or
sum / total momentum (of a system of objects) is constant
 if no (resultant) external force / for an isolated system [2]
- (b) (i) $3m \times 4 = m \times v \sin \theta$
 $(v \sin \theta = 12)$
 $2m \times 6 = m \times v \cos \theta$
 $(v \cos \theta = 12)$
 therefore $\sin \theta = \cos \theta$ or $\tan \theta = 1$
 $\theta = 45^\circ$ [3]
 (ii) $mv \times \cos 45^\circ = 12m$
 or

$$mv \times \sin 45^\circ = 12m$$

or

$$(mv)^2 = (3m \times 4)^2 + (2m \times 6)^2$$

$$v = 17 \text{ m s}^{-1}$$

[2]

(c) (i) (chemical energy) = $0.0050 \times 700 = 3.5 \text{ (J)}$

or

(chemical energy) = $5.0 \times 0.700 = 3.5 \text{ (J)}$

[1]

(ii) $E = \frac{1}{2}mv^2$

$$\text{total } E = (0.5 \times 3m \times 4^2) + (0.5 \times 2m \times 6^2) + (0.5 \times m \times 17^2)$$

$$3.5 = 204m$$

$$m = 0.017 \text{ kg}$$

[3]

38. MAR 23/P22/Q4

(a) $E = \frac{1}{2}mv^2$

$$p = mv$$

$$m = 0.37^2 / (2 \times 0.30) \text{ or } 0.37 / 1.6 \text{ or } (0.30 \times 2) / 1.6^2$$

$$= 0.23 \text{ kg}$$

[3]

(b) $0.37 - 0.65 = -0.13 - p$

$$p = 0.15 \text{ kg m s}^{-1}$$

[1]

(c) $7.7 = (0.13 + 0.37) / (\Delta)t$

or

$$7.7 = (0.65 - 0.15) / (\Delta)t$$

$$\text{time} = 0.065 \text{ s}$$

[2]

39. M/J 22/P22/Q4

(a) sum/total momentum before = sum/total momentum after

or

sum/total momentum (of a system of objects) is constant

if no (resultant) external force/for a closed system

[2]

(b) (i) $(3.0 \times 4.0 \times \cos \theta) \text{ or } (2.5 \times 4.8 \times \cos \theta) \text{ or } (5.5 \times 3.7)$

$$(3.0 \times 4.0 \times \cos \theta) + (2.5 \times 4.8 \times \cos \theta) = (5.5 \times 3.7)$$

$$\theta = 32^\circ$$

[3]

(ii) (initial $E_K = \frac{1}{2} \times 3.0 \times 4.0^2 + \frac{1}{2} \times 2.5 \times 4.8^2 = 53 \text{ (J)}$)

or

$$(\text{final } E_K = \frac{1}{2} \times 5.5 \times 3.7^2 = 38 \text{ (J)})$$

values of initial E_K and final E_K both correct **and** inelastic stated

[2]

40. O/N 21/P21/Q2

(a) mass $\times$ velocity

[1]

(b) (i) kinetic energy = $\frac{1}{2}mv^2$

$$= \frac{1}{2} \times 0.24 \times 2.3^2$$

$$= 0.63 \text{ J}$$

[3]

(ii) change in momentum = $\frac{1}{2} \times 240 \times 5.0 \times 10^{-3}$

$$= 0.60 \text{ N s}$$

[2]

(iii) (change in velocity of Y) = $0.60 / 0.12$
 (= 5.0 m s^{-1})

final velocity of Y = $5.0 - 2.3$
 = 2.7 m s^{-1}

or

(final momentum of Y) = $0.60 - 0.12 \times 2.3$
 (= 0.324 N s)

final velocity of Y = $0.324 / 0.12$
 = 2.7 m s^{-1}

[2]

- (c) sloping straight line from (0, 0) to $t = 3.0 \text{ ms}$ and another straight line continuous with the first from $t = 3.0 \text{ ms}$ to (5.0, 0)
 lines showing maximum force of magnitude 240 N
 lines wholly in the negative F region of the graph

[3]

41. M/J 21/P22/Q2

(a) $v^2 = u^2 + 2as$

$u^2 = 8.7^2 - (2 \times 9.81 \times 1.5)$

$u = 6.8 \text{ m s}^{-1}$

[2]

- (b) (magnitude of) force on ball (by ground) equal to force on ground (by ball)
 (direction of) force on ball (by ground) opposite to force on ground (by ball)

[2]

- (c) (i) ($p =$) 0.059×8.7 or 0.059×5.4
 change in momentum = $0.059 (8.7 + 5.4)$
 = 0.83 N s

[2]

- (ii) resultant force = $0.83 / 0.091$ or $0.059 [(8.7 + 5.4) / 0.091]$
 = 9.1 N

[1]

(iii) ($W =$) 0.059×9.81

($W =$) 0.58 (N)

force = $9.1 + 0.58$
 = 9.7 N

[2]

- (d) straight line with a positive gradient and starting from a non-zero value of speed at $t = 0$ and ending when $t = T$

[1]

- (e) air resistance increases
 resultant force/acceleration decreases so gradient (of curve) decreases

[2]

42. M/J 21/P21/Q2

(a) (i) $P = Fv$
 = 18×1.4
 = 25 W

[2]

(ii) $a = F / m$
 $a = 18 / 72 = 0.25 \text{ (ms}^{-2}\text{)}$
 $t = 1.4 / 0.25$
 = 5.6 s

[3]

(b) (i) $a = (54 - 18) / 72$ or $36 / 72 (= 0.50 \text{ ms}^{-2})$
 $v^2 = 2 \times 0.50 \times 9.5$
 $v = 3.1 \text{ ms}^{-1}$

[3]

(ii) $W = 54 \times 9.5$
 = 510 J

[1]

- (iii) curved line from the origin
gradient of line increases [2]
- (c) (force due to) air resistance increases/changes/not constant
or air resistance increases with speed [1]

43. O/N 20/P21/Q3

- (a) (force =) rate of change of momentum [1]
- (b) (i) $E = \frac{1}{2}mv^2$ or $\frac{1}{2} \times 0.062 \times 3.8^2$ or $\frac{1}{2} \times 0.062 \times 1.7^2$
loss of KE = $\frac{1}{2} \times 0.062 \times (3.8^2 - 1.7^2)$
= 0.36 J [2]
- (ii) $p = mv$ or 0.062×3.8 or 0.062×1.7
change in momentum = $0.062 \times (1.7 + 3.8)$
= 0.34 N s [2]
- (iii) (average resultant force =) $0.34 / 0.081 = 4.2$ (N)
or
(average resultant force =) $0.062 \times (1.7 + 3.8) / 0.081 = 4.2$ (N) [1]
- (iv) 1. average force = $4.2 + (0.062 \times 9.81)$
= 4.8 N
2. average force = 4.8 N [2]

44. M/J 20/P23/Q2

- (a) a body continues at (rest or) constant velocity unless
acted upon by a resultant force [1]
- (b) (i) distance = $[\frac{1}{2} \times (2.0 + 4.4) \times 3.0] + [4.4 \times 2.0]$
= 9.6 + 8.8
= 18 m [2]
- (ii) $a = (v - u) / t$ or gradient or $\Delta v / (\Delta) t$
e.g. $a = (4.4 - 2.0) / 3.0 = 0.80 \text{ m s}^{-2}$ [2]
- (iii) 1. force = $240 \cos 28^\circ$ or $240 \sin 62^\circ$
= 210 N [1]
2. resultant force = 89×0.80 (= 71.2 N)
 $R = 210 - 71$
= 140 N [2]
- (iv) $T \sin 45^\circ = mg$
 $T = (89 \times 9.81) / \sin 45^\circ$
= 1200 N [2]

45. O/N 19/P22/Q3

- (a) force on body A (by body B) is equal (in magnitude) to force on body B (by body A)
force on body A (by body B) is opposite (in direction) to force on body B (by body A) [2]
- (b) (i) $m_X \times 5v$ or $(m_X + m_Y) \times v$
 $m_X \times 5v = (m_X + m_Y) \times v$ (so) $m_Y / m_X = 4$ [2]
- (ii) $(E =) \frac{1}{2}mv^2$
ratio $= [\frac{1}{2} \times (m_X + m_Y) \times v^2] / [\frac{1}{2} \times m_X \times (5v)^2]$
 $= 0.2$ [3]
- (iii) ratio = 1 [1]
- (c) (i) 1. (magnitude of resultant force is) zero
2. (magnitude of resultant force is) constant
(direction of resultant force is) opposite to the momentum [3]
- (ii) horizontal line from (0 ms, 0 squares) ending at (20 ms, 0 squares)
straight line from (20 ms, 0 squares) ending at (40 ms, 4.0 squares [= 4.0 cm vertically])
horizontal line from (40 ms, 4.0 squares) ending at (60 ms, 4.0 squares) [3]

46. O/N 19/P23/Q4/a

- (a) (i) $p = mv$
 $= 0.2(00) \times 6.(00) \times \sin 60(.0)^\circ$ or $0.2(00) \times 6.(00) \times \cos 30(.0)^\circ$
 $= 1.04 \text{ kg ms}^{-1}$ [2]
- (ii) $0.300 \times v_x \times \sin 60.0^\circ = 1.04$
 $v_x = 4.00 \text{ ms}^{-1}$ [1]
- (iii) $0.30 \times 4.0 \times \cos 60^\circ$ or $0.20 \times 6.0 \times \cos 60^\circ$ or $(0.30 + 0.20)v$ or $0.50v$
 $0.30 \times 4.0 \times \cos 60^\circ + 0.20 \times 6.0 \times \cos 60^\circ = (0.30 + 0.20)v$ or $0.50v$
so $v = 2.4 \text{ ms}^{-1}$ [2]

47. M/J 19/P22/Q2/a,b

- (a) (resultant) force proportional/equal to/is rate of change of momentum [1]
- (b) (i) distance = area under graph or $s = \frac{1}{2}(u + v)t$
 $= \frac{1}{2} \times (9 + 13) \times 10$
or
 $s = ut + \frac{1}{2}at^2$
 $= (9 \times 10) + (\frac{1}{2} \times 0.40 \times 10^2)$
or
 $s = vt - \frac{1}{2}at^2$
 $= (13 \times 10) - (\frac{1}{2} \times 0.40 \times 10^2)$
or
 $v^2 = u^2 + 2as$
 $13^2 = 9^2 + (2 \times 0.40 \times s)$
distance = 110 m [2]
- (ii) 1. $a = \text{gradient}$ or $a = (v - u)/t$ or $a = \Delta v / (\Delta t)$
e.g. $a = (14 - 9) / 12.5$ or $(13 - 9) / 10$
 $a = 0.40 \text{ m s}^{-2}$ [2]
2. resultant force = 850×0.40
 $= 340 \text{ N}$ [1]
3. $(F =) 510 + 440 + 340 = 1300 \text{ (N)}$ [1]
4. $P = Fv$
 $= 1300 \times 13$
 $= 1.7 \times 10^4 \text{ W}$ [2]

48. M/J 19/P21/Q2

- (a) (momentum =) mass $\times$ velocity [1]
 (b) (i) time = 40ms [1]
 (ii) 1. (the magnitude of the acceleration is) constant
 2. (the magnitude of the acceleration is) zero [2]
 (c) $F = \Delta p / (\Delta)t$ or $F = \text{gradient}$
 e.g. $F = 0.50 / 40 \times 10^{-3}$
 $= 13 \text{ N}$ [2]
 (d) horizontal line from (0, 0.40) to (60, 0.40)
 straight line from (60, 0.40) to (100, -0.10)
 horizontal line from (100, -0.10) to (160, -0.10) [3]

49. MAR 19/P22/Q3

Answer from the official mark scheme

3(a)	$(m \times 3.0)$ or $(2.5 \times 9.6 \times \cos 60^\circ)$	C1
	$(m \times 3.0) - (2.5 \times 9.6 \times \cos 60^\circ) = 0$ so $m = 4.0 \text{ (kg)}$	A1
3(b)	$2.5 \times 9.6 \times \sin 60^\circ = (4.0 + 2.5) \times V$	C1
	$V = 3.2 \text{ m s}^{-1}$	A1
	or use of momentum vector triangle: $(4.0 \times 3.0)^2 + [(4.0 + 2.5) \times V]^2 = (2.5 \times 9.6)^2$	(C1)
	$V = 3.2 \text{ m s}^{-1}$	(A1)
3(c)	$E = \frac{1}{2}mv^2$ difference in $E_k = \frac{1}{2} \times 2.5 \times (9.6)^2 - \frac{1}{2} \times 4.0 \times (3.0)^2$	C1
	$= 97 \text{ J}$	A1

50. O/N 18/P22/Q3

Answer from the official mark scheme

3(a)	<u>sum/total</u> momentum (of a system of bodies) is constant or <u>sum/total</u> momentum before = <u>sum/total</u> momentum after	M1
	for an isolated system or no (resultant) external force	A1
3(b)(i)	$m = \rho V$	C1
	$= 1.3 \times \pi \times 0.045^2 \times 1.8 \times 2.0 = 0.030 \text{ (kg)}$	A1
3(b)(ii)	1. $(\Delta)p = (\Delta)mv$	C1
	$= 0.030 \times 1.8$	A1
	$= 0.054 \text{ N s}$	
	2. $F = 0.054 / 2.0$ or $0.030 \times 1.8 / 2.0$	A1
3(b)(iii)	$= 0.027 \text{ N}$	
	force on air (by propeller) equal to force on propeller (by air)	M1
	and opposite (in direction)	A1
3(b)(iv)	resultant force $= 0.20 \times 0.075 (= 0.015 \text{ N})$	C1
	frictional force $= 0.027 - 0.015$	
	$= 0.012 \text{ N}$	A1

51. O/N 18/P21/Q2

(a) $(p =) mv$ or 4.0×45 or 2.0×85 or $89v$

$$(4.0 \times 45) - (2.0 \times 85) = 89v$$

$$v = 0.11 \text{ ms}^{-1}$$

[2]

(b) (i) 1. speed of approach = 47 ms^{-1}
and

2. speed of separation = 0

[1]

(ii) speed of separation less than/not equal to speed of approach and so inelastic collision

[1]

(c) force is equal to rate of change of momentum

force on ball (by block) equal and opposite to force on block (by ball)

so rates of change of momentum are equal and opposite

or

force on ball (by block) equal and opposite to force on block (by ball)

force is equal to rate of change of momentum so rates of change of momentum are equal and opposite

[2]

52. O/N 18/P23/Q3

(a) (resultant) force proportional/equal to rate of change of momentum

[1]

(b) (i) $\rho = m/V$

$$V = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 (= 4.59 \times 10^{-4} \text{ m}^3)$$

$$m = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 \times 1000 = 0.46 \text{ kg}$$

[2]

(ii) 1. $(\Delta)p = (\Delta m)v$

$$(\Delta)p = 0.46 \times 13$$

$$= 6.0 \text{ N s}$$

2. $F = 6.0/0.20$

$$= 30 \text{ N}$$

[3]

(iii) force on water (by rocket/nozzle) equal to force on rocket/nozzle (by water) in the opposite direction

[2]

(iv) 1. mass = $0.40 + 0.70 - 0.46 = 0.64 \text{ kg}$

2. acceleration = $[30 - (0.64 \times 9.81)]/0.64$ or $30/0.64 - 9.81$
 $= 37 \text{ ms}^{-2}$

[3]

53. M/J 18/P22/Q3

(a) time = 12 s [1]

(b) distance (up slope) = $\frac{1}{2} \times 12 \times 18$ (= 108)

distance (down slope) = $\frac{1}{2} \times 12 \times 6$ (= 36)

displacement from A = 108 – 36

$$= 72\text{m}$$

[3]

(c) $v = u + at$ or $a = \text{gradient}$ or $a = \Delta v / (\Delta)t$

$a = 6 / 12 = 0.50 \text{ (ms}^{-2}\text{)}$ (other points from the line may be used)

or

$$v^2 = u^2 + 2as \text{ and } u = 0$$

or

$$v^2 = 2as$$

$$a = 6.0^2 / (2 \times 36) = 0.50 \text{ (ms}^{-2}\text{)}$$

or

$$s = ut + \frac{1}{2}at^2 \text{ and } u = 0$$

or

$$s = \frac{1}{2}at^2$$

$$a = 2 \times 36 / 12^2 = 0.50 \text{ (ms}^{-2}\text{)} \text{ or}$$

$$s = vt - \frac{1}{2}at^2$$

$$a = 2 \times (6 \times 12 - 36) / 12^2 = 0.50 \text{ (ms}^{-2}\text{)}$$

[2]

(d) (i) $F = 70 \times 0.50$ (= 35)

frictional force = 80 – 35

$$= 45\text{N}$$

[2]

(ii) $\sin \theta = 80 / (70 \times 9.81)$

$$\theta = 6.7^\circ$$

[2]

(e) (i) $f_0 = (900 \times 340) / (340 + 12)$

$$= 870\text{Hz}$$

[2]

(ii) speed/velocity (of sledge) decreases and (so) frequency increases

[1]

54. M/J 18/P21/Q3

(a) mass is the property (of a body/object) resisting changes in motion

or

mass is the quantity of matter (in a body)

[1]

(b) (i) force on A (by B) equal and opposite to force on B (by A)

or both A and B exert equal and opposite forces on each other

force is rate of change of momentum and time (of contact) is same

[2]

(ii) $p = mv$ or $3M \times 0.40$ or $M \times 0.25$ or $3M \times 0.2$ or Mv

$$(3M \times 0.40) - (M \times 0.25) = (3M \times 0.2) + Mv$$

$$v = (3 \times 0.40) - 0.25 - (3 \times 0.2)$$

$$= 0.35\text{ms}^{-1}$$

[3]

(iii) 1. relative speed of approach = 0.40 + 0.25

$$= 0.65\text{ms}^{-1}$$

2. relative speed of separation = 0.35 – 0.20

$$= 0.15\text{ms}^{-1}$$

[2]

(iv) (relative) speed of separation not equal to/less than (relative) speed of approach

or answers (to (b)(iii)) are not equal

and so inelastic collision

[1]

55. O/N 17/P23/Q3

- (a) sum/total momentum (of system of bodies) is constant
or
sum/total momentum before = sum/total momentum after
for an isolated system/no (resultant) external force [2]
- (b) (i) $p = mv$
 $(4.0 \times 6.0 \times \sin \theta) - (12 \times 3.5 \times \sin 30^\circ) = 0$
or
 $(m_A v_A \times \sin \theta) - (m_B v_B \times \sin 30^\circ) = 0$
 $\theta = 61^\circ$ [3]
- (ii) shows the horizontal momentum component of ball A or of ball B as
 $(4.0 \times 6.0 \times \cos \theta)$ or $(12 \times 3.5 \times \cos 30^\circ)$
 $(4.0 \times 6.0 \times \cos 61^\circ) + (12 \times 3.5 \times \cos 30^\circ) = 4.0v$ so $v = 12 \text{ (ms}^{-1}\text{)}$ [2]
- (iii) initial $E_K (= \frac{1}{2} \times 4.0 \times 12^2) = 290 \text{ (288) (J)}$
final $E_K (= \frac{1}{2} \times 4.0 \times 6.0^2 + \frac{1}{2} \times 12 \times 3.5^2) = 150 \text{ (145.5) (J)}$
(initial $E_K >$ final E_K) so inelastic [both M1 marks required to award this mark] [3]

56. M/J 17/P22/Q4

- (a) a body/mass/object continues (at rest or) at constant/uniform velocity
unless acted on by a resultant force [1]
- (b) (i) initial momentum = final momentum
 $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$
 $0.60 \times 100 - 0.80 \times 200 = -0.40 \times 100 + v \times 200$
 $v = (-) 0.3(0) \text{ ms}^{-1}$ [2]
- (ii) kinetic energy is not conserved/is lost (but) total energy is conserved/constant
or
some of the (initial) kinetic energy is transformed into other forms of energy [1]

57. M/J 17/P21/Q2

- (a) resultant force (in any direction) is zero
resultant torque/moment (about any point) is zero [2]
- (b) (i) $a = (v - u) / t$ or gradient or $\Delta v / (\Delta t)$
e.g. $a = (8.8 - 4.6) / (7.0 - 4.0) = 1.4 \text{ ms}^{-2}$ [2]
- (ii) $s = 4.6 \times 4 + [(8.8 + 4.6) / 2] \times 3$
 $= 18.4 + 20.1$
 $= 39 \text{ (38.5) m}$ [2]
- (iii) $\Delta E = \frac{1}{2} \times 95 [(8.8)^2 - (4.6)^2]$
 $= 3678 - 1005$
 $= 2700 \text{ (2673) J}$ [2]
- (iv) 1. weight = $95 \times 9.81 (= 932 \text{ N})$
vertical tension force = $280 \sin 25^\circ$ or $280 \cos 65^\circ (= 118.3 \text{ N})$
 $F = 932 + 118$
 $= 1100 \text{ (1050) N}$ [3]
2. horizontal tension force = $280 \cos 25^\circ$ or $280 \sin 65^\circ (= 253.8 \text{ N})$
resultant force = $95 \times 1.4 (= 133 \text{ N})$
 $133 = 253.8 - R$
 $R = 120 \text{ (120.8) N}$ [3]

58. M/J 17/P23/Q2

- (a) (resultant) force is proportional/equal to the rate of change of momentum [1]
- (b) (i) change in momentum = $m(v_2 - v_1)$
 $= 0.84 \times (8.8 - 4.2)$
 $= 3.9 \text{ (3.86) kg m s}^{-1}$ [2]
- (ii) $F = (3.9 / 4.0) = 0.97 \text{ (0.965) N}$ [1]
- (c) (i) change in momentum for A: $0.84 \times (4.7 - 8.8) = -3.4 \text{ (3.44)}$
 change in momentum for B: $0.73 \times (4.7 - 0) = 3.4 \text{ (3.43)}$
 change in momentum for B is equal and opposite to A [2]
- (ii) change in momentum equal (for A and B)
 force is change in momentum / time and time (of collision) is the same
 hence force on A and B equal and opposite as for Newton's third law [2]
- (iii) inelastic as relative speed of approach not equal to relative speed of separation [1]

59. O/N 16/P22/Q2

- (a) $\Delta E = mg\Delta h$
 $= 0.030 \times 9.81 \times (-)0.31$
 $= (-)0.091 \text{ J}$ [2]
- (b) $E = \frac{1}{2}mv^2$
 (initial) $E = \frac{1}{2} \times 0.030 \times 1.3^2 (= 0.0254)$
 $0.5 \times 0.030 \times v^2 = (0.5 \times 0.030 \times 1.3^2) + (0.030 \times 9.81 \times 0.31)$ so $v = 2.8 \text{ ms}^{-1}$
 or
 $0.5 \times 0.030 \times v^2 = (0.0254) + (0.091)$ so $v = 2.8 \text{ ms}^{-1}$ [3]
- (c) (i) $0.096 = 0.030(v + 2.8)$
 $v = 0.40 \text{ ms}^{-1}$ [2]
- (ii) $F = \Delta p / (\Delta)t$ or $F = ma$
 $= 0.096 / 20 \times 10^{-3}$ or $0.030 (0.40 + 2.8) / 20 \times 10^{-3}$
 $= 4.8 \text{ N}$ [2]
- (d) kinetic energy (of ball and wall) decreases/changes/not conserved, so inelastic
 or
 (relative) speed of approach (of ball and wall) not equal to/greater than (relative)
 speed of separation, so inelastic. [1]
- (e) force = work done / distance moved
 $= (0.091 - 0.076) / 0.60$
 $= 0.025 \text{ N}$ [2]

60. M/J 16/P22/Q1

- (a) acceleration = change in velocity / time (taken) or rate of change of velocity [1]
- (b) (i) $v = 0 + at$ or $v = at$
 $(a = 36/19 =) 1.9 \text{ (1.8947) ms}^{-2}$ [2]
- (ii) $s = \frac{1}{2}(u + v)t$ or $s = v^2/2a$ or $s = \frac{1}{2}at^2$
 $= \frac{1}{2} \times 36 \times 19 = 36^2/(2 \times 1.89) = \frac{1}{2} \times 1.89 \times 19^2$
 $= 340 \text{ m (342 m/343 m/341 m)}$ [1]
- (iii) 1. $(\Delta KE =) \frac{1}{2} \times 95 \times (36)^2$
 $= 62\,000 \text{ (61\,560) J}$ [2]
 2. $(\Delta PE =) 95 \times 9.81 \times 340 \sin 40^\circ$ or $95 \times 9.81 \times 218.5$
 $= 200\,000 \text{ J}$ [2]
- (iv) work done (by frictional force) = $\Delta PE - \Delta KE$
 or
 work done = $200\,000 - 62\,000$ (values from 1b(iii) 1. and 2.)
 (frictional force = $138\,000/340 =$) 410 (406) N [420 N if full figures used] [2]
- (v) $-ma = mg \sin 20^\circ - f$ or $ma = -mg \sin 20^\circ + f$
 $-95 \times 3.0 = 95 \times 3.36 - f$
 $f = 600 \text{ (604) N}$ [2]

61. M/J 16/P21/Q3

- (a) (i) force (= $mg = 0.15 \times 9.81$) = 1.5 (1.47) N [1]
 (ii) resultant force (on ball) is zero so normal contact force = weight
 or
 the forces are in opposite directions so normal contact force = weight
 or
 normal contact force up = weight down [1]
- (b) (i) (resultant) force proportional/equal to rate of change of momentum [1]
 (ii) change in momentum = $0.15 \times (6.2 + 2.5)$ (= 1.305 N s)
 magnitude of force = $1.305/0.12$
 $= 11 \text{ (10.9) N}$
 or
 (average) acceleration = $(6.2 + 2.5) / 0.12$ (= 72.5 ms^{-2})
 magnitude of force = 0.15×72.5
 $= 11 \text{ (10.9) N}$
 (direction of force is) upwards/up [3]
- (iii) there is a change/gain in momentum of the floor
 this is equal (and opposite) to the change/loss in momentum of the ball so
 momentum is conserved [2]
 or
 change of (total) momentum of ball and floor is zero
 so momentum is conserved
 or
 (total) momentum of ball and floor before is equal to the (total) momentum
 of ball and floor after
 so momentum is conserved

62. M/J 16/P23/Q5

- (a) the total momentum of a system (of colliding particles) remains constant provided there is no resultant external force acting on the system/isolated or closed system [2]
- (b) (i) the total kinetic energy before (the collision) is equal to the total kinetic energy after (the collision) [1]
- (ii) $p (= mv = 1.67 \times 10^{-27} \times 500) = 8.4 (8.35) \times 10^{-25} \text{ N s}$ [1]
- (iii) 1. $mv_A \cos 60^\circ + mv_B \cos 30^\circ$ or $m(v_A^2 + v_B^2)^{1/2}$
 2. $mv_A \sin 60^\circ + mv_B \sin 30^\circ$ [2]
- (iv) 8.35×10^{-25} or $500m = mv_A \cos 60^\circ + mv_B \cos 30^\circ$
 and
 $0 = mv_A \sin 60^\circ + mv_B \sin 30^\circ$
 or using a vector triangle
 $v_A = 250 \text{ m s}^{-1}$
 $v_B = 430 (433) \text{ m s}^{-1}$ [3]

63. O/N 15/P23/Q4

- (a) (i) reaction/vertical force = weight – $P \cos 60^\circ$
 $= 180 - 35 \cos 60^\circ$
 $= 160 (163) \text{ N}$ [2]
- (ii) work done = $35 \sin 60^\circ \times 20$
 $= 610 (606) \text{ J}$ [2]
- (b) (i) work done by force P = work done against frictional force [1]
- (ii) horizontal component of P is equal and opposite to frictional force
 vertical component of P + normal reaction force equal and opposite to weight [2]

64. M/J 15/P22/Q4

- (a) ($p =$) mv
 $\Delta p (= -6.64 \times 10^{-27} \times 1250 - 6.64 \times 10^{-27} \times 1250) = 1.66 \times 10^{-23} \text{ N s}$ [2]
- (b) (i) molecule collides with wall/container and there is a change in momentum
 change in momentum / time is force or $\Delta p = Ft$
many/all/sum of molecular collisions over surface/area of container produces pressure [3]
- (ii) more collisions per unit time so greater pressure [1]

65. M/J 15/P21/Q3

- (a) $4.5 \times 50 - 2.8 \times M (= \dots)$
 $(\dots) = -1.8 \times 50 + 1.4 \times M$
 $(M =) 75 \text{ g}$ [3]
- (b) total initial kinetic energy/KE not equal to the total final kinetic energy/KE
 or relative speed of approach is not equal to relative speed of separation
 so not elastic or is inelastic [1]
- (c) force on X is equal and opposite to force on Y (Newton III)
 force equals/is proportional to rate of change of momentum (Newton II)
 time of collision same for both balls hence change in momentum is the same [3]

66. O/N 14/P22/Q1/c

- (a) (i) arrow drawn up to the left of 7.5 N force
approximately 5° to 40° to west of north [1]
- (ii) 1. correct vector triangle or working to show
magnitude of resultant force = 6.6 N
allow 6.5 to 6.7 N if scale diagram [1]
2. magnitude of acceleration = $6.6 / 0.75$
[scale diagram: (6.5 to 6.7) / 0.75]
= 8.8 m s^{-2} [scale diagram: 8.7 – 8.9 m s^{-2}] [2]
- (iii) 19° [use of scale diagram allow 17° to 21° (a diagram must be seen)] [1]